

GEOMETRIC MEASURE THEORY

IZ TECÚM

ABSTRACT. SLMath notes.

CONTENTS

Part 1. Lecture A¹	3
1. Multilinear Algebra	3
1.1. Tensors, exterior powers, and contractions	3
1.2. Bases, Euclidean structure, and simple multivectors	5
1.3. Differential forms, pullback, and Stokes' theorem	8
2. Currents	10
2.1. Test forms, currents, and the boundary operator	11
2.2. Mass, support, products, and pushforward	12
2.3. Regularization and the constancy theorem	15
2.4. Normal, flat, and rectifiable currents	16
3. Slicing, Deformation, and Area Minimization	19
3.1. Slicing and the coarea formula	19
3.2. Closure, boundary rectifiability, and integrality criteria	22
3.3. Area minimization and regularity	23
3.4. Cubical deformation and polyhedral approximation	25
Part 2. Lecture B²	29
4. Radon Measures, Covering, and Differentiation	29
4.1. Outer measures, Carathéodory measurability, and Radon measures	29
4.2. Signed measures and Riesz representation	31
4.3. Covering theorems	32
4.4. Differentiation of Radon measures	34
5. Hausdorff Measures, Lipschitz Maps, and Rectifiability	36
5.1. Hausdorff measure, dimension, and densities	36
5.2. Lipschitz maps, Rademacher's theorem, and Whitney extension	39
5.3. The area formula and rectifiable sets	41
5.4. Approximate tangent planes and the cone criterion	44
6. Varifolds, First Variation, and Allard Regularity	46
6.1. Varifolds and the first variation	46
6.2. Monotonicity, tangent cones, and Allard regularity	49
Part 3. Exercises	53
7. Exercises	53

Date: September 4, 2026.

¹Based on lectures by Camillo De Lellis [7].

²Based on lectures by Annalisa Massaccesi [24].

8. Homework	54
References	57

Part 1. Lecture A³

1. MULTILINEAR ALGEBRA

Exterior algebra encodes the oriented planes and volume elements used for currents [15, 22, 28]. Throughout

V finite-dimensional real, $n := \dim V$, $V^* := (V, \mathbb{R})$,
 $\langle \cdot, \cdot \rangle$ fixed whenever metric quantities occur, $\mathcal{H}^k$ Hausdorff measure, $\mathcal{L}^n$ Lebesgue measure,
 $\mathbf{v}, \mathbf{e}_j, \mathbf{q}_i \in V$, τ, η orientations, $\sigma \in \mathfrak{S}_k$.

1.1. Tensors, exterior powers, and contractions.

Definition 1.1.1 (Tensor spaces and tensor products). For $k, \ell \geq 0$, set

$$T_\ell^k(V) := \left\{ F : V^k \times (V^*)^\ell \longrightarrow \mathbb{R} : F \text{ multilinear} \right\}, \quad T^k(V) := T_0^k(V), \quad T_\ell(V) := T_\ell^0(V),$$

$$T^0(V) = \mathbb{R} = T_0(V), \quad T^1(V) = V^*, \quad T_1(V) = V^{**} \cong V.$$

$$F \in T^k(V) \text{ a } k\text{-covector}, \quad G \in T_\ell(V) \text{ an } \ell\text{-vector}.$$

For $F \in T_\ell^k(V)$ and $G \in T_q^p(V)$, set

$$F \otimes G \in T_{\ell+q}^{k+p}(V),$$

$$(F \otimes G)(\mathbf{v}_1, \dots, \mathbf{v}_{k+p}; \xi_1, \dots, \xi_{\ell+q}) := F(\mathbf{v}_1, \dots, \mathbf{v}_k; \xi_1, \dots, \xi_\ell) \cdot G(\mathbf{v}_{k+1}, \dots, \mathbf{v}_{k+p}; \xi_{\ell+1}, \dots, \xi_{\ell+q}).$$

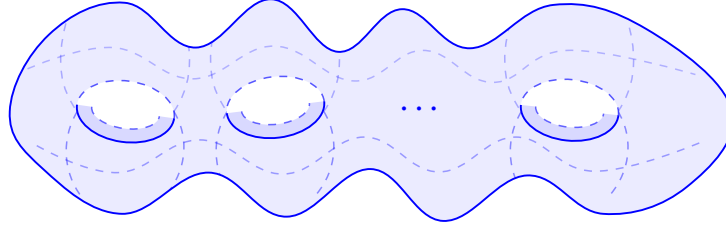


FIGURE 1. Cross-sectional axes resolve tensor data on a smooth genus- g surface.

Definition 1.1.2 (Alternating tensors and the exterior product). For $F \in T^k(V)$ and $G \in T_\ell(V)$, set

$$F \text{ alternating} \iff F(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = \text{sgn}(\sigma) F(\mathbf{v}_1, \dots, \mathbf{v}_k) \quad (\sigma \in \mathfrak{S}_k).$$

$$G \text{ alternating} \iff G(\xi_{\sigma(1)}, \dots, \xi_{\sigma(\ell)}) = \text{sgn}(\sigma) G(\xi_1, \dots, \xi_\ell) \quad (\sigma \in \mathfrak{S}_\ell).$$

$$\Lambda^k(V) := \{ F \in T^k(V) : F \text{ alternating} \}, \quad \Lambda_\ell(V) := \{ G \in T_\ell(V) : G \text{ alternating} \},$$

$$\Lambda^0(V) = \Lambda_0(V) = \mathbb{R}, \quad \Lambda^1(V) = V^*, \quad \Lambda_1(V) = V^{**} \cong V.$$

For $A \in \Lambda^k(V)$ and $B \in \Lambda^p(V)$, set

$$(A \wedge B)(\mathbf{v}_1, \dots, \mathbf{v}_{k+p}) := \frac{1}{k!p!} \sum_{\sigma \in \mathfrak{S}_{k+p}} \text{sgn}(\sigma) A(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) B(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+p)}),$$

$$\wedge : \Lambda^k(V) \times \Lambda^p(V) \longrightarrow \Lambda^{k+p}(V), \quad \wedge : \Lambda_k(V) \times \Lambda_p(V) \longrightarrow \Lambda_{k+p}(V).$$

³Based on lectures by Camillo De Lellis [7].

Proposition 1.1.3 (Exterior-product identities). *For $A_1, A_2 \in \Lambda^k(V)$, $B \in \Lambda^p(V)$, $C \in \Lambda^q(V)$, and $a, b \in \mathbb{R}$, $(aA_1 + bA_2) \wedge B = a(A_1 \wedge B) + b(A_2 \wedge B)$, $(A_1 \wedge B) \wedge C = A_1 \wedge (B \wedge C)$, $A_1 \wedge B = (-1)^{kp} B \wedge A_1$, and, for $\xi_1, \dots, \xi_k \in V^*$ and $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$,*

$$(\xi_1 \wedge \dots \wedge \xi_k)(\mathbf{v}_1, \dots, \mathbf{v}_k) = \det[\langle \xi_i, \mathbf{v}_j \rangle]_{i,j=1}^k.$$

These identities hold equally in $\Lambda_(V)$ [15, Chapter 1] [22, Chapter 1].*

Proof. Permutation shuffles isolate the sign and order data in the exterior product.

$$k_a \in \mathbb{N}_0, \quad s_0 := 0, \quad s_a := \sum_{b=1}^a k_b, \quad \text{Sh}(k_1, \dots, k_r) := \{\sigma \in \mathfrak{S}_r : \sigma(s_{a-1} + 1) < \dots < \sigma(s_a), 1 \leq a \leq r\},$$

$$(A \wedge B)(\mathbf{v}_1, \dots, \mathbf{v}_{k+p}) = \sum_{\sigma \in \text{Sh}(k,p)} \text{sgn}(\sigma) A(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) B(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+p)}),$$

$$(aA_1 + bA_2) \wedge B = a(A_1 \wedge B) + b(A_2 \wedge B), \quad A \wedge B = (-1)^{kp} B \wedge A,$$

$$((A \wedge B) \wedge C)(\mathbf{v}_1, \dots, \mathbf{v}_{k+p+q}) = \sum_{\sigma \in \text{Sh}(k,p,q)} \text{sgn}(\sigma) A(\mathbf{v}_{\sigma(1)}, \dots) B(\mathbf{v}_{\sigma(k+1)}, \dots) C(\mathbf{v}_{\sigma(k+p+1)}, \dots),$$

$$((A \wedge B) \wedge C)(\mathbf{v}_1, \dots, \mathbf{v}_{k+p+q}) = (A \wedge (B \wedge C))(\mathbf{v}_1, \dots, \mathbf{v}_{k+p+q}),$$

$$(\xi_1 \wedge \dots \wedge \xi_k)(\mathbf{v}_1, \dots, \mathbf{v}_k) = \sum_{\sigma \in \mathfrak{S}_k} \text{sgn}(\sigma) \prod_{i=1}^k \langle \xi_i, \mathbf{v}_{\sigma(i)} \rangle = \det[\langle \xi_i, \mathbf{v}_j \rangle]_{i,j=1}^k.$$

The same reindexing applies after exchanging V and V^* . □

Definition 1.1.4 (Exterior pairing and interior products). By Proposition 1.1.3, set

$$\langle \xi_1 \wedge \dots \wedge \xi_k, \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k \rangle := \det[\langle \xi_i, \mathbf{v}_j \rangle]_{i,j=1}^k, \quad \langle \cdot, \cdot \rangle : \Lambda^k(V) \times \Lambda_k(V) \longrightarrow \mathbb{R} \text{ nondegenerate.}$$

For $\alpha \in \Lambda^r(V)$, $\mathbf{u} \in \Lambda_k(V)$, and $0 \leq r \leq k$, set

$$\alpha \lrcorner \mathbf{u} \in \Lambda_{k-r}(V), \quad \langle \beta, \alpha \lrcorner \mathbf{u} \rangle := \langle \alpha \wedge \beta, \mathbf{u} \rangle \quad (\beta \in \Lambda^{k-r}(V)).$$

For $\alpha \in \Lambda^k(V)$, $\mathbf{u} \in \Lambda_r(V)$, and $0 \leq r \leq k$, set

$$\alpha \lrcorner \mathbf{u} \in \Lambda^{k-r}(V), \quad \langle \alpha \lrcorner \mathbf{u}, \mathbf{v} \rangle := \langle \alpha, \mathbf{u} \wedge \mathbf{v} \rangle \quad (\mathbf{v} \in \Lambda_{k-r}(V)).$$

For $\mathbf{v}, \mathbf{w} \in V$, set

$$\langle \mathbf{v}^\flat, \mathbf{w} \rangle := \langle \mathbf{v}, \mathbf{w} \rangle, \quad \mathbf{v} \lrcorner \mathbf{u} := \mathbf{v}^\flat \lrcorner \mathbf{u}, \quad \langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij} \implies \mathbf{v}_1 \lrcorner (\mathbf{v}_1 \wedge \mathbf{v}_2) = \mathbf{v}_2.$$

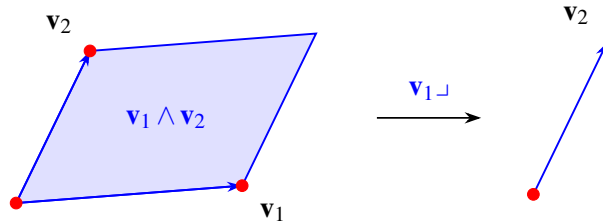


FIGURE 2. Exterior multiplication and interior contraction by $\mathbf{v}_1$.

Lemma 1.1.5 (Vanishing criterion). *For $1 \leq k \leq n$,*

$$\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k = 0 \iff \mathbf{v}_1, \dots, \mathbf{v}_k \text{ dependent}, \quad \xi_1 \wedge \dots \wedge \xi_k = 0 \iff \xi_1, \dots, \xi_k \text{ dependent}.$$

Proof. Dependence creates a repeated exterior factor, while independence supplies a dual frame.

$$\begin{aligned} \sum_{i=1}^k c_i \mathbf{v}_i &= 0, \quad c_j \neq 0 \implies \mathbf{v}_j = -\sum_{i \neq j} \frac{c_i}{c_j} \mathbf{v}_i, \\ \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k &= -\sum_{i \neq j} \frac{c_i}{c_j} \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_{j-1} \wedge \mathbf{v}_i \wedge \mathbf{v}_{j+1} \wedge \cdots \wedge \mathbf{v}_k = 0, \\ \mathbf{v}_1, \dots, \mathbf{v}_k \text{ linearly independent} &\implies \exists \mathbf{v}_{k+1}, \dots, \mathbf{v}_n, \xi_1, \dots, \xi_n : (\mathbf{v}_1, \dots, \mathbf{v}_n) \text{ basis}, \quad \langle \xi_i, \mathbf{v}_j \rangle = \delta_{ij}, \\ \langle \xi_i, \mathbf{v}_j \rangle = \delta_{ij} &\xrightarrow{1.1.3} \langle \xi_1 \wedge \cdots \wedge \xi_k, \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k \rangle = \det[\delta_{ij}]_{i,j=1}^k = 1 \implies \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k \neq 0, \\ V \longleftrightarrow V^* &\implies \xi_1 \wedge \cdots \wedge \xi_k = 0 \iff \xi_1, \dots, \xi_k \text{ linearly dependent.} \end{aligned}$$

Thus each wedge vanishes exactly when its factors are dependent. $\square$

1.2. Bases, Euclidean structure, and simple multivectors.

Definition 1.2.1 (Increasing multi-indices and induced bases). For $0 \leq k \leq n$, set

$$J(k, n) := \{\mathbf{i} = (i_1, \dots, i_k) : 1 \leq i_1 < \cdots < i_k \leq n\}, \quad J(0, n) := \{\emptyset\}, \quad J(k, n) := \emptyset \quad (k > n).$$

Fix

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}, \quad \langle dx_i, \mathbf{e}_j \rangle = \delta_{ij}, \quad 1 \leq i, j \leq n.$$

For $\mathbf{i} = (i_1, \dots, i_k) \in J(k, n)$, set

$$\mathbf{e}_{\mathbf{i}} := \mathbf{e}_{i_1} \wedge \cdots \wedge \mathbf{e}_{i_k}, \quad dx_{\mathbf{i}} := dx_{i_1} \wedge \cdots \wedge dx_{i_k}, \quad \mathbf{e}_{\emptyset} := 1, \quad dx_{\emptyset} := 1.$$

Proposition 1.2.2 (Exterior bases). For $0 \leq k \leq n$,

$$\{\mathbf{e}_{\mathbf{i}} : \mathbf{i} \in J(k, n)\} \text{ and } \{dx_{\mathbf{i}} : \mathbf{i} \in J(k, n)\} \text{ dual bases of } \Lambda_k(V) \text{ and } \Lambda^k(V),$$

$$\dim \Lambda_k(V) = \dim \Lambda^k(V) = \binom{n}{k}, \quad \Lambda_k(V) = \Lambda^k(V) = \{0\} \quad (k > n).$$

Proof. The determinant pairing identifies the coordinates of both exterior powers.

$$\langle dx_i, \mathbf{e}_j \rangle = \delta_{ij} \xrightarrow{1.1.3} \langle dx_{\mathbf{i}}, \mathbf{e}_{\mathbf{j}} \rangle = \det[\delta_{i_a j_b}]_{a,b=1}^k = \delta_{\mathbf{i}\mathbf{j}},$$

$$\omega = \sum_{\mathbf{i} \in J(k, n)} \langle \omega, \mathbf{e}_{\mathbf{i}} \rangle dx_{\mathbf{i}}, \quad \mathbf{u} = \sum_{\mathbf{i} \in J(k, n)} \langle dx_{\mathbf{i}}, \mathbf{u} \rangle \mathbf{e}_{\mathbf{i}},$$

$$|J(k, n)| = \binom{n}{k} \implies \dim \Lambda_k(V) = \dim \Lambda^k(V) = \binom{n}{k},$$

$$k > n \implies k\text{-tuples in } V \text{ are linearly dependent} \implies \Lambda_k(V) = \Lambda^k(V) = \{0\}.$$

Thus the displayed families are dual bases with the stated dimensions. $\square$

Definition 1.2.3 (Induced Euclidean structures and the exterior bundle). For the bases in Proposition 1.2.2, set

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\Lambda_k(V)} := \sum_{\mathbf{i} \in J(k, n)} \langle dx_{\mathbf{i}}, \mathbf{u} \rangle \langle dx_{\mathbf{i}}, \mathbf{v} \rangle, \quad |\mathbf{u}|^2 := \langle \mathbf{u}, \mathbf{u} \rangle_{\Lambda_k(V)},$$

$$\langle \alpha, \beta \rangle_{\Lambda^k(V)} := \sum_{\mathbf{i} \in J(k, n)} \langle \alpha, \mathbf{e}_{\mathbf{i}} \rangle \langle \beta, \mathbf{e}_{\mathbf{i}} \rangle, \quad |\alpha|^2 := \langle \alpha, \alpha \rangle_{\Lambda^k(V)}, \quad \text{independent of the orthonormal basis.}$$

For a smooth n -manifold M , set

$$\Lambda^k T^*M := \bigsqcup_{p \in M} \Lambda^k(T_p M), \quad \pi : \Lambda^k T^*M \longrightarrow M, \quad \pi^{-1}(p) = \Lambda^k(T_p M),$$

$$\Omega^k(M) := \{\omega \in C^\infty(M, \Lambda^k T^*M) : \pi \circ \omega = \text{id}_M\}.$$

Definition 1.2.4 (Simple alternating tensors and parallelotopes). For $1 \leq k \leq n$, set

$$\omega \text{ simple} \iff \omega = \xi_1 \wedge \cdots \wedge \xi_k, \quad \mathbf{u} \text{ simple} \iff \mathbf{u} = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k,$$

$$\mathcal{S}_k(V) := \{\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k : \mathbf{v}_1, \dots, \mathbf{v}_k \in V\}, \quad \mathcal{S}_k^1(V) := \{\mathbf{u} \in \mathcal{S}_k(V) : |\mathbf{u}| = 1\}, \quad \mathcal{S}_0(V) := \mathcal{S}_0^1(V) := \{1\}.$$

For $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$, set

$$\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k) := \left\{ \sum_{i=1}^k \lambda_i \mathbf{v}_i : 0 \leq \lambda_i \leq 1 \right\}.$$

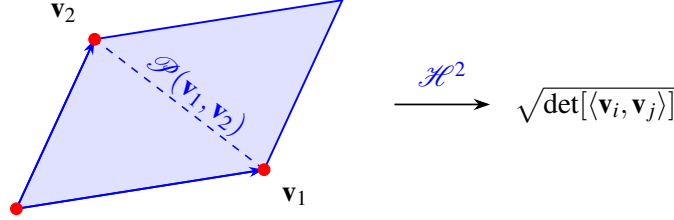


FIGURE 3. The oriented parallelotope $\mathcal{P}(\mathbf{v}_1, \mathbf{v}_2)$ and its Gram volume.

Proposition 1.2.5 (Gram norm and parallelotope volume). For $V = \mathbb{R}^n$, $1 \leq k \leq n$, and $\mathbf{u} = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$,

$$|\mathbf{u}|^2 = \det[\langle \mathbf{v}_i, \mathbf{v}_j \rangle]_{i,j=1}^k, \quad |\mathbf{u}| = \mathcal{H}^k(\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k)).$$

Proof. Cauchy–Binet converts exterior coordinates to the Gram determinant, while a reduced QR factorization computes volume.

$$A := [\mathbf{v}_1 \ \cdots \ \mathbf{v}_k] \in \mathbb{R}^{n \times k}, \quad A_i := [A_{ab}]_{a,b=1}^k, \quad A = QR, \quad Q^\top Q = I_k,$$

$$\mathbf{u} = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k \xrightarrow{1.1.3} \langle d\mathbf{x}_i, \mathbf{u} \rangle = \det(A_i), \quad |\mathbf{u}|^2 = \sum_{\mathbf{i} \in J(k,n)} \det(A_i)^2,$$

$$\sum_{\mathbf{i} \in J(k,n)} \det(A_i)^2 = \det(A^\top A) = \det[\langle \mathbf{v}_i, \mathbf{v}_j \rangle]_{i,j=1}^k,$$

$$\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k) = Q(R([0, 1]^k)), \quad \mathcal{H}^k(\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k)) = |\det R| = \sqrt{\det(A^\top A)} = |\mathbf{u}|.$$

Figure 3 depicts the resulting identity for $k = 2$. □

Corollary 1.2.6 (Change of frame). For $M \in \text{GL}(k, \mathbb{R})$ and linearly independent $\mathbf{v}_1, \dots, \mathbf{v}_k$, set

$$\tilde{\mathbf{v}}_j := \sum_{i=1}^k M_{ij} \mathbf{v}_i, \quad 1 \leq j \leq k.$$

Then

$$\tilde{\mathbf{v}}_1 \wedge \cdots \wedge \tilde{\mathbf{v}}_k = \det(M) \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k, \quad \mathcal{H}^k(\mathcal{P}(\tilde{\mathbf{v}}_1, \dots, \tilde{\mathbf{v}}_k)) = |\det(M)| \mathcal{H}^k(\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k)).$$

Proof. Multilinearity turns a frame change into its determinant, and the Gram formula removes its sign.

$$\tilde{A} := AM, \quad \tilde{\mathbf{v}}_1 \wedge \cdots \wedge \tilde{\mathbf{v}}_k = \sum_{i_1, \dots, i_k=1}^k M_{i_1 1} \cdots M_{i_k k} \mathbf{v}_{i_1} \wedge \cdots \wedge \mathbf{v}_{i_k},$$

$$\sum_{i_1, \dots, i_k=1}^k M_{i_1 1} \cdots M_{i_k k} \mathbf{v}_{i_1} \wedge \cdots \wedge \mathbf{v}_{i_k} = \det(M) \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k,$$

$$\tilde{A}^\top \tilde{A} = M^\top A^\top A M, \quad \det(\tilde{A}^\top \tilde{A}) = (\det M)^2 \det(A^\top A),$$

$$\det(\tilde{A}^T \tilde{A}) = (\det M)^2 \det(A^T A) \xrightarrow{1.2.5} \mathcal{H}^k(\mathcal{P}(\tilde{\mathbf{v}}_1, \dots, \tilde{\mathbf{v}}_k)) = |\det M| \mathcal{H}^k(\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k)).$$

Thus orientation records $\text{sgn}(\det M)$, while volume records $|\det M|$. $\square$

Definition 1.2.7 (Oriented Grassmannian). For $1 \leq k \leq n$, set

$$\text{Fr}_k(V) := \{(\mathbf{v}_1, \dots, \mathbf{v}_k) \in V^k : \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k \neq 0\},$$

$$(\mathbf{v}_1, \dots, \mathbf{v}_k) \sim_+ (\mathbf{w}_1, \dots, \mathbf{w}_k) \iff \mathbf{w}_j = \sum_{i=1}^k M_{ij} \mathbf{v}_i, \quad M \in \text{GL}(k, \mathbb{R}), \quad \det M > 0,$$

$$\text{Gr}_k^+(V) := \text{Fr}_k(V) / \sim_+.$$

Corollary 1.2.8 (Oriented Grassmannian). For $1 \leq k \leq n$, the map

$$\Phi : \mathcal{S}_k^1(V) \longrightarrow \text{Gr}_k^+(V), \quad \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k \longmapsto [(\mathbf{v}_1, \dots, \mathbf{v}_k)]_+,$$

Φ bijective.

Proof. Positive frames select one of the two unit exterior representatives of each k -plane.

$$\mathbf{u} = \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k = \mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_k, \quad \mathbf{w}_j = \sum_{i=1}^k M_{ij} \mathbf{v}_i \xrightarrow{1.2.6} \det M = 1 \implies [(\mathbf{v}_1, \dots, \mathbf{v}_k)]_+ = [(\mathbf{w}_1, \dots, \mathbf{w}_k)]_+,$$

$$\mathcal{Q} \in \text{Gr}_k^+(V) \implies \exists (\mathbf{q}_1, \dots, \mathbf{q}_k) \in \mathcal{Q} : \langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij}, \quad \mathbf{u}_{\mathcal{Q}} := \mathbf{q}_1 \wedge \dots \wedge \mathbf{q}_k \in \mathcal{S}_k^1(V), \quad \Phi(\mathbf{u}_{\mathcal{Q}}) = \mathcal{Q},$$

$$\Phi(\mathbf{u}) = \Phi(\tilde{\mathbf{u}}) \implies \tilde{\mathbf{v}}_j = \sum_{i=1}^k M_{ij} \mathbf{v}_i, \quad \det M > 0, \quad |\mathbf{u}| = |\tilde{\mathbf{u}}| = 1,$$

$$|\mathbf{u}| = |\tilde{\mathbf{u}}| = 1 \xrightarrow{1.2.6} |\det M| = 1, \quad \det M > 0 \implies \det M = 1 \implies \mathbf{u} = \tilde{\mathbf{u}}.$$

Thus Φ is a bijection, and Figure 4 displays its two orientations. $\square$

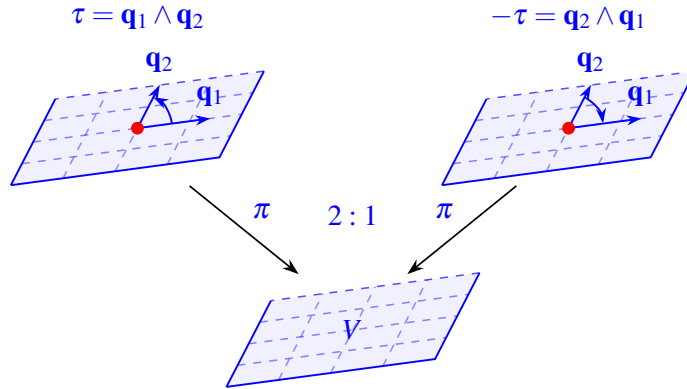


FIGURE 4. Each k -plane has the two orientations $\pm \tau$.

Definition 1.2.9 (Comass and mass norms). For $\omega \in \Lambda^k(\mathbb{R}^n)$ and $\mathbf{v} \in \Lambda_k(\mathbb{R}^n)$, set

$$\|\omega\|^* := \sup\{|\langle \omega, \mathbf{u} \rangle| : \mathbf{u} \in \mathcal{S}_k^1(\mathbb{R}^n)\}, \quad \|\mathbf{v}\| := \sup\{|\langle \omega, \mathbf{v} \rangle| : \omega \in \Lambda^k(\mathbb{R}^n), \|\omega\|^* \leq 1\},$$

$$\|\cdot\|^* \text{ comass norm}, \quad \|\cdot\| \text{ mass norm}.$$

These dual norms follow the conventions of [15, 28].

Proposition 1.2.10 (Mass of a simple multivector). For $\mathbf{v} = \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k \in \mathcal{S}_k(\mathbb{R}^n)$,

$$\|\mathbf{v}\| = |\mathbf{v}| = \mathcal{H}^k(\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k)).$$

Proof. An orthonormal representative exposes both sides of the dual norm.

$$\begin{aligned}
\mathbf{v} = 0 &\implies \|\mathbf{v}\| = |\mathbf{v}| = \mathcal{H}^k(\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k)) = 0, \\
\mathbf{v} \neq 0 &\implies \exists \mathbf{q}_1, \dots, \mathbf{q}_k : \langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij}, \quad \mathbf{v} = |\mathbf{v}| \mathbf{q}_1 \wedge \dots \wedge \mathbf{q}_k, \quad \boldsymbol{\omega}_0 := \mathbf{q}_1^b \wedge \dots \wedge \mathbf{q}_k^b, \\
\mathbf{u} \in \mathcal{S}_k^1(\mathbb{R}^n) &\implies \mathbf{u} = \mathbf{r}_1 \wedge \dots \wedge \mathbf{r}_k, \quad \langle \mathbf{r}_i, \mathbf{r}_j \rangle = \delta_{ij}, \quad |\langle \boldsymbol{\omega}_0, \mathbf{u} \rangle| = |\det[\langle \mathbf{q}_i, \mathbf{r}_j \rangle]_{i,j=1}^k| \leq 1, \\
\langle \boldsymbol{\omega}_0, \mathbf{q}_1 \wedge \dots \wedge \mathbf{q}_k \rangle &= 1 \implies \|\boldsymbol{\omega}_0\|^* = 1, \quad \langle \boldsymbol{\omega}_0, \mathbf{v} \rangle = |\mathbf{v}| \implies |\mathbf{v}| \leq \|\mathbf{v}\|, \\
\|\boldsymbol{\omega}\|^* \leq 1 &\implies |\langle \boldsymbol{\omega}, \mathbf{v} \rangle| = |\mathbf{v}| \left| \left\langle \boldsymbol{\omega}, \frac{\mathbf{v}}{|\mathbf{v}|} \right\rangle \right| \leq |\mathbf{v}| \implies \|\mathbf{v}\| \leq |\mathbf{v}|, \\
\|\mathbf{v}\| &= |\mathbf{v}| \xrightarrow{1,2,5} \|\mathbf{v}\| = \mathcal{H}^k(\mathcal{P}(\mathbf{v}_1, \dots, \mathbf{v}_k)).
\end{aligned}$$

Thus Euclidean length, mass, and parallelotope volume agree on simple multivectors. $\square$

1.3. Differential forms, pullback, and Stokes' theorem.

Definition 1.3.1 (Differential forms on Euclidean domains). For $U \subset \mathbb{R}^n$ open, $0 \leq k \leq n$, and $0 \leq r \leq \infty$, set

$$\begin{aligned}
C^r(U, \Lambda^k(\mathbb{R}^n)) &:= \left\{ \boldsymbol{\omega} : U \rightarrow \Lambda^k(\mathbb{R}^n) : \boldsymbol{\omega} = \sum_{\mathbf{i} \in J(k,n)} \omega_{\mathbf{i}} dx_{\mathbf{i}}, \omega_{\mathbf{i}} \in C^r(U) \right\}, \\
\text{spt}(\boldsymbol{\omega}) &:= \overline{\{x \in U : \boldsymbol{\omega}(x) \neq 0\}}^U. \\
C_c^r(U, \Lambda^k(\mathbb{R}^n)) &:= \{\boldsymbol{\omega} \in C^r(U, \Lambda^k(\mathbb{R}^n)) : \text{spt}(\boldsymbol{\omega}) \Subset U\}.
\end{aligned}$$

Definition 1.3.2 (Exterior and interior derivatives). For $\boldsymbol{\omega} \in C^r(U, \Lambda^k(\mathbb{R}^n))$ and $r \geq 1$, set

$$\begin{aligned}
D\boldsymbol{\omega}(x) &:= \sum_{\mathbf{i} \in J(k,n)} \sum_{j=1}^n \frac{\partial \omega_{\mathbf{i}}}{\partial x_j}(x) dx_j \otimes dx_{\mathbf{i}} \in \Lambda^1(\mathbb{R}^n) \otimes \Lambda^k(\mathbb{R}^n), \\
d\boldsymbol{\omega} &:= \sum_{\mathbf{i} \in J(k,n)} \sum_{j=1}^n \frac{\partial \omega_{\mathbf{i}}}{\partial x_j} dx_j \wedge dx_{\mathbf{i}} \in C^{r-1}(U, \Lambda^{k+1}(\mathbb{R}^n)), \\
\nabla \lrcorner \boldsymbol{\omega} &:= \sum_{\mathbf{i} \in J(k,n)} \sum_{j=1}^n \frac{\partial \omega_{\mathbf{i}}}{\partial x_j} (dx_{\mathbf{i}} \lrcorner \mathbf{e}_j) \in C^{r-1}(U, \Lambda^{k-1}(\mathbb{R}^n)), \quad \nabla \lrcorner \boldsymbol{\omega} := 0 \quad (k=0).
\end{aligned}$$

For $\boldsymbol{\omega} \in C^1(U, \Lambda^k(\mathbb{R}^n))$ and $1 \leq k \leq n$, set

$$\boldsymbol{\omega} \text{ closed} \iff d\boldsymbol{\omega} = 0, \quad \boldsymbol{\omega} \text{ exact} \iff \exists \boldsymbol{\psi} \in C^1(U, \Lambda^{k-1}(\mathbb{R}^n)) : \boldsymbol{\omega} = d\boldsymbol{\psi}.$$

Proposition 1.3.3 (Nilpotence of the exterior derivative). For $\boldsymbol{\omega} \in C^2(U, \Lambda^k(\mathbb{R}^n))$,

$$d(d\boldsymbol{\omega}) = 0.$$

Proof. Symmetry of second derivatives cancels against antisymmetry of the exterior product.

$$\begin{aligned}
d(d\boldsymbol{\omega}) &= \sum_{\mathbf{i} \in J(k,n)} \sum_{j,\ell=1}^n \frac{\partial^2 \omega_{\mathbf{i}}}{\partial x_{\ell} \partial x_j} dx_{\ell} \wedge dx_j \wedge dx_{\mathbf{i}}, \\
\sum_{j,\ell=1}^n \frac{\partial^2 \omega_{\mathbf{i}}}{\partial x_{\ell} \partial x_j} dx_{\ell} \wedge dx_j &= - \sum_{j,\ell=1}^n \frac{\partial^2 \omega_{\mathbf{i}}}{\partial x_j \partial x_{\ell}} dx_{\ell} \wedge dx_j, \\
d(d\boldsymbol{\omega}) &= \frac{1}{2} \sum_{\mathbf{i} \in J(k,n)} \sum_{j,\ell=1}^n \left(\frac{\partial^2 \omega_{\mathbf{i}}}{\partial x_{\ell} \partial x_j} - \frac{\partial^2 \omega_{\mathbf{i}}}{\partial x_j \partial x_{\ell}} \right) dx_{\ell} \wedge dx_j \wedge dx_{\mathbf{i}}, \\
\frac{\partial^2 \omega_{\mathbf{i}}}{\partial x_{\ell} \partial x_j} &= \frac{\partial^2 \omega_{\mathbf{i}}}{\partial x_j \partial x_{\ell}} \implies d(d\boldsymbol{\omega}) = 0.
\end{aligned}$$

Thus $d^2 = 0$ on C^2 -forms. $\square$

Corollary 1.3.4 (Exact forms are closed). *For $1 \leq k \leq n$ and $\psi \in C^2(U, \Lambda^{k-1}(\mathbb{R}^n))$,*

$$\omega = d\psi \xrightarrow{1.3.3} d\omega = d(d\psi) = 0.$$

Definition 1.3.5 (Pullback). *For $\Omega \subset \mathbb{R}^m$ and $U \subset \mathbb{R}^n$ open, $F \in C^1(\Omega, U)$, and $\omega \in C^0(U, \Lambda^k(\mathbb{R}^n))$, set*

$$F^*\omega \in C^0(\Omega, \Lambda^k(\mathbb{R}^m)), \quad \langle F^*\omega(x), \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k \rangle := \langle \omega(F(x)), DF(x)\mathbf{v}_1 \wedge \cdots \wedge DF(x)\mathbf{v}_k \rangle,$$

$$F^*\omega = \sum_{\mathbf{i}=(i_1, \dots, i_k) \in J(k, n)} (\omega_{\mathbf{i}} \circ F) dF_{i_1} \wedge \cdots \wedge dF_{i_k}, \quad dF_a = \sum_{b=1}^m \frac{\partial F_a}{\partial x_b} dx_b, \quad F^*\omega = \omega \circ F \quad (k=0).$$

$$\begin{array}{ccccc} U & \xrightarrow{f} & V & & \\ Df_x \downarrow & & \downarrow \omega & & \\ T_x U & \xrightarrow{Df_x} & T_{f(x)} V & \xrightarrow{\omega_{f(x)}} & \mathbb{R} \end{array}$$

$$(f^*\omega)_x(\mathbf{v}_1, \dots, \mathbf{v}_k) = \omega_{f(x)}(Df_x \mathbf{v}_1, \dots, Df_x \mathbf{v}_k).$$

FIGURE 5. Pullback of a differential form and naturality of d .

Proposition 1.3.6 (Naturality of the exterior derivative). *For $F \in C^2(\Omega, U)$ and $\omega \in C^1(U, \Lambda^k(\mathbb{R}^n))$,*

$$d(F^*\omega) = F^*(d\omega).$$

Proof. The chain rule differentiates the coefficients, while $d^2 F_i = 0$ removes every remaining product term.

$$F^*\omega = \sum_{\mathbf{i} \in J(k, n)} (\omega_{\mathbf{i}} \circ F) dF_{i_1} \wedge \cdots \wedge dF_{i_k},$$

$$d(F^*\omega) = \sum_{\mathbf{i} \in J(k, n)} d(\omega_{\mathbf{i}} \circ F) \wedge dF_{i_1} \wedge \cdots \wedge dF_{i_k},$$

$$F_{i_a} \in C^2(\Omega) \xrightarrow{1.3.3} d(dF_{i_a}) = 0, \quad d(\omega_{\mathbf{i}} \circ F) = \sum_{j=1}^n \left(\frac{\partial \omega_{\mathbf{i}}}{\partial x_j} \circ F \right) dF_j,$$

$$d(F^*\omega) = \sum_{\mathbf{i} \in J(k, n)} \sum_{j=1}^n \left(\frac{\partial \omega_{\mathbf{i}}}{\partial x_j} \circ F \right) dF_j \wedge dF_{i_1} \wedge \cdots \wedge dF_{i_k},$$

$$d(F^*\omega) = F^* \left(\sum_{\mathbf{i} \in J(k, n)} \sum_{j=1}^n \frac{\partial \omega_{\mathbf{i}}}{\partial x_j} dx_j \wedge dx_{\mathbf{i}} \right) = F^*(d\omega).$$

Thus pullback commutes with d , as represented in Figure 5. $\square$

Definition 1.3.7 (Orientation and induced boundary orientation). *For a C^1 embedded k -submanifold $S \subset \mathbb{R}^n$, set*

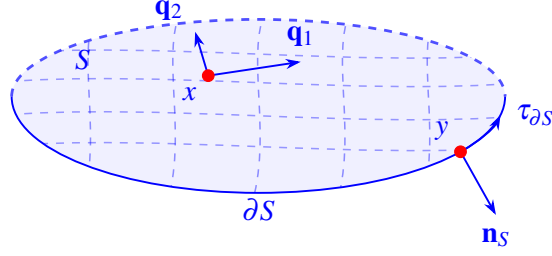
$$\tau_S : S \longrightarrow \mathcal{S}_k^1(\mathbb{R}^n), \quad \tau_S \text{ continuous}, \quad \tau_S(x) \in \Lambda_k(T_x S) \quad (x \in S), \quad (S, \tau_S) \text{ oriented.}$$

For $x \in \partial S$, choose

$$\mathbf{n}_S(x) \in T_x S \cap T_x(\partial S)^\perp, \quad |\mathbf{n}_S(x)| = 1, \quad \mathbf{n}_S(x) \text{ outward.}$$

Set

$$\tau_{\partial S}(x) \in \mathcal{S}_{k-1}^1(\mathbb{R}^n), \quad \mathbf{n}_S(x) \wedge \tau_{\partial S}(x) = \tau_S(x), \quad x \in \partial S.$$

FIGURE 6. The outward-first convention orients ∂S .

Definition 1.3.8 (Integration of forms). For $U \subset \mathbb{R}^n$ open, assume

$$S \subset U, \quad (S, \tau_S) \text{ oriented embedded } k\text{-submanifold of class } C^1, \quad \mathcal{H}^k \llcorner S \text{ locally finite.}$$

For $\omega \in C_c^0(U, \Lambda^k(\mathbb{R}^n))$, set

$$\int_S \omega := \int_S \langle \omega(x), \tau_S(x) \rangle d\mathcal{H}^k(x).$$

Theorem 1.3.9 (Stokes). For $U \subset \mathbb{R}^n$ open, $1 \leq k \leq n$, $S \subset U$ an oriented C^2 embedded k -submanifold with boundary, and $\psi \in C^1(U, \Lambda^{k-1}(\mathbb{R}^n))$, assume

$$\text{spt}(\psi) \cap S \subseteq S.$$

Then

$$\int_{\partial S} \psi = \int_S d\psi.$$

This is the compactly supported Stokes formula [22, Chapter 3] [28, §26].

Proof. An oriented atlas and a partition of unity reduce the identity to Euclidean Stokes on open subsets of the half-space.

$$\begin{aligned} \text{spt}(\psi) \cap S &\subset \bigcup_{a=1}^N F_a(\Omega_a), \quad F_a : \Omega_a \subset \mathbb{H}^k \longrightarrow S, \quad \sum_{a=1}^N \chi_a = 1 \text{ near } \text{spt}(\psi) \cap S, \\ \int_S d\psi &= \sum_{a=1}^N \int_S d(\chi_a \psi) = \sum_{a=1}^N \int_{\Omega_a} F_a^*(d(\chi_a \psi)), \\ \chi_a \psi &\in C^1(S, \Lambda^{k-1}(\mathbb{R}^n)) \xrightarrow{1.3.6} F_a^*(d(\chi_a \psi)) = d\alpha_a, \quad \alpha_a := F_a^*(\chi_a \psi) \in C_c^1(\Omega_a, \Lambda^{k-1}(\mathbb{R}^k)), \\ \int_S d\psi &= \sum_{a=1}^N \int_{\Omega_a} d\alpha_a = \sum_{a=1}^N \int_{\partial \mathbb{H}^k \cap \Omega_a} \alpha_a = \int_{\partial S} \left(\sum_{a=1}^N \chi_a \right) \psi = \int_{\partial S} \psi. \end{aligned}$$

The outward-first convention in Figure 6 supplies the boundary sign and completes the proof. $\square$

2. CURRENTS

Currents extend distributions and integration over oriented submanifolds [14, Chapters 1–3], [17, Chapter 9]. Throughout

$$\begin{aligned} U &\subset \mathbb{R}^n \text{ open}, & A \Subset U &\iff \bar{A} \subset U \text{ compact}, \\ \mathbf{M} &\text{ mass}, & \mathbf{N} &\text{ normal mass}, & \mathbf{F} &\text{ flat seminorm}, & \|T\| &\text{ Radon measure}, \\ [\![\cdot]\!] &\text{ integration current}, & \rightharpoonup &\text{ weak-* convergence.} \end{aligned}$$

2.1. Test forms, currents, and the boundary operator.

Definition 2.1.1 (Smooth compact-open topology). For

$$K \Subset U, \quad r \in \mathbb{N}_0, \quad \omega \in C^\infty(U, \Lambda^k(\mathbb{R}^n)),$$

Set

$$[\omega]_{K,r} := \max_{0 \leq j \leq r} \sup_{x \in K} |D^j \omega(x)|.$$

Choose

$$K_0 \Subset K_1 \Subset \cdots \Subset U, \quad \bigcup_{h=0}^{\infty} K_h = U.$$

Set

$$d_{C^\infty}(\omega, \eta) := \sum_{r=0}^{\infty} \sum_{h=0}^{\infty} 2^{-r-h-2} \frac{[\omega - \eta]_{K_h, r}}{1 + [\omega - \eta]_{K_h, r}},$$

$$\mathcal{U}(\psi; K, r, \varepsilon) := \left\{ \omega \in C^\infty(U, \Lambda^k(\mathbb{R}^n)) : [\omega - \psi]_{K, r} < \varepsilon \right\},$$

d_{C^∞} compatible with the smooth compact-open topology, $\bigcap_{i=1}^N \mathcal{U}(\psi; K_i, r_i, \varepsilon_i)$ a neighborhood basis $(N \in \mathbb{N})$.

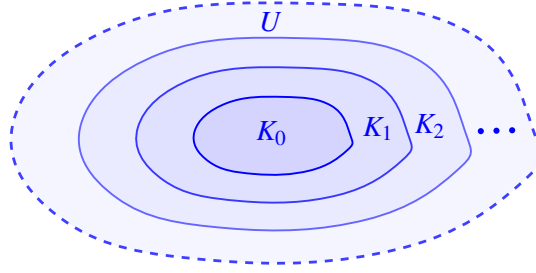


FIGURE 7. Nested compact sets exhaust U .

Definition 2.1.2 (Test forms and currents). For $K \Subset U$, set

$$\mathcal{D}_K^k(U) := \left\{ \omega \in C_c^\infty(U, \Lambda^k(\mathbb{R}^n)) : \text{spt}(\omega) \subset K \right\}.$$

$$\mathcal{D}^k(U) := C_c^\infty(U, \Lambda^k(\mathbb{R}^n)) = \varinjlim_{K \Subset U} \mathcal{D}_K^k(U) \quad \text{locally convex inductive limit.}$$

For $(\omega_v) \subset \mathcal{D}^k(U)$, write

$$\omega_v \longrightarrow 0 \text{ in } \mathcal{D}^k(U) \iff \begin{cases} \exists K \Subset U : \text{spt}(\omega_v) \subset K & (v \in \mathbb{N}), \\ [\omega_v]_{K, r} \longrightarrow 0 & (r \in \mathbb{N}_0). \end{cases}$$

Set

$$\mathcal{D}_k(U) := (\mathcal{D}^k(U))', \quad T_v \rightharpoonup T \iff T_v(\omega) \longrightarrow T(\omega) \quad (\omega \in \mathcal{D}^k(U)).$$

Proposition 2.1.3 (Continuity criterion). For a linear $T : \mathcal{D}^k(U) \rightarrow \mathbb{R}$,

$$T \in \mathcal{D}_k(U) \iff \left[\forall K \Subset U \exists r_K \in \mathbb{N}_0 \exists C_K < \infty : |T(\omega)| \leq C_K [\omega]_{K, r_K} \quad (\omega \in \mathcal{D}_K^k(U)) \right].$$

Proof. [15, §4.1.7], [28, §26]. □

Example 2.1.4 (Distributions and measure currents). Set

$$\mathcal{D}_0(U) = \mathcal{D}'(U).$$

For a locally finite signed Radon measure μ on U , set

$$T_\mu^0(\varphi) := \int_U \varphi d\mu \quad (\varphi \in \mathcal{D}^0(U)), \quad T_\mu^0 \in \mathcal{D}_0(U),$$

$$T_\mu^n(\omega) := \int_U \langle \omega(x), \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n \rangle d\mu(x) \quad (\omega \in \mathcal{D}^n(U)), \quad T_\mu^n \in \mathcal{D}_n(U).$$

Example 2.1.5 (Integration current). Let

$S \subset U$ oriented embedded C^1 k -submanifold, τ_S unit orientation, $\mathcal{H}^k(S \cap K) < \infty$ ($K \Subset U$).

Set

$$[S](\omega) := \int_S \langle \omega(x), \tau_S(x) \rangle d\mathcal{H}^k(x) \quad (\omega \in \mathcal{D}^k(U)).$$

For $K \Subset U$ and $\omega \in \mathcal{D}_K^k(U)$,

$$|[S](\omega)| \leq \int_{S \cap K} |\omega(x)| d\mathcal{H}^k(x) \leq \mathcal{H}^k(S \cap K)[\omega]_{K,0} \xrightarrow{2.1.3} [S] \in \mathcal{D}_k(U).$$

Definition 2.1.6 (Boundary of a current). For $T \in \mathcal{D}_k(U)$, set

$$\mathcal{D}_{-1}(U) := \{0\}, \quad \partial T(\psi) := T(d\psi) \quad (k \geq 1, \psi \in \mathcal{D}^{k-1}(U)), \quad \partial T := 0 \in \mathcal{D}_{-1}(U) \quad (k = 0).$$

Proposition 2.1.7 (Boundary identities). For

$$T \in \mathcal{D}_k(U), \quad S \subset U \text{ relatively closed oriented embedded } C^2 \text{ } k\text{-submanifold with boundary,}$$

$$\partial^2 T = 0, \quad \partial[S] = [[\partial S]].$$

Proof. For T, S as above, nilpotence of d and Stokes's theorem supply the two identities.

$$\partial^2 T(\psi) = T(d(d\psi)) \xrightarrow{1.3.3} 0 \quad (\psi \in \mathcal{D}^{k-2}(U)).$$

$$\partial[S](\psi) = \int_S d\psi \xrightarrow{1.3.9} \int_{\partial S} \psi = [[\partial S]](\psi) \quad (\psi \in \mathcal{D}^{k-1}(U)).$$

Thus the algebraic and geometric boundaries agree. $\square$

2.2. Mass, support, products, and pushforward.

Definition 2.2.1 (Mass and support). For $T \in \mathcal{D}_k(U)$ and open $W \subset U$, set

$$\mathbf{M}_W(T) := \sup \left\{ |T(\omega)| : \omega \in \mathcal{D}^k(U), \text{spt}(\omega) \subset W, \sup_{x \in U} \|\omega(x)\|^* \leq 1 \right\}, \quad \mathbf{M}(T) := \mathbf{M}_U(T),$$

$$T \text{ locally finite mass} \iff \mathbf{M}_W(T) < \infty \quad (W \Subset U \text{ open}),$$

$$\text{spt}(T) := U \setminus \bigcup \{W \subset U : W \text{ open, } T(\omega) = 0 \text{ if } \text{spt}(\omega) \subset W\}.$$

Proposition 2.2.2 (Radon representation). Let $T \in \mathcal{D}_k(U)$ have locally finite mass.

$\exists \mu_T$ Radon measure $\exists \tau_T : U \rightarrow \Lambda_k(\mathbb{R}^n)$ μ_T -measurable : $\|\tau_T(x)\| = 1$ for μ_T -almost every x ,

$$T(\omega) = \int_U \langle \omega(x), \tau_T(x) \rangle d\mu_T(x), \quad \|T\| := \mu_T,$$

$$\mathbf{M}_W(T) = \|T\|(W) \quad (W \subset U \text{ open}), \quad \text{spt}(T) = \text{spt}(\|T\|).$$

Proof. [15, §4.1.5], [28, §26]. $\square$

Definition 2.2.3 (Products, restrictions, and pushforward). For

$$T \in \mathcal{D}_k(U), \quad \mathbf{X} \in C^\infty(U, \Lambda_h(\mathbb{R}^n)), \quad \psi \in C^\infty(U, \Lambda^h(\mathbb{R}^n)), \quad 0 \leq h \leq k,$$

Set

$$(T \wedge \mathbf{X})(\phi) := T(\phi \lrcorner \mathbf{X}) \quad (\phi \in \mathcal{D}^{k+h}(U)), \quad (T \lrcorner \psi)(\phi) := T(\psi \wedge \phi) \quad (\phi \in \mathcal{D}^{k-h}(U)).$$

For T as in Proposition 2.2.2 and $A \subset U$ $\|T\|$ -measurable, set

$$(T \lrcorner A)(\omega) := \int_A \langle \omega(x), \tau_T(x) \rangle d\|T\|(x).$$

For

$$\Omega \subset \mathbb{R}^m \text{ open}, \quad F : U \rightarrow \Omega \text{ smooth}, \quad F|_{\text{spt}(T)} \text{ proper},$$

Choose

$$\omega \in \mathcal{D}^k(\Omega), \quad \chi \in C_c^\infty(U), \quad \chi = 1 \text{ near } \text{spt}(T) \cap F^{-1}(\text{spt}(\omega)),$$

Set

$$F_\# T(\omega) := T(\chi F^* \omega), \quad F_\# T(\omega) = T(F^* \omega) \quad (F \text{ proper}).$$

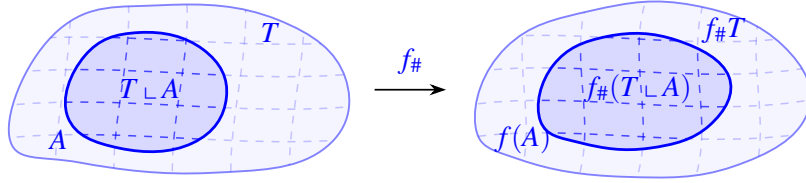


FIGURE 8. Restriction localizes T ; proper pushforward transports it.

Proposition 2.2.4 (Boundary commutes with pushforward). For

$$T \in \mathcal{D}_k(U), \quad \Omega \subset \mathbb{R}^m \text{ open}, \quad F : U \rightarrow \Omega \text{ smooth}, \quad F|_{\text{spt}(T)} \text{ proper},$$

$$\partial(F_\# T) = F_\#(\partial T).$$

Proof. For T, F as above, one cutoff serves both currents and naturality moves d through pullback.

$$\text{spt}(\partial T) \subset \text{spt}(T), \quad \psi \in \mathcal{D}^{k-1}(\Omega), \quad \chi = 1 \text{ near } \text{spt}(T) \cap F^{-1}(\text{spt}(\psi)).$$

$$F \text{ smooth} \xrightarrow{1.3.6} d(F^* \psi) = F^*(d\psi), \quad d(\chi F^* \psi) = d\chi \wedge F^* \psi + \chi F^*(d\psi).$$

$$\text{spt}(d\chi \wedge F^* \psi) \cap \text{spt}(T) = \emptyset \implies T(d\chi \wedge F^* \psi) = 0.$$

$$F_\#(\partial T)(\psi) = T(d(\chi F^* \psi)) = T(\chi F^*(d\psi)) = F_\# T(d\psi) = \partial(F_\# T)(\psi).$$

Thus boundary and proper pushforward commute. $\square$

Proposition 2.2.5 (Mass identities and lower semicontinuity). For

$$S \subset U \text{ relatively closed oriented embedded } C^1 \text{ } k\text{-submanifold}, \quad \mathcal{H}^k(S \cap K) < \infty \quad (K \Subset U),$$

$$\text{spt}(\llbracket S \rrbracket) = S, \quad \mathbf{M}_W(\llbracket S \rrbracket) = \mathcal{H}^k(S \cap W) \quad (W \subset U \text{ open}).$$

For $T_h \rightharpoonup T$ in $\mathcal{D}_k(U)$,

$$\mathbf{M}_W(T) \leq \liminf_{h \rightarrow \infty} \mathbf{M}_W(T_h) \quad (W \subset U \text{ open}).$$

Proof. [15, §4.1.7, §4.1.12], [28, §26]. $\square$

Definition 2.2.6 (Cartesian product of currents). Let

$$U_i \subset \mathbb{R}^{n_i} \text{ open, } T_i \in \mathcal{D}_{k_i}(U_i) \quad (i = 1, 2), \quad \omega \in \mathcal{D}^{k_1+k_2}(U_1 \times U_2),$$

$$\omega = \sum_{r+s=k_1+k_2} \sum_{\mathbf{i} \in J(r, n_1)} \sum_{\mathbf{j} \in J(s, n_2)} g_{\mathbf{ij}}(x, y) dx_{\mathbf{i}} \wedge dy_{\mathbf{j}}.$$

Set

$$(T_1 \times T_2)(\omega) := \sum_{\mathbf{i} \in J(k_1, n_1)} \sum_{\mathbf{j} \in J(k_2, n_2)} T_1(T_2(g_{\mathbf{ij}}(x, \cdot) dy_{\mathbf{j}}) dx_{\mathbf{i}}),$$

$$T_1 \times T_2 \in \mathcal{D}_{k_1+k_2}(U_1 \times U_2).$$

Proposition 2.2.7 (Boundary of a Cartesian product). For $T_i \in \mathcal{D}_{k_i}(U_i)$,

$$\partial(T_1 \times T_2) = (\partial T_1) \times T_2 + (-1)^{k_1} T_1 \times (\partial T_2).$$

Proof. [15, §4.1.8], [28, §26]. □

Theorem 2.2.8 (Homotopy formula). Let

$$\Omega \subset \mathbb{R}^m \text{ open, } f, g : U \rightarrow \Omega \text{ smooth, } I := [0, 1],$$

$$H : I \times U \rightarrow \Omega \text{ smooth, } H(0, \cdot) = f, \quad H(1, \cdot) = g, \quad H|_{I \times \text{spt}(T)} \text{ proper, } T \in \mathcal{D}_k(U).$$

$$g_{\#}T - f_{\#}T = \partial H_{\#}([I] \times T) + H_{\#}([I] \times \partial T).$$

Assume

$$H(t, x) = (1-t)f(x) + tg(x), \quad \mathbf{M}(T) < \infty.$$

$$\mathbf{M}(H_{\#}([I] \times T)) \leq \sup_{x \in \text{spt}(T)} |f(x) - g(x)| \cdot \sup_{(t, x) \in I \times \text{spt}(T)} |(1-t)Df(x) + tDg(x)|^k \cdot \mathbf{M}(T).$$

Proof. For T, H as above, the product boundary records the endpoint slices and pushforward transports the chain identity.

$$\partial[I] = [\{1\}] - [\{0\}] \xrightarrow{2.2.7} \partial([I] \times T) = ([\{1\}] - [\{0\}]) \times T - [I] \times \partial T.$$

$$\partial H_{\#}([I] \times T) \xrightarrow{2.2.4} H_{\#}\partial([I] \times T).$$

$$H_{\#}([\{1\}] - [\{0\}] \times T) = g_{\#}T - f_{\#}T, \quad H_{\#}\partial([I] \times T) = g_{\#}T - f_{\#}T - H_{\#}([I] \times \partial T).$$

$$g_{\#}T - f_{\#}T = \partial H_{\#}([I] \times T) + H_{\#}([I] \times \partial T).$$

The affine estimate is the corresponding $(k+1)$ -Jacobian bound [28, §26], and Figure 9 records its chain geometry. □

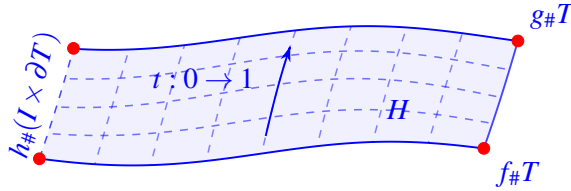


FIGURE 9. The homotopy current joins $f_{\#}T$ to $g_{\#}T$.

2.3. Regularization and the constancy theorem.

Definition 2.3.1 (Distributional derivatives of a current). For

$$T \in \mathcal{D}_k(U), \quad 1 \leq j \leq n, \quad \omega = \sum_{\mathbf{i} \in J(k,n)} \omega_{\mathbf{i}} dx_{\mathbf{i}} \in \mathcal{D}^k(U),$$

Set

$$D_{x_j} T(\omega) := -T \left(\sum_{\mathbf{i} \in J(k,n)} \frac{\partial \omega_{\mathbf{i}}}{\partial x_j} dx_{\mathbf{i}} \right), \quad D_{x_j} T \in \mathcal{D}_k(U).$$

Proposition 2.3.2 (Coordinate boundary identity). For $T \in \mathcal{D}_k(U)$ and $1 \leq k \leq n$,

$$\partial T = - \sum_{j=1}^n (D_{x_j} T) \lrcorner dx_j.$$

For $k = n$,

$$\partial T = 0 \iff D_{x_j} T = 0 \quad (1 \leq j \leq n).$$

Proof. For T as above, the coordinate formula for d transfers test-form derivatives to distributional derivatives of T .

$$\begin{aligned} \psi &\in \mathcal{D}^{k-1}(U), \quad d\psi = \sum_{j=1}^n dx_j \wedge \frac{\partial \psi}{\partial x_j}. \\ - \sum_{j=1}^n ((D_{x_j} T) \lrcorner dx_j)(\psi) &= \sum_{j=1}^n T \left(dx_j \wedge \frac{\partial \psi}{\partial x_j} \right) = T(d\psi) = \partial T(\psi). \end{aligned}$$

For $k = n$, choose

$$\omega_j := \varphi dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n, \quad \varphi \in C_c^\infty(U).$$

$$\partial T(\omega_j) = (-1)^{j-1} T \left(\frac{\partial \varphi}{\partial x_j} dx_1 \wedge \cdots \wedge dx_n \right) = (-1)^j D_{x_j} T(\varphi dx_1 \wedge \cdots \wedge dx_n).$$

$$\partial T = 0 \implies D_{x_j} T = 0 \quad (1 \leq j \leq n), \quad D_{x_j} T = 0 \quad (1 \leq j \leq n) \implies \partial T = 0.$$

Thus the coordinate identity yields the top-dimensional equivalence. $\square$

Definition 2.3.3 (Mollification of a current). Fix

$$\rho \in C_c^\infty(B_1(0)), \quad \rho(-x) = \rho(x), \quad \rho \geq 0, \quad \int_{\mathbb{R}^n} \rho(x) dx = 1.$$

Set

$$\rho_\varepsilon(x) := \varepsilon^{-n} \rho(x/\varepsilon), \quad U_\varepsilon := \{x \in U : \text{dist}(x, \partial U) > \varepsilon\}.$$

For $T \in \mathcal{D}_k(U)$ and $\omega \in \mathcal{D}^k(U_\varepsilon)$, set

$$(\rho_\varepsilon * \omega)(x) := \int_{U_\varepsilon} \rho_\varepsilon(x-y) \omega(y) dy, \quad T_\varepsilon(\omega) := T(\rho_\varepsilon * \omega), \quad T_\varepsilon \in \mathcal{D}_k(U_\varepsilon).$$

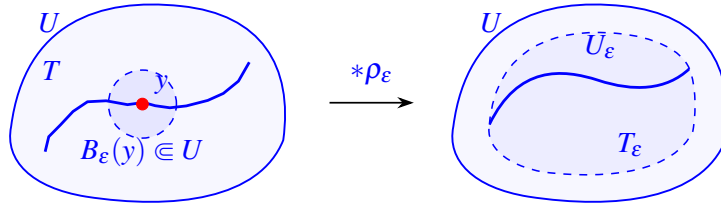


FIGURE 10. Mollification smooths T on U_ε .

Lemma 2.3.4 (Regularization). *For $T \in \mathcal{D}_k(U)$,*

$$T_\varepsilon(\omega) \longrightarrow T(\omega) \quad (\omega \in \mathcal{D}^k(U), \varepsilon \downarrow 0 \text{ sufficiently small}), \quad D_{x_j} T_\varepsilon = (D_{x_j} T)_\varepsilon \quad (1 \leq j \leq n).$$

For

$$f_{\varepsilon, \mathbf{i}}(y) := T(\rho_\varepsilon(y - \cdot) dx_{\mathbf{i}}), \quad \tau_\varepsilon(y) := \sum_{\mathbf{i} \in J(k, n)} f_{\varepsilon, \mathbf{i}}(y) \mathbf{e}_{\mathbf{i}},$$

$$\tau_\varepsilon \in C^\infty(U_\varepsilon, \Lambda_k(\mathbb{R}^n)), \quad T_\varepsilon(\omega) = \int_{U_\varepsilon} \langle \omega(y), \tau_\varepsilon(y) \rangle dy.$$

For $k = n$, set

$$F_\varepsilon(y) := T(\rho_\varepsilon(y - \cdot) dx_1 \wedge \cdots \wedge dx_n),$$

$$T_\varepsilon(\varphi dx_1 \wedge \cdots \wedge dx_n) = \int_{U_\varepsilon} \varphi(y) F_\varepsilon(y) dy.$$

Proof. Exercise 8.4, [22, Chapter 7], [28, §26]. □

Theorem 2.3.5 (Constancy theorem). *For*

$$U \subset \mathbb{R}^n \text{ open connected}, \quad T \in \mathcal{D}_n(U), \quad \partial T = 0,$$

$$\exists c \in \mathbb{R} : T = c \llbracket U \rrbracket, \quad \llbracket U \rrbracket(\varphi dx_1 \wedge \cdots \wedge dx_n) := \int_U \varphi(x) dx.$$

Proof. For T, U as above, regularization converts the boundary condition into local constancy of a smooth density.

$$B \Subset U \text{ open connected}, \quad 0 < \varepsilon < \text{dist}(\bar{B}, \partial U), \quad \eta_B \in C_c^\infty(B), \quad \int_B \eta_B dx = 1.$$

$$\partial T = 0 \xLeftrightarrow{2.3.2} D_{x_j} T = 0 \quad (1 \leq j \leq n) \xRightarrow{2.3.4} \nabla F_\varepsilon = 0 \implies F_\varepsilon \equiv c_{B, \varepsilon} \quad \text{on } B.$$

$$c_{B, \varepsilon} = T_\varepsilon(\eta_B dx_1 \wedge \cdots \wedge dx_n) \longrightarrow T(\eta_B dx_1 \wedge \cdots \wedge dx_n) =: c_B.$$

For $\varphi \in C_c^\infty(B)$,

$$T(\varphi dx_1 \wedge \cdots \wedge dx_n) = \lim_{\varepsilon \downarrow 0} c_{B, \varepsilon} \int_B \varphi dx = c_B \int_B \varphi dx.$$

$$B_1 \cap B_2 \neq \emptyset \implies c_{B_1} = c_{B_2}, \quad U \text{ connected} \implies c_B = c \quad (B \Subset U \text{ open connected}).$$

Hence $T = c \llbracket U \rrbracket$ on every compactly supported top form. □

2.4. Normal, flat, and rectifiable currents.

Definition 2.4.1 (Normal currents). For $T \in \mathcal{D}_k(U)$ and open $W \subset U$, set

$$\mathbf{N}_W(T) := \mathbf{M}_W(T) + \mathbf{M}_W(\partial T), \quad \mathbf{N}(T) := \mathbf{N}_U(T).$$

Set

$$\mathcal{N}_k(U) := \{T \in \mathcal{D}_k(U) : \mathbf{N}(T) < \infty\}, \quad \mathcal{N}_{k, \text{loc}}(U) := \{T \in \mathcal{D}_k(U) : \mathbf{N}_W(T) < \infty \quad (W \Subset U \text{ open})\}.$$

For $T \in \mathcal{N}_k(U)$, Proposition 2.2.2 gives

$$T(\omega) = \int_U \langle \omega, \tau_T \rangle d\|T\|, \quad \partial T(\psi) = \int_U \langle \psi, \tau_{\partial T} \rangle d\|\partial T\|.$$

Proposition 2.4.2 (Compactness and the direct method). *For*

$$K \subset U \text{ compact}, \quad (T_h) \subset \mathcal{N}_k(U), \quad \text{spt}(T_h) \subset K, \quad \sup_{h \geq 1} \mathbf{N}(T_h) < \infty,$$

$$\exists (h_j) \subset \mathbb{N} \text{ strictly increasing} \quad \exists T \in \mathcal{N}_k(U) : \quad T_{h_j} \rightharpoonup T, \quad \text{spt}(T) \subset K.$$

$$\mathbf{M}(T) \leq \liminf_{j \rightarrow \infty} \mathbf{M}(T_{h_j}), \quad \mathbf{M}(\partial T) \leq \liminf_{j \rightarrow \infty} \mathbf{M}(\partial T_{h_j}).$$

For

$$\Gamma \in \mathcal{N}_{k-1}(U), \quad \partial \Gamma = 0, \quad \mathcal{A}_{\Gamma, K} := \{T \in \mathcal{N}_k(U) : \partial T = \Gamma, \text{spt}(T) \subset K\} \neq \emptyset, \\ \exists T_* \in \mathcal{A}_{\Gamma, K} : \mathbf{M}(T_*) = \inf_{T \in \mathcal{A}_{\Gamma, K}} \mathbf{M}(T).$$

Proof. Under the stated support and normal-mass bounds, normal-current compactness gives weak limits for T_h and ∂T_h .

$$T_h \rightharpoonup T, \quad \partial T_h \rightharpoonup S, \quad S(\psi) = \lim_{h \rightarrow \infty} T_h(d\psi) = T(d\psi) = \partial T(\psi).$$

$$\text{spt}(T_h) \subset K \implies \text{spt}(T) \subset K, \quad S = \partial T.$$

$$T_h \rightharpoonup T, \quad \partial T_h \rightharpoonup \partial T \xrightarrow{2.2.5} \mathbf{M}(T) \leq \liminf_{h \rightarrow \infty} \mathbf{M}(T_h), \quad \mathbf{M}(\partial T) \leq \liminf_{h \rightarrow \infty} \mathbf{M}(\partial T_h).$$

$$(T_h) \subset \mathcal{A}_{\Gamma, K} \text{ minimizing} \implies \partial T = \Gamma, \quad \text{spt}(T) \subset K, \quad \inf_{R \in \mathcal{A}_{\Gamma, K}} \mathbf{M}(R) \leq \mathbf{M}(T) \leq \lim_{h \rightarrow \infty} \mathbf{M}(T_h) = \inf_{R \in \mathcal{A}_{\Gamma, K}} \mathbf{M}(R).$$

Thus the weak limit is an admissible mass minimizer [23, Part I]. $\square$

Definition 2.4.3 (Flat seminorm and flat chains). For $T \in \mathcal{D}_k(U)$ and open $W \subset U$, set

$$\mathbf{F}_W(T) := \inf \{ \mathbf{M}_W(R) + \mathbf{M}_W(S) : T = R + \partial S \text{ in } W, R \in \mathcal{D}_k(U), S \in \mathcal{D}_{k+1}(U) \}, \quad \mathbf{F}(T) := \mathbf{F}_U(T).$$

Set

$$\mathcal{N}_{k,c}(U) := \{T \in \mathcal{N}_k(U) : \text{spt}(T) \Subset U\}, \quad \mathcal{F}_k(U) := \overline{\mathcal{N}_{k,c}(U)}^{\mathbf{F}}.$$

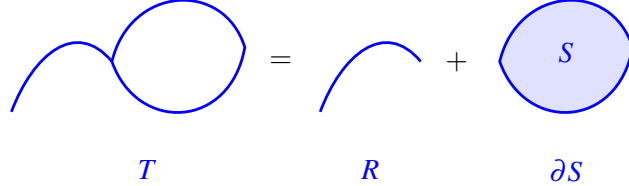


FIGURE 11. A flat decomposition $T = R + \partial S$.

Definition 2.4.4 (Polyhedral chains). For

$$\sigma = [\mathbf{v}_0, \dots, \mathbf{v}_k] \Subset U \text{ oriented affine } k\text{-simplex}, \quad \mathbb{A} \in \{\mathbb{Z}, \mathbb{R}\},$$

Set

$$\mathcal{P}_k^{\mathbb{A}}(U) := \left\{ \sum_{i=1}^N a_i [\sigma_i] : N \in \mathbb{N}, a_i \in \mathbb{A}, \sigma_i \Subset U \right\},$$

$$\mathcal{P}_k^{\mathbb{A}, \text{loc}}(U) := \left\{ \sum_{i=1}^{\infty} a_i [\sigma_i] : a_i \in \mathbb{A}, \{\sigma_i\}_{i=1}^{\infty} \text{ locally finite in } U \right\}.$$

Write

$$\mathcal{P}_k(U) := \mathcal{P}_k^{\mathbb{R}}(U), \quad \mathcal{P}_k^{\text{loc}}(U) := \mathcal{P}_k^{\mathbb{R}, \text{loc}}(U).$$

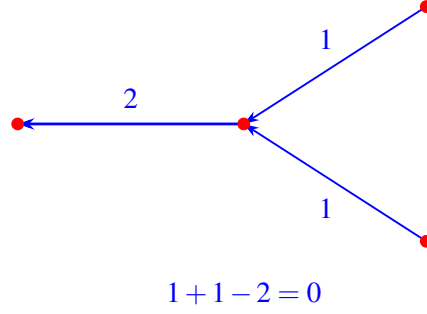


FIGURE 12. An integer polyhedral one-chain whose boundary vanishes at the junction.

Definition 2.4.5 (Integer rectifiable, integral, and integer flat currents). Let

$$M \subset U \text{ countably } k\text{-rectifiable}, \quad \theta \in L^1(\mathcal{H}^k \llcorner M; \mathbb{N}),$$

$$\tau : M \rightarrow \Lambda_k(\mathbb{R}^n) \text{ measurable unit simple}, \quad \text{span}(\tau(x)) = T_x M \quad \text{for } \mathcal{H}^k\text{-almost every } x \in M.$$

Set

$$T(\omega) := \int_M \theta(x) \langle \omega(x), \tau(x) \rangle d\mathcal{H}^k(x) \quad (\omega \in \mathcal{D}^k(U)), \quad \mathcal{R}_k(U) := \{T : T \text{ has this representation}\}.$$

Set

$$\begin{aligned} \mathbf{I}_k(U) &:= \mathcal{R}_k(U) \cap \mathcal{N}_k(U) = \{T \in \mathcal{R}_k(U) : \partial T \in \mathcal{R}_{k-1}(U)\}, \\ \mathcal{F}_k^{\mathbb{Z}}(U) &:= \{R + \partial S : R \in \mathcal{R}_k(U), S \in \mathcal{R}_{k+1}(U)\}, \\ \mathbf{F}^{\mathbb{Z}}(T) &:= \inf \{\mathbf{M}(R) + \mathbf{M}(S) : T = R + \partial S, R \in \mathcal{R}_k(U), S \in \mathcal{R}_{k+1}(U)\}, \\ \mathbf{F}^{\mathbb{Z}}(T_1 + T_2) &\leq \mathbf{F}^{\mathbb{Z}}(T_1) + \mathbf{F}^{\mathbb{Z}}(T_2). \end{aligned}$$

Proposition 2.4.6 (Density representation). For T as in Proposition 2.2.2, assume

$$\mathbf{M}(T) < \infty, \quad \omega_k := \mathcal{L}^k(B_1(0)), \quad \Theta^k(\|T\|, x) := \lim_{r \downarrow 0} \frac{\|T\|(B_r(x))}{\omega_k r^k}.$$

$$T \in \mathcal{R}_k(U) \xLeftrightarrow{2.4.5} \exists \Sigma \subset U \text{ countably } k\text{-rectifiable}$$

$$\|T\|(U \setminus \Sigma) = 0, \quad \Theta^k(\|T\|, x) \in \mathbb{N} \quad \text{for } \|T\|\text{-almost every } x,$$

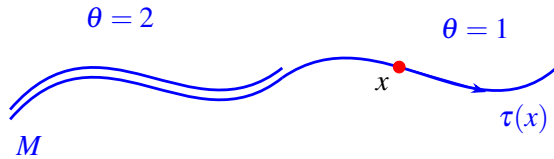
$$\tau_T(x) = \mathbf{v}_1(x) \wedge \cdots \wedge \mathbf{v}_k(x), \quad T_x \Sigma = \text{span}(\mathbf{v}_1(x), \dots, \mathbf{v}_k(x)) \quad \text{for } \|T\|\text{-almost every } x.$$

Set

$$\theta(x) := \Theta^k(\|T\|, x).$$

$$\|T\| = \theta \mathcal{H}^k \llcorner \Sigma, \quad T(\omega) = \int_{\Sigma} \theta(x) \langle \omega(x), \tau_T(x) \rangle d\mathcal{H}^k(x), \quad \mathbf{M}(T) = \int_{\Sigma} \theta(x) d\mathcal{H}^k(x).$$

Proof. [15, §4.1.28], [28, §27], [8, Chapters 1–2]. □

FIGURE 13. An integer rectifiable current is a carrier M , an orientation τ , and an integer multiplicity θ .

Proposition 2.4.7 (Current-class inclusions). *For compactly supported chains,*

$$\mathcal{P}_k^{\mathbb{Z}}(U) \subset \mathbf{I}_k(U) \subset \mathcal{R}_k(U) \subset \mathcal{F}_k^{\mathbb{Z}}(U) \subset \mathcal{F}_k(U), \quad \mathcal{P}_k(U) \subset \mathcal{N}_k(U) \subset \mathcal{F}_k(U).$$

Proof. [15, §4.1.24], [22, Chapter 7], [28, §27]. $\square$

3. SLICING, DEFORMATION, AND AREA MINIMIZATION

Throughout, $1 \leq h \leq k$, u is Lipschitz with constant $\text{Lip}(u)$, $\langle T, u, y \rangle$ is the slice of T over $y \in \mathbb{R}^h$, a superscript S denotes restriction to $T_x S$, $c_{n,k}$ depends only on dimension and may change from line to line, ω_k is the volume of the unit k -ball, $\dim_{\mathcal{H}}$ is Hausdorff dimension, and the notation of §2 and §5 is used [8], [25, Chapters 15–17], [14, Chapter 3].

3.1. Slicing and the coarea formula.

Proposition 3.1.1 (Smooth level sets). *Let $S \subset \mathbb{R}^n$ be a smooth k -dimensional submanifold, $1 \leq h \leq k$, and $u : S \rightarrow \mathbb{R}^h$ smooth. For almost every $y \in \mathbb{R}^h$, the set $S_y := S \cap u^{-1}(y)$ is empty or a smooth $(k-h)$ -dimensional submanifold, and*

$$T_x S_y = \ker(D^S u(x)) = T_x S \cap \bigcap_{i=1}^h \ker(du_i(x)) \quad (x \in S_y).$$

Proof. Sard's theorem and the regular-level-set theorem give the claim [15, §3.1], [22, Chapter 5]. $\square$

Definition 3.1.2 (Tangential Jacobian and rectifiable slices). For $1 \leq h \leq k$, $T = \llbracket S, \tau, \theta \rrbracket \in \mathcal{R}_k(\mathbb{R}^n)$, and Lipschitz $u = (u_1, \dots, u_h) : \mathbb{R}^n \rightarrow \mathbb{R}^h$, write $D^S u(x) := Du(x)|_{T_x S}$ wherever $T_x S$ and $Du(x)$ exist, and

$$\nabla^S u_i(x) := P_{T_x S} \nabla u_i(x), \quad J_h^S u(x) := |\nabla^S u_1(x) \wedge \dots \wedge \nabla^S u_h(x)|,$$

with $J_h^S u(x) := 0$ elsewhere. Let $S_g \subset S$ be the set on which $T_x S$ and $D^S u(x)$ exist and $J_h^S u(x) > 0$, and let $S_y := S_g \cap u^{-1}(y)$. For $x \in S_g$,

$$\eta_u(x) := \frac{\nabla^S u_1(x) \wedge \dots \wedge \nabla^S u_h(x)}{J_h^S u(x)}, \quad \eta_u(x) \wedge \tau_y(x) = \tau(x), \quad |\tau_y(x)| = 1,$$

and for almost every $y \in \mathbb{R}^h$,

$$\langle T, u, y \rangle := \llbracket S_y, \tau_y, \theta|_{S_y} \rrbracket \in \mathcal{R}_{k-h}(\mathbb{R}^n).$$

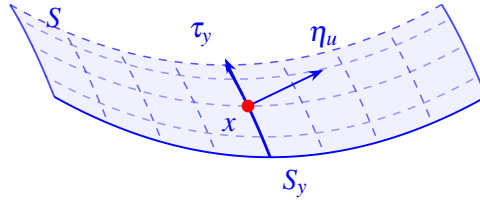


FIGURE 14. The slice carries the induced orientation.

Proposition 3.1.3 (Rectifiable coarea and slice identities). *For the data of Definition 3.1.2 and every nonnegative Borel $f : S \rightarrow [0, \infty]$,*

$$\int_{\mathbb{R}^h} \int_{S_y} f(x) d\mathcal{H}^{k-h}(x) dy = \int_S f(x) J_h^S u(x) d\mathcal{H}^k(x),$$

and, for almost every $y \in \mathbb{R}^h$,

$$\mathcal{H}^{k-h}((S \cap u^{-1}(y)) \setminus S_g) = 0, \quad \text{spt}(\langle T, u, y \rangle) \subset \text{spt}(T) \cap u^{-1}(y),$$

$$\begin{aligned} \int_{\mathbb{R}^h} \mathbf{M}(\langle T, u, y \rangle) dy &= \int_S \theta(x) J_h^S u(x) d\mathcal{H}^k(x) \leq \text{Lip}(u)^h \mathbf{M}(T), \\ \int_{\mathbb{R}^h} \langle T, u, y \rangle(\omega) dy &= T(du_1 \wedge \cdots \wedge du_h \wedge \omega) \quad (\omega \in \mathcal{D}^{k-h}(\mathbb{R}^n)). \end{aligned}$$

Proof. Apply the coarea formula on Lipschitz charts to $S \setminus S_g$ and θ ; Figure 14 fixes the orientation [15, §3.2].

$$\begin{aligned} \int_{\mathbb{R}^h} \mathcal{H}^{k-h}((S \setminus S_g) \cap u^{-1}(y)) dy &= \int_{S \setminus S_g} J_h^S u d\mathcal{H}^k = 0, \quad \int_{\mathbb{R}^h} \mathbf{M}(\langle T, u, y \rangle) dy = \int_S \theta J_h^S u d\mathcal{H}^k, \\ J_h^S u &= |\nabla^S u_1 \wedge \cdots \wedge \nabla^S u_h| \xrightarrow{1.2.5} J_h^S u \leq \prod_{i=1}^h |\nabla^S u_i| \leq \text{Lip}(u)^h, \\ \eta_u \wedge \tau_y &= \tau \xrightarrow{3.1.2} \langle du_1 \wedge \cdots \wedge du_h \wedge \omega, \tau \rangle = J_h^S u \langle \omega, \tau_y \rangle, \\ T(du_1 \wedge \cdots \wedge du_h \wedge \omega) &= \int_S \theta J_h^S u \langle \omega, \tau_y \rangle d\mathcal{H}^k = \int_{\mathbb{R}^h} \langle T, u, y \rangle(\omega) dy. \end{aligned}$$

□

Corollary 3.1.4 (Euclidean coarea and polar coordinates). *For Lipschitz $u : \mathbb{R}^n \rightarrow \mathbb{R}^h$ and Borel $f : \mathbb{R}^n \rightarrow [0, \infty]$,*

$$\begin{aligned} \int_{\mathbb{R}^n} f(x) J_h u(x) dx &= \int_{\mathbb{R}^h} \int_{u^{-1}(y)} f(x) d\mathcal{H}^{n-h}(x) dy, \\ u(x) = |x|, \quad |\nabla u(x)| = 1 \ (x \neq 0) &\implies \int_{\mathbb{R}^n} f(x) dx = \int_0^\infty \int_{\partial B_r(0)} f(x) d\mathcal{H}^{n-1}(x) dr. \end{aligned}$$

Proof. Proposition 3.1.3 with $S = \mathbb{R}^n$, and with $h = 1$ and $u(x) = |x|$ for the second identity, gives both claims. □

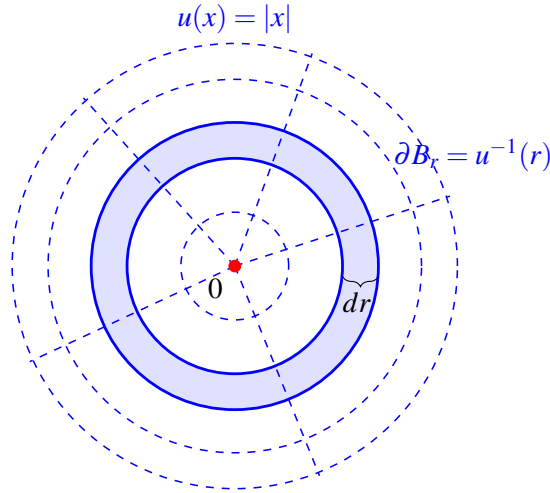


FIGURE 15. Polar coordinates decompose $\mathbb{R}^n$ into the level spheres of $u(x) = |x|$.

Definition 3.1.5 (Slices of normal currents). For $T \in \mathcal{N}_k(\mathbb{R}^n)$, Lipschitz $u : \mathbb{R}^n \rightarrow \mathbb{R}$, and almost every $r \in \mathbb{R}$,

$$\langle T, u, r \rangle := \partial(T \llcorner \{u < r\}) - (\partial T) \llcorner \{u < r\} \in \mathcal{N}_{k-1}(\mathbb{R}^n),$$

while for Lipschitz $u = (u_1, \dots, u_h) : \mathbb{R}^n \rightarrow \mathbb{R}^h$ and almost every $y = (y_1, \dots, y_h) \in \mathbb{R}^h$,

$$\langle T, u, y \rangle := \langle \cdots \langle \langle T, u_1, y_1 \rangle, u_2, y_2 \rangle \cdots, u_h, y_h \rangle.$$

Theorem 3.1.6 (Slicing theorem). *For $T \in \mathcal{N}_k(\mathbb{R}^n)$, Lipschitz $u : \mathbb{R}^n \rightarrow \mathbb{R}^h$, $1 \leq h \leq k$, and almost every $y \in \mathbb{R}^h$,*

$$\text{spt}(\langle T, u, y \rangle) \subset \text{spt}(T) \cap u^{-1}(y), \quad h < k \implies \partial \langle T, u, y \rangle = (-1)^h \langle \partial T, u, y \rangle, \quad h = k \implies \partial \langle T, u, y \rangle = 0,$$

$$\int_{\mathbb{R}^h} \mathbf{M}(\langle T, u, y \rangle) dy \leq \text{Lip}(u)^h \mathbf{M}(T), \quad \int_{\mathbb{R}^h} \mathbf{M}(\partial \langle T, u, y \rangle) dy \leq \text{Lip}(u)^h \mathbf{M}(\partial T),$$

$$\int_{\mathbb{R}^h} \langle T, u, y \rangle(\omega) dy = T(du_1 \wedge \cdots \wedge du_h \wedge \omega) \quad (\omega \in \mathcal{D}^{k-h}(\mathbb{R}^n)).$$

For $T \in \mathcal{R}_k(\mathbb{R}^n)$, this agrees almost everywhere with Definition 3.1.2; for $T \in \mathbf{I}_k(\mathbb{R}^n)$, one has $\langle T, u, y \rangle \in \mathbf{I}_{k-h}(\mathbb{R}^n)$ for almost every $y \in \mathbb{R}^h$.

Proof. The one-dimensional boundary identity iterates in h , and the remaining assertions are the normal-current slicing theorem [15, §4.2.1 and §4.3.1], [28, §28].

$$h = 1, k > 1 \xrightarrow{3.1.5} \partial \langle T, u, r \rangle = -\partial((\partial T) \llcorner \{u < r\}) = -\langle \partial T, u, r \rangle, \quad k = 1 \implies \partial \langle T, u, r \rangle = 0.$$

□

Theorem 3.1.7 (Compactness of slices). *Let $(T_j)_{j \geq 1} \subset \mathcal{N}_k(\mathbb{R}^n)$ converge weakly to T with $\sup_{j \geq 1} (\mathbf{M}(T_j) + \mathbf{M}(\partial T_j)) < \infty$, and let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be Lipschitz. A nonrelabeled subsequence depending on u satisfies*

$$\langle T_j, u, r \rangle \rightharpoonup \langle T, u, r \rangle \quad \text{for almost every } r \in \mathbb{R}, \quad \sup_{j \geq 1} \int_{\mathbb{R}} (\mathbf{M}(\langle T_j, u, r \rangle) + \mathbf{M}(\partial \langle T_j, u, r \rangle)) dr \leq \text{Lip}(u) \sup_{j \geq 1} \mathbf{N}(T_j).$$

Proof. Let $\mu_j := \|T_j\| + \|\partial T_j\|$ and pass to a weakly convergent subsequence.

$$\sup_{j \geq 1} \mu_j(\mathbb{R}^n) < \infty \implies \mu_j \rightharpoonup \mu,$$

$$\mu(u^{-1}(r)) = 0 \text{ for a.e. } r \implies T_j \llcorner \{u < r\} \rightharpoonup T \llcorner \{u < r\}, \quad (\partial T_j) \llcorner \{u < r\} \rightharpoonup (\partial T) \llcorner \{u < r\}.$$

$$T_j \llcorner \{u < r\} \rightharpoonup T \llcorner \{u < r\}, \quad (\partial T_j) \llcorner \{u < r\} \rightharpoonup (\partial T) \llcorner \{u < r\} \xrightarrow{3.1.5} \langle T_j, u, r \rangle \rightharpoonup \langle T, u, r \rangle \quad \text{for almost every } r.$$

$$T_j \in \mathcal{N}_k(\mathbb{R}^n) \xrightarrow{3.1.6} \sup_{j \geq 1} \int_{\mathbb{R}} (\mathbf{M}(\langle T_j, u, r \rangle) + \mathbf{M}(\partial \langle T_j, u, r \rangle)) dr \leq \text{Lip}(u) \sup_{j \geq 1} \mathbf{N}(T_j).$$

□

Corollary 3.1.8 (Lower semicontinuity of sliced mass). *For the sequence and subsequence of Theorem 3.1.7,*

$$\int_{\mathbb{R}} \mathbf{M}(\langle T, u, r \rangle) dr \leq \liminf_{j \rightarrow \infty} \int_{\mathbb{R}} \mathbf{M}(\langle T_j, u, r \rangle) dr \leq \text{Lip}(u) \liminf_{j \rightarrow \infty} \mathbf{M}(T_j).$$

Proof.

$$\langle T_j, u, r \rangle \rightharpoonup \langle T, u, r \rangle \xrightarrow{2.2.5} \mathbf{M}(\langle T, u, r \rangle) \leq \liminf_{j \rightarrow \infty} \mathbf{M}(\langle T_j, u, r \rangle) \quad \text{for almost every } r.$$

$$\mathbf{M}(\langle T, u, r \rangle) \leq \liminf_{j \rightarrow \infty} \mathbf{M}(\langle T_j, u, r \rangle) \xrightarrow{\text{Fatou}} \int_{\mathbb{R}} \mathbf{M}(\langle T, u, r \rangle) dr \leq \liminf_{j \rightarrow \infty} \int_{\mathbb{R}} \mathbf{M}(\langle T_j, u, r \rangle) dr.$$

$$T_j \in \mathcal{N}_k(\mathbb{R}^n) \xrightarrow{3.1.6} \int_{\mathbb{R}} \mathbf{M}(\langle T_j, u, r \rangle) dr \leq \text{Lip}(u) \mathbf{M}(T_j) \implies \liminf_{j \rightarrow \infty} \int_{\mathbb{R}} \mathbf{M}(\langle T_j, u, r \rangle) dr \leq \text{Lip}(u) \liminf_{j \rightarrow \infty} \mathbf{M}(T_j).$$

□

3.2. Closure, boundary rectifiability, and integrality criteria.

Theorem 3.2.1 (Closure theorem for integral currents). *Let $(T_j)_{j \geq 1} \subset \mathbf{I}_k(\mathbb{R}^n)$ satisfy $T_j \rightarrow T$ and $\sup_{j \geq 1} (\mathbf{M}(T_j) + \mathbf{M}(\partial T_j)) < \infty$.*

$$T \in \mathbf{I}_k(\mathbb{R}^n).$$

Proof. [15, §4.2.17], [28, §27]. □

Lemma 3.2.2 (Slice criterion for rectifiability). *Let $T \in \mathcal{N}_{k,\text{loc}}(\mathbb{R}^n)$ satisfy $\partial T = 0$, and assume every Lipschitz $u : \mathbb{R}^n \rightarrow \mathbb{R}$ has $\langle T, u, r \rangle \in \mathcal{R}_{k-1,\text{loc}}(\mathbb{R}^n)$ for almost every $r \in \mathbb{R}$.*

$$T \in \mathcal{R}_{k,\text{loc}}(\mathbb{R}^n).$$

Proof. [15, §4.2.15], [18, §2.2]. □

Theorem 3.2.3 (Boundary rectifiability). *For $T \in \mathcal{R}_k(\mathbb{R}^n) \cap \mathcal{N}_k(\mathbb{R}^n)$,*

$$\partial T \in \mathcal{R}_{k-1}(\mathbb{R}^n), \quad T \in \mathbf{I}_k(\mathbb{R}^n).$$

Proof. [15, §4.2.16], [16], [18, §2.2]. □

Theorem 3.2.4 (Bounded variation of slice maps). *For $T \in \mathcal{N}_k(\mathbb{R}^n)$, $1 \leq h \leq k$, and Lipschitz $u : \mathbb{R}^n \rightarrow \mathbb{R}^h$, the slice map $\Phi_{T,u} : \mathbb{R}^h \rightarrow \mathcal{F}_{k-h}(\mathbb{R}^n)$, $\Phi_{T,u}(y) := \langle T, u, y \rangle$, has a representative satisfying*

$$\Phi_{T,u} \in BV_{\text{loc}}(\mathbb{R}^h; \mathcal{F}_{k-h}(\mathbb{R}^n)).$$

Proof. [21], [1, Chapter 3]. □

Theorem 3.2.5 (Projection criterion for integer rectifiability). *Let $T \in \mathcal{N}_k(\mathbb{R}^n)$, and write $\pi_P : \mathbb{R}^n \rightarrow P$ for orthogonal projection onto a coordinate k -plane. If $\langle T, \pi_P, y \rangle \in \mathbf{I}_0(\mathbb{R}^n)$ for almost every $y \in P$ and every such P ,*

$$T \in \mathcal{R}_k(\mathbb{R}^n).$$

Proof. [4]. □

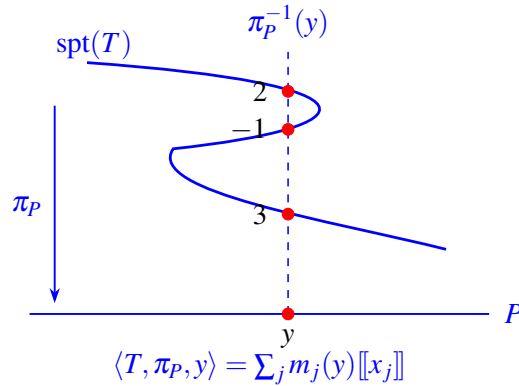


FIGURE 16. Projection onto P reduces T to integer-weighted points on each fibre.

3.3. Area minimization and regularity.

Definition 3.3.1 (Area minimization and regular sets). For a smooth Riemannian manifold M^n , an open $\Omega \subset M$, and $T \in \mathbf{I}_{k,\text{loc}}(M)$, the current T is locally area minimizing in Ω if, for every open $W \Subset \Omega$ and $Q \in \mathbf{I}_{k+1,\text{loc}}(M)$ with $\text{spt}(Q) \Subset W$,

$$\mathbf{M}_W(T) \leq \mathbf{M}_W(T + \partial Q).$$

A point $x \in \text{spt}(T) \setminus \text{spt}(\partial T)$ lies in $\text{Reg}_i(T)$ if a neighborhood $W \ni x$, an integer $m > 0$, and a smooth connected oriented embedded k -submanifold $\Sigma \subset W$ satisfy

$$T \llcorner W = m[\![\Sigma]\!].$$

$$\text{Sing}_i(T) := (\text{spt}(T) \setminus \text{spt}(\partial T)) \setminus \text{Reg}_i(T).$$

A point $x \in \text{spt}(\partial T)$ lies in $\text{Reg}_b(T)$ if a neighborhood $W \ni x$, an integer $m > 0$, and a smooth connected oriented embedded k -submanifold $\Sigma \subset W$ with boundary satisfy

$$T \llcorner W = m[\![\Sigma]\!], \quad \partial T \llcorner W = m[\![\partial \Sigma]\!].$$

$$\text{Sing}_b(T) := \text{spt}(\partial T) \setminus \text{Reg}_b(T).$$

Theorem 3.3.2 (Existence in integral homology classes). For compact M , every $\alpha \in H_k(M; \mathbb{Z})$ represented by an integral cycle admits $T_\alpha \in \mathbf{I}_k(M)$ satisfying

$$\partial T_\alpha = 0, \quad [T_\alpha] = \alpha, \quad \mathbf{M}(T_\alpha) = \inf \{ \mathbf{M}(S) : S \in \mathbf{I}_k(M), \partial S = 0, [S] = \alpha \}.$$

Proof. Let m_α be the displayed infimum; integral flat compactness gives a minimizing sequence $(S_j)_{j \geq 1}$ and T_α with $\mathbf{F}(S_j - T_\alpha) \rightarrow 0$ [16], [15, §5.1.6], [19, Chapter 1].

$$\begin{aligned} \mathbf{F}(S_j - T_\alpha) \rightarrow 0 &\implies S_j \rightarrow T_\alpha \xrightarrow{3.2.1} T_\alpha \in \mathbf{I}_k(M), \\ \mathbf{F}(S_j - T_\alpha) \rightarrow 0, \quad \partial S_j = 0, \quad [S_j] = \alpha &\implies \partial T_\alpha = 0, \quad [T_\alpha] = \alpha. \\ \mathbf{M}(S_j) \rightarrow m_\alpha, \quad m_\alpha \leq \mathbf{M}(T_\alpha) &\xrightarrow{2.2.5} \mathbf{M}(T_\alpha) \leq \liminf_{j \rightarrow \infty} \mathbf{M}(S_j) = m_\alpha. \end{aligned}$$

□

Theorem 3.3.3 (Interior singular sets). Let $T \in \mathbf{I}_k(M^n)$ be locally area minimizing in a smooth metric. For $n - k \geq 2$,

$$\begin{aligned} k = 1 &\implies \text{Sing}_i(T) = \emptyset, \\ k \geq 2 &\implies \dim_{\mathcal{H}}(\text{Sing}_i(T)) \leq k - 2, \quad \text{Sing}_i(T) \text{ is countably } (k - 2)\text{-rectifiable}, \\ k = 2 &\implies \text{Sing}_i(T) \text{ is locally discrete.} \end{aligned}$$

For $n - k = 1$,

$$\begin{aligned} k \leq 6 &\implies \text{Sing}_i(T) = \emptyset, \\ k \geq 7 &\implies \dim_{\mathcal{H}}(\text{Sing}_i(T)) \leq k - 7. \\ k \geq 7 &\implies \text{Sing}_i(T) \text{ is countably } (k - 7)\text{-rectifiable.} \\ k \geq 7 &\implies \mathcal{H}^{k-7}(\text{Sing}_i(T) \cap K) < \infty \quad (K \Subset M). \end{aligned}$$

Proof. [3, 6, 11, 26], [28, §37].

□

Example 3.3.4 (Sharp interior models). The calibrated two-dimensional complex curve and seven-dimensional Simons hypersurface cone are

$$\Sigma_{\text{cusp}} := \{(z, w) \in \mathbb{C}^2 : z^2 = w^3\} \subset \mathbb{C}^2 \cong \mathbb{R}^4, \quad \mathcal{C} := \{x \in \mathbb{R}^8 : x_1^2 + x_2^2 + x_3^2 + x_4^2 = x_5^2 + x_6^2 + x_7^2 + x_8^2\}$$

and define locally area-minimizing currents satisfying

$$[\Sigma_{\text{cusp}}] \in \mathbf{I}_{2,\text{loc}}(\mathbb{R}^4), \quad [\mathcal{C}] \in \mathbf{I}_{7,\text{loc}}(\mathbb{R}^8),$$

$$\text{Sing}_i([\Sigma_{\text{cusp}}]) = \text{Sing}_i([\mathcal{C}]) = \{0\}.$$

Local finiteness of $\mathcal{H}^{k-2} \llcorner \text{Sing}_i(T)$ remains open in higher codimension.

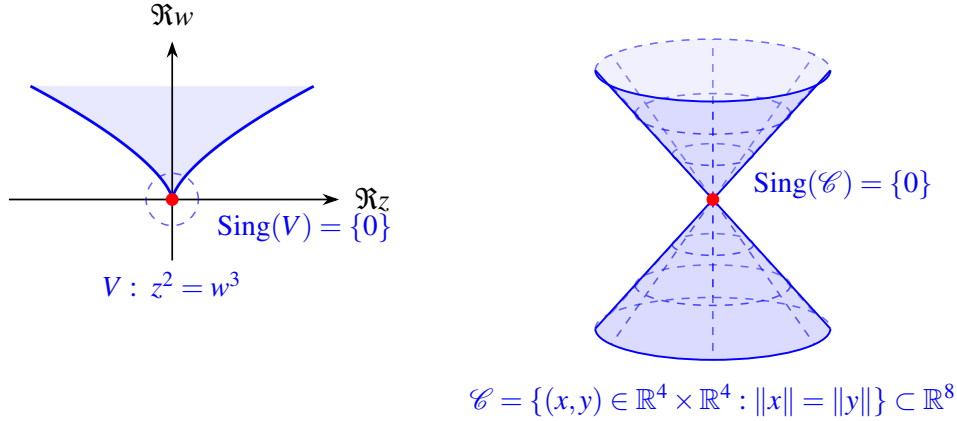


FIGURE 17. The real trace $V \cap \mathbb{R}^2 = \{(t^3, t^2) : t \in \mathbb{R}\}$ of the Kähler-calibrated complex cusp $V = \{(z, w) \in \mathbb{C}^2 : z^2 = w^3\}$ and a two-dimensional section of the seven-dimensional Simons cone $\mathcal{C} = \{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 : \|x\| = \|y\|\} \subset \mathbb{R}^8$ both have singular set $\{0\}$.

Definition 3.3.5 (Area minimization modulo p). For a smooth Riemannian manifold M^n , $p \in \mathbb{N}$, $p \geq 2$, and $T \in \mathcal{F}_k^{\mathbb{Z}}(M)$, write $[T]_p \in \mathcal{F}_k(M; \mathbb{Z}_p)$. For $S \in \mathcal{F}_k^{\mathbb{Z}}(M)$ and open $W \subset M$,

$$[S]_p = [T]_p \iff \exists Q \in \mathcal{F}_k^{\mathbb{Z}}(M), \quad S - T = pQ,$$

$$\mathbf{M}_{p,W}([T]_p) := \inf \left\{ \mathbf{M}_W(S) : S \in \mathcal{F}_k^{\mathbb{Z}}(M), [S]_p = [T]_p \right\}.$$

The class $[T]_p$ is locally area minimizing modulo p if, for every open $W \Subset M$ and $Q \in \mathcal{F}_{k+1}^{\mathbb{Z}}(M)$ with $\text{spt}(Q) \Subset W$,

$$\mathbf{M}_{p,W}([T]_p) \leq \mathbf{M}_{p,W}([T + \partial Q]_p).$$

A point $x \in \text{spt}([T]_p) \setminus \text{spt}(\partial[T]_p)$ lies in $\text{Reg}_i([T]_p)$ if a neighborhood $W \ni x$, an integer $m \notin p\mathbb{Z}$, and a smooth connected oriented embedded k -submanifold $\Sigma \subset W$ satisfy

$$[T \llcorner W]_p = [m[\Sigma]]_p.$$

$$\text{Sing}_i([T]_p) := (\text{spt}([T]_p) \setminus \text{spt}(\partial[T]_p)) \setminus \text{Reg}_i([T]_p).$$

Theorem 3.3.6 (Area minimizers modulo p). For locally area-minimizing $[T]_p \in \mathcal{F}_k(M^n; \mathbb{Z}_p)$, $p \in \mathbb{N}$, $p \geq 2$,

$$\dim_{\mathcal{H}}(\text{Sing}_i([T]_p)) \leq k - 1.$$

For odd p ,

$$\text{Sing}_i([T]_p) \text{ is countably } (k-1)\text{-rectifiable}, \quad \mathcal{H}^{k-1}(\text{Sing}_i([T]_p) \cap K) < \infty \quad (K \Subset M).$$

Proof. [10]. □

Theorem 3.3.7 (Boundary regularity regimes). *Let $T \in \mathbf{I}_k(M^n)$ be locally area minimizing with $\partial T = \llbracket \Gamma \rrbracket$, where $\Gamma \subset M$ is a smooth closed embedded oriented $(k-1)$ -submanifold of multiplicity one.*

$$\overline{\text{Reg}_b(T)} = \text{spt}(\partial T).$$

If either $\Gamma \subset \partial\Omega$ for a uniformly convex $\Omega \subset M$ with $\text{spt}(T) \subset \overline{\Omega}$, or $n-k=1$,

$$\text{Sing}_b(T) = \emptyset.$$

Higher-codimensional boundary singularities may occur without the barrier hypothesis.

Proof. [2, 20, 9, 12]. □

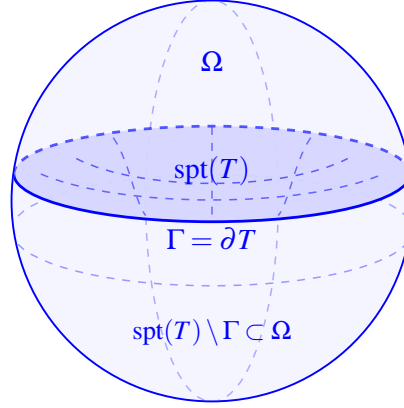


FIGURE 18. The horizontal depth ellipse displays the uniformly convex domain Ω as a transparent sphere containing the recessed meshed support $\text{spt}(T)$, whose closed rim is the prescribed boundary $\Gamma = \partial T \subset \partial\Omega$.

3.4. Cubical deformation and polyhedral approximation.

Definition 3.4.1 (Primal and dual cubical skeleta). For $C = [0, 1]^n$, let $\mathcal{Q}_j$ be its j -faces, and give each $F \in \mathcal{Q}_j$ its coordinate orientation $\llbracket F \rrbracket$, $0 \leq j \leq n$. Define

$$\mathbb{R}^n = \bigcup_{z \in \mathbb{Z}^n} (z + C), \quad \mathcal{L}_j := \bigcup_{z \in \mathbb{Z}^n} \bigcup_{F \in \mathcal{Q}_j} (z + F), \quad \overline{\mathcal{L}}_j := \left(\frac{1}{2}, \dots, \frac{1}{2}\right) + \mathcal{L}_j,$$

$$\mathcal{L}_j \cap \overline{\mathcal{L}}_{n-j-1} = \emptyset \quad (0 \leq j < n), \quad \mathcal{Q}_j^{\delta, a} := \{a + \delta(z + F) : z \in \mathbb{Z}^n, F \in \mathcal{Q}_j\},$$

Every scaled face inherits the orientation of F , and

$$\mathcal{L}_j^{\delta, a} := a + \delta \mathcal{L}_j, \quad \overline{\mathcal{L}}_j^{\delta, a} := a + \delta \overline{\mathcal{L}}_j \quad (\delta > 0, a \in \mathbb{R}^n).$$

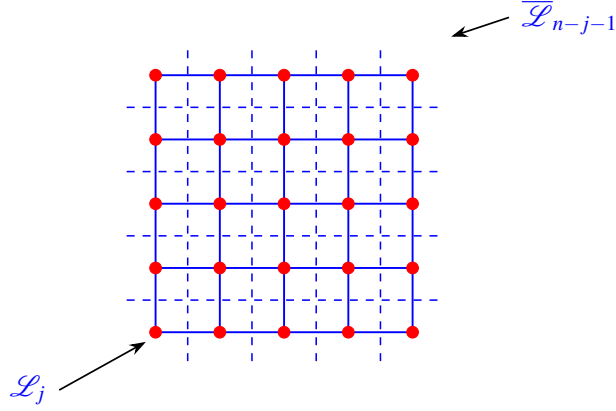


FIGURE 19. The primal skeleton $\mathcal{L}_j$ and the half-integer translate $\overline{\mathcal{L}}_{n-j-1}$ it never meets.

Lemma 3.4.2 (Cubical retraction). *For $0 \leq k < n$ and $\rho(x) := \text{dist}(x, \overline{\mathcal{L}}_{n-k-1})$, there exists a locally Lipschitz retraction $\psi : \mathbb{R}^n \setminus \overline{\mathcal{L}}_{n-k-1} \rightarrow \mathcal{L}_k$ satisfying*

$$\begin{aligned} \psi|_{\mathcal{L}_k} &= \text{id}, & \psi(z+x) &= z + \psi(x) \quad (z \in \mathbb{Z}^n), \\ |D\psi(x)| &\leq \frac{c_{n,k}}{\rho(x)} \quad \text{for almost every } x, & \int_C |D\psi(x)|^k dx &\leq c_{n,k}. \end{aligned}$$

Proof. For $0 \leq j \leq n-k-1$, radially project each open $(n-j)$ -face from its dual j -face in Figure 19.

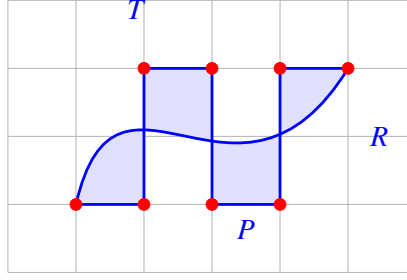
$$\begin{aligned} \psi_j : \mathcal{L}_{n-j} \setminus \overline{\mathcal{L}}_j &\rightarrow \mathcal{L}_{n-j-1}, & \psi_j|_{\mathcal{L}_{n-j-1}} &= \text{id}, & |D\psi_j(x)| &\leq \frac{c_n}{\text{dist}(x, \overline{\mathcal{L}}_j)}, \\ \psi &:= \psi_{n-k-1} \circ \cdots \circ \psi_1 \circ \psi_0 \implies \psi : \mathbb{R}^n \setminus \overline{\mathcal{L}}_{n-k-1} \rightarrow \mathcal{L}_k, & |D\psi(x)| &\leq \frac{c_{n,k}}{\rho(x)}, \\ \text{codim}(\overline{\mathcal{L}}_{n-k-1}) &= k+1 \implies \mathcal{L}^n(C \cap \{\rho < r\}) \leq c_{n,k} r^{k+1} \quad (0 < r < 1), \\ \int_C \rho(x)^{-k} dx &\leq \sum_{q=0}^{\infty} 2^{k(q+1)} \mathcal{L}^n(C \cap \{\rho < 2^{-q}\}) + \mathcal{L}^n(C) \leq c_{n,k} \sum_{q=0}^{\infty} 2^{-q} + 1 < \infty, \\ \int_C \rho^{-k} dx < \infty &\implies \int_C |D\psi|^k dx \leq c_{n,k} \int_C \rho^{-k} dx \leq c_{n,k}. \end{aligned}$$

□

Theorem 3.4.3 (Federer–Fleming deformation). *For $0 \leq k \leq n$, $T \in \mathcal{N}_k(\mathbb{R}^n)$, and $\delta > 0$, there exist $a \in [0, \delta]^n$, $P \in \mathcal{P}_k^{\text{loc}}(\mathbb{R}^n)$, $R \in \mathcal{N}_{k+1}(\mathbb{R}^n)$, and $S \in \mathcal{N}_k(\mathbb{R}^n)$ satisfying*

$$\begin{aligned} T &= P + \partial R + S, & \text{spt}(P) &\subset \mathcal{L}_k^{\delta, a}, \\ \mathbf{M}(P) &\leq c_{n,k} \mathbf{M}(T), & \mathbf{M}(\partial P) &\leq c_{n,k} \mathbf{M}(\partial T), & \mathbf{M}(R) &\leq c_{n,k} \delta \mathbf{M}(T), & \mathbf{M}(S) &\leq c_{n,k} \delta \mathbf{M}(\partial T), \\ \text{spt}(P) \cup \text{spt}(R) &\subset \{x : \text{dist}(x, \text{spt}(T)) \leq 2\sqrt{n}\delta\}, & \text{spt}(\partial P) \cup \text{spt}(S) &\subset \{x : \text{dist}(x, \text{spt}(\partial T)) \leq 2\sqrt{n}\delta\}. \\ T \in \mathcal{R}_k(\mathbb{R}^n) &\implies P = \sum_{Q \in \mathcal{Q}_k^{\delta, a}} m_Q [Q], & m_Q &\in \mathbb{Z}, & R &\in \mathcal{R}_{k+1}(\mathbb{R}^n), & \partial T \in \mathcal{R}_{k-1}(\mathbb{R}^n) &\implies S \in \mathcal{R}_k(\mathbb{R}^n), \\ \text{spt}(T) \in \mathbb{R}^n &\implies P \in \mathcal{P}_k(\mathbb{R}^n), & T \in \mathcal{R}_k(\mathbb{R}^n), \text{spt}(T) \in \mathbb{R}^n &\implies P \in \mathcal{P}_k^{\mathbb{Z}}(\mathbb{R}^n). \end{aligned}$$

Proof. The averaged translated cubical retractions and homotopies give the decomposition in Figure 20 [15, §4.2.9], [28, §29], [22, Chapter 7]. □


 FIGURE 20. The current T is pushed onto the grid skeleton to a polyhedral P , sweeping R .

Theorem 3.4.4 (Isoperimetric inequality for integral cycles). *Every $B \in \mathbf{I}_m(\mathbb{R}^n)$, $1 \leq m < n$, with $\text{spt}(B) \Subset \mathbb{R}^n$ and $\partial B = 0$, admits $R \in \mathbf{I}_{m+1}(\mathbb{R}^n)$ satisfying*

$$\partial R = B, \quad \mathbf{M}(R) \leq c_{n,m} \mathbf{M}(B)^{\frac{m+1}{m}}, \quad \text{spt}(R) \subset \left\{ x : \text{dist}(x, \text{spt}(B)) \leq c_{n,m} \mathbf{M}(B)^{1/m} \right\}.$$

Proof. The case $B = 0$ has $R = 0$; assume $B \neq 0$, let C dominate the constants in Theorem 3.4.3, and put $\delta := (2C\mathbf{M}(B))^{1/m}$.

$$\partial B = 0 \xrightarrow{3.4.3} B = P + \partial R + S, \quad \mathbf{M}(P) \leq C\mathbf{M}(B), \quad \mathbf{M}(R) \leq C\delta\mathbf{M}(B), \quad \mathbf{M}(S) \leq C\delta\mathbf{M}(\partial B) = 0,$$

$$P = \sum_{Q \in \mathcal{Q}_m^{\delta,a}} m_Q [[Q]], \quad m_Q \in \mathbb{Z}, \quad P \neq 0 \xrightarrow{3.4.1} \delta^m \leq \mathbf{M}(P),$$

$$2C\mathbf{M}(B) = \delta^m \leq \mathbf{M}(P) \leq C\mathbf{M}(B) \implies P = 0,$$

$$\mathbf{M}(S) = 0 \implies S = 0,$$

$$B = P + \partial R + S, \quad P = S = 0 \implies \partial R = B.$$

$$B \in \mathbf{I}_m(\mathbb{R}^n) \xrightarrow{3.4.3} R \in \mathcal{R}_{m+1}(\mathbb{R}^n).$$

$$\partial R = B \in \mathcal{R}_m(\mathbb{R}^n) \xrightarrow{2.4.5} R \in \mathbf{I}_{m+1}(\mathbb{R}^n),$$

$$\mathbf{M}(R) \leq C(2C)^{1/m} \mathbf{M}(B)^{\frac{m+1}{m}} \leq c_{n,m} \mathbf{M}(B)^{\frac{m+1}{m}},$$

$$B \xrightarrow{3.4.3} \text{spt}(R) \subset \left\{ x : \text{dist}(x, \text{spt}(B)) \leq 2\sqrt{n}\delta \right\} \subset \left\{ x : \text{dist}(x, \text{spt}(B)) \leq c_{n,m} \mathbf{M}(B)^{1/m} \right\}.$$

□

Theorem 3.4.5 (Weak polyhedral approximation). *Every $T \in \mathcal{N}_k(\mathbb{R}^n)$ admits $P_j \in \mathcal{P}_k^{\text{loc}}(\mathbb{R}^n)$ satisfying*

$$P_j \rightharpoonup T, \quad \partial P_j \rightharpoonup \partial T, \quad \sup_{j \geq 1} \mathbf{M}(P_j) \leq c_{n,k} \mathbf{M}(T), \quad \sup_{j \geq 1} \mathbf{M}(\partial P_j) \leq c_{n,k} \mathbf{M}(\partial T).$$

$$T \in \mathbf{I}_k(\mathbb{R}^n) \implies P_j \in \mathcal{P}_k^{\mathbb{Z}, \text{loc}}(\mathbb{R}^n), \quad \text{spt}(T) \Subset \mathbb{R}^n \implies P_j \in \mathcal{P}_k(\mathbb{R}^n), \quad \bigcup_{j \geq 1} \text{spt}(P_j) \Subset \mathbb{R}^n.$$

Proof. Fix $\delta_j \downarrow 0$ and $\omega \in \mathcal{D}^k(\mathbb{R}^n)$.

$$T \xrightarrow{3.4.3} T = P_j + \partial R_j + S_j, \quad \mathbf{M}(R_j) \leq c_{n,k} \delta_j \mathbf{M}(T), \quad \mathbf{M}(S_j) \leq c_{n,k} \delta_j \mathbf{M}(\partial T),$$

$$\mathbf{M}(P_j) \leq c_{n,k} \mathbf{M}(T), \quad \mathbf{M}(\partial P_j) \leq c_{n,k} \mathbf{M}(\partial T),$$

$$|(P_j - T)(\omega)| \leq \mathbf{M}(R_j) \sup_{\mathbb{R}^n} \|d\omega\|^* + \mathbf{M}(S_j) \sup_{\mathbb{R}^n} \|\omega\|^* \longrightarrow 0 \implies P_j \rightharpoonup T,$$

$$k = 0 \implies \partial P_j = \partial T = 0.$$

For $k \geq 1$ and $\eta \in \mathcal{D}^{k-1}(\mathbb{R}^n)$,

$$S_j \xrightarrow{3.4.3} \partial(P_j - T) = -\partial S_j, \quad \mathbf{M}(S_j) \rightarrow 0 \implies |\partial(P_j - T)(\eta)| \leq \mathbf{M}(S_j) \sup_{\mathbb{R}^n} \|d\eta\|^* \rightarrow 0 \implies \partial P_j \rightarrow \partial T,$$

$$T \in \mathbf{I}_k(\mathbb{R}^n) \xrightarrow{3.4.3} P_j \in \mathcal{P}_k^{\mathbb{Z}, \text{loc}}(\mathbb{R}^n), \quad \text{spt}(T) \in \mathbb{R}^n \xrightarrow{3.4.3} P_j \in \mathcal{P}_k(\mathbb{R}^n), \quad \bigcup_{j \geq 1} \text{spt}(P_j) \in \mathbb{R}^n.$$

□

Theorem 3.4.6 (Strong polyhedral approximation). *For $T \in \mathbf{I}_k(\mathbb{R}^n)$ and $\varepsilon > 0$, there exist a C^1 bi-Lipschitz diffeomorphism $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $P \in \mathcal{P}_k^{\mathbb{Z}}(\mathbb{R}^n)$ satisfying*

$$\mathbf{M}(T - f_{\#}P) + \mathbf{M}(\partial T - f_{\#}\partial P) < \varepsilon, \quad \sup_{x \in \mathbb{R}^n} (|Df(x) - I| + |f(x) - x|) < \varepsilon, \quad f(x) = x \quad (\text{dist}(x, \text{spt}(T)) \geq \varepsilon).$$

Proof. [15, §4.2.20], [22, Chapter 7].

□

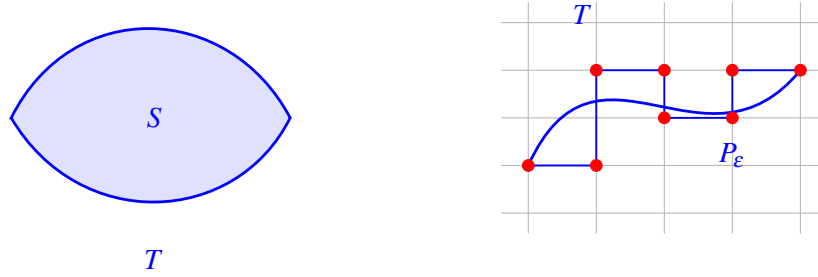


FIGURE 21. An integral cycle T bounds S , and is approximated on a fine grid by P_ε .

Part 2. Lecture B⁴**4. RADON MEASURES, COVERING, AND DIFFERENTIATION**

Throughout, X is a set, $\mathcal{P}(X)$ is its power set, $\mathcal{B}(\mathbb{R}^n)$ is the Borel σ -algebra, every $B_r(x)$ is closed with $r = r(B) > 0$, $tB_r(x) := B_{tr}(x)$, outer measures carry a star, $N(n)$ is the Besicovitch overlap number, and $f\mu$ denotes $A \mapsto \int_A f d\mu$ [17, 14, 25, 15, 27].

4.1. Outer measures, Carathéodory measurability, and Radon measures.

Definition 4.1.1 (Outer measure and Carathéodory measurability). An outer measure on X is a map $\mu^* : \mathcal{P}(X) \rightarrow [0, +\infty]$ with

$$\mu^*(\emptyset) = 0, \quad A \subset B \implies \mu^*(A) \leq \mu^*(B), \quad A \subset \bigcup_{j=1}^{\infty} A_j \implies \mu^*(A) \leq \sum_{j=1}^{\infty} \mu^*(A_j).$$

A set $E \subset X$ is μ^* -measurable when

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \setminus E) \quad (A \subset X), \quad \mathcal{M}_{\mu^*} := \{E \subset X : E \text{ is } \mu^*\text{-measurable}\}.$$

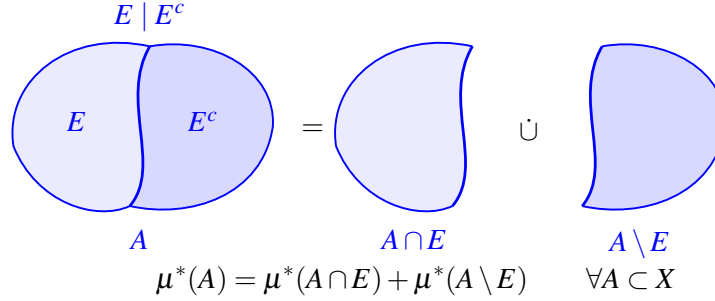


FIGURE 22. Every test set splits additively across E .

Theorem 4.1.2 (Carathéodory construction). Every outer measure μ^* on X determines a σ -algebra $\mathcal{M}_{\mu^*}$ and a complete measure $\mu := \mu^*|_{\mathcal{M}_{\mu^*}}$.

Proof. Figure 22 records the splitting identity; for disjoint $E_j \in \mathcal{M}_{\mu^*}$, write $F_N := \bigcup_{j=1}^N E_j$.

$$\begin{aligned} \mu^*(A \cap E) + \mu^*(A \setminus E) &= \mu^*(A \cap E^c) + \mu^*(A \setminus E^c) \implies (E \in \mathcal{M}_{\mu^*} \iff E^c \in \mathcal{M}_{\mu^*}), \\ E, F \in \mathcal{M}_{\mu^*} \implies \mu^*(A) &= \mu^*(A \cap E) + \mu^*(A \cap (F \setminus E)) + \mu^*(A \setminus (E \cup F)) \geq \mu^*(A \cap (E \cup F)) + \mu^*(A \setminus (E \cup F)), \\ \mu^*(A) &\leq \mu^*(A \cap (E \cup F)) + \mu^*(A \setminus (E \cup F)) \implies E \cup F \in \mathcal{M}_{\mu^*}, \\ F_N \in \mathcal{M}_{\mu^*}, \quad \mu^*(A) &= \sum_{j=1}^N \mu^*(A \cap E_j) + \mu^*(A \setminus F_N) \geq \sum_{j=1}^N \mu^*(A \cap E_j) + \mu^*\left(A \setminus \bigcup_{j=1}^{\infty} E_j\right), \\ N \rightarrow \infty, \quad E := \bigcup_{j=1}^{\infty} E_j \implies \mu^*(A) &\geq \sum_{j=1}^{\infty} \mu^*(A \cap E_j) + \mu^*(A \setminus E) \geq \mu^*(A \cap E) + \mu^*(A \setminus E) \implies E \in \mathcal{M}_{\mu^*}, \\ A = E, \quad \mu^*(E) &\leq \sum_{j=1}^{\infty} \mu^*(E_j) \implies \mu(A) = \sum_{j=1}^{\infty} \mu(E_j), \\ N \subset Z, \quad \mu(Z) = 0 \implies \mu^*(A \cap N) &= 0, \quad \mu^*(A) = \mu^*(A \setminus N) \implies N \in \mathcal{M}_{\mu^*}, \quad \mu(N) = 0. \end{aligned}$$

□

⁴Based on lectures by Annalisa Massaccesi [24].

Definition 4.1.3 (Metric, Borel, and Borel regular outer measures). On a metric space (X, d) , an outer measure μ^* is metric when

$$\text{dist}(A, B) > 0 \implies \mu^*(A \cup B) = \mu^*(A) + \mu^*(B) \quad (A, B \subset X);$$

it is Borel when every Borel set is μ^* -measurable, and Borel regular when moreover every $A \subset X$ admits a Borel $B \supset A$ with $\mu^*(B) = \mu^*(A)$.

Theorem 4.1.4 (Borel measurability of metric outer measures). *Every metric outer measure on a metric space is Borel.*

Proof. Fix a closed $F \subset X$ and $A \subset X$, and put $A_0 := \emptyset$, $A_j := \{x \in A : \text{dist}(x, F) \geq j^{-1}\}$, and $C_j := A_j \setminus A_{j-1}$.

$$\begin{aligned} \mu^*(A \setminus F) = +\infty &\implies \mu^*(A) = +\infty = \mu^*(A \cap F) + \mu^*(A \setminus F), \\ \mu^*(A \setminus F) < +\infty, \quad |i - j| > 1 &\implies \text{dist}(C_i, C_j) > 0 \implies \sum_{j=2}^{\infty} \mu^*(C_j) \leq 2\mu^*(A \setminus F) < +\infty, \\ \mu^*(A) &\geq \mu^*(A_j) + \mu^*(A \cap F) \geq \mu^*(A \setminus F) - \sum_{i=j+1}^{\infty} \mu^*(C_i) + \mu^*(A \cap F) \xrightarrow{j \rightarrow \infty} \mu^*(A \setminus F) + \mu^*(A \cap F); \\ \mu^*(A) &\leq \mu^*(A \setminus F) + \mu^*(A \cap F) \leq \mu^*(A) \implies F \in \mathcal{M}_{\mu^*}, \\ \{F \subset X : F \text{ closed}\} &\subset \mathcal{M}_{\mu^*} \xrightarrow{4.1.2} \mathcal{B}(X) \subset \mathcal{M}_{\mu^*}. \end{aligned}$$

□

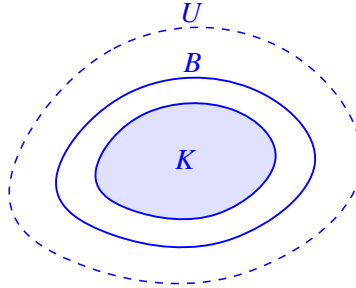
Definition 4.1.5 (Radon measure). On a locally compact Hausdorff space X , a Borel measure μ is Radon when

$$\mu(K) < +\infty \quad (K \text{ compact}), \quad \mu(B) = \inf_{\substack{U \supset B \\ U \text{ open}}} \mu(U) \quad (B \text{ Borel}), \quad \mu(U) = \sup_{\substack{K \subset U \\ K \text{ compact}}} \mu(K) \quad (U \text{ open}).$$

Theorem 4.1.6 (Regularity on Euclidean space). *Every Borel measure μ on $\mathbb{R}^n$ finite on compact sets is Radon, and every Borel $B \subset \mathbb{R}^n$ satisfies*

$$\mu(B) = \sup_{\substack{K \subset B \\ K \text{ compact}}} \mu(K) = \inf_{\substack{U \supset B \\ U \text{ open}}} \mu(U).$$

Proof. This is the regularity theorem for locally finite Borel measures on locally compact second-countable spaces [17, Chapter 7]. □



$$\mu(B) = \sup_{K \in \mathcal{K}} \mu(K) = \inf_{U \supset B} \mu(U)$$

FIGURE 23. A Radon measure is squeezed between compact sets from inside and open sets from outside.

4.2. Signed measures and Riesz representation.

Definition 4.2.1 (Signed Radon measure and total variation). A signed Radon measure on $\mathbb{R}^n$ is a countably additive $\nu : \mathcal{B}(\mathbb{R}^n) \rightarrow [-\infty, +\infty]$, finite on compact sets and taking at most one of the values $\pm\infty$. For $B \in \mathcal{B}(\mathbb{R}^n)$, its positive, negative, and total variations are

$$\nu^+(B) := \sup_{\substack{A \subset B \\ A \in \mathcal{B}(\mathbb{R}^n)}} \nu(A), \quad \nu^-(B) := - \inf_{\substack{A \subset B \\ A \in \mathcal{B}(\mathbb{R}^n)}} \nu(A), \quad |\nu| := \nu^+ + \nu^-.$$

Theorem 4.2.2 (Hahn–Jordan decomposition). *For every signed measure ν on $\mathbb{R}^n$, there exist $P, N \in \mathcal{B}(\mathbb{R}^n)$ satisfying*

$$P \cap N = \emptyset, \quad P \cup N = \mathbb{R}^n, \quad \nu(A) \geq 0 \quad (A \in \mathcal{B}(\mathbb{R}^n), A \subset P), \quad \nu(A) \leq 0 \quad (A \in \mathcal{B}(\mathbb{R}^n), A \subset N),$$

$$\nu = \nu^+ - \nu^-, \quad |\nu| = \nu^+ + \nu^-, \quad \nu^+ \perp \nu^-.$$

The Hahn decomposition $\mathbb{R}^n = P \sqcup N$ is unique up to $|\nu|$ -null sets, and the Jordan decomposition is unique.

Proof. See [17, Chapter 3] and [27, Chapter 11]. $\square$

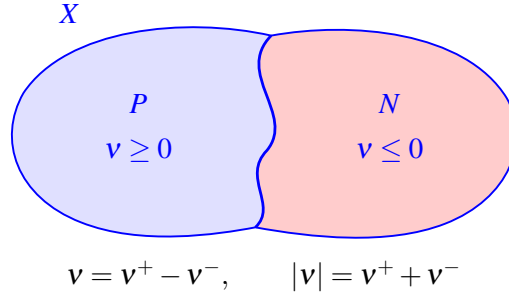


FIGURE 24. A smooth schematic of the Hahn decomposition $X = P \sqcup N$ carries ν^+ on P and ν^- on N , so $|\nu| = \nu^+ + \nu^-$ records no cancellation.

Proposition 4.2.3 (Variation as a supremum and polar decomposition). *For every signed Radon measure ν and $B \in \mathcal{B}(\mathbb{R}^n)$,*

$$|\nu|(B) = \sup \left\{ \sum_{j=1}^{\infty} |\nu(B_j)| : B = \bigsqcup_{j=1}^{\infty} B_j, \quad B_j \in \mathcal{B}(\mathbb{R}^n) \right\},$$

and there is a unique $|\nu|$ -measurable $\sigma : \mathbb{R}^n \rightarrow \{-1, 1\}$, up to $|\nu|$ -null sets, satisfying

$$\nu(B) = \int_B \sigma d|\nu| \quad (B \in \mathcal{B}(\mathbb{R}^n)).$$

Proof. Let $\mathbb{R}^n = P \sqcup N$ be the Hahn decomposition in Theorem 4.2.2.

$$B = \bigsqcup_{j=1}^{\infty} B_j \implies \sum_{j=1}^{\infty} |\nu(B_j)| \leq \sum_{j=1}^{\infty} (\nu^+(B_j) + \nu^-(B_j)) = |\nu|(B).$$

$$B = (B \cap P) \sqcup (B \cap N) \xrightarrow{4.2.2} |\nu|(B) = |\nu(B \cap P)| + |\nu(B \cap N)|.$$

$$\sigma := \mathbf{1}_P - \mathbf{1}_N \implies \int_B \sigma d|\nu| = \nu^+(B) - \nu^-(B) = \nu(B), \quad \tau |\nu| = \nu \implies \sigma = \tau \quad |\nu| \text{-a.e..}$$

$\square$

Theorem 4.2.4 (Polar decomposition of a vector measure). *For every $\mathbb{R}^m$ -valued Radon measure $\nu = (\nu_1, \dots, \nu_m)$ on $\mathbb{R}^n$, there exist a positive Radon measure $|\nu|$ and a Borel $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfying*

$$|\sigma(x)| = 1 \quad |\nu|\text{-a.e.}, \quad \nu(B) = \int_B \sigma d|\nu| \quad (B \in \mathcal{B}(\mathbb{R}^n)),$$

$$|\nu|(B) = \sup \left\{ \sum_{j=1}^{\infty} |\nu(B_j)| : B = \bigsqcup_{j=1}^{\infty} B_j, \quad B_j \in \mathcal{B}(\mathbb{R}^n) \right\} \quad (B \in \mathcal{B}(\mathbb{R}^n)).$$

The measure $|\nu|$ is unique, and σ is unique $|\nu|$ -almost everywhere.

Proof. See [14, §1.9] and [17, Chapter 3]. □

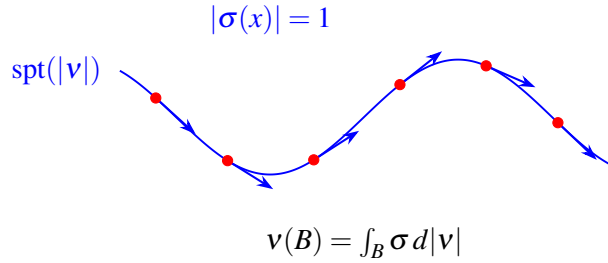


FIGURE 25. The mass $|\nu|$ lives on the carrier and σ records its unit polar direction.

Theorem 4.2.5 (Riesz representation theorem). *Let $L : C_c(\mathbb{R}^n) \rightarrow \mathbb{R}$ be linear and suppose $\|L\|_K := \sup\{|L(f)| : f \in C_c(\mathbb{R}^n), |f| \leq 1, \text{spt}(f) \subset K\} < +\infty$ for every compact $K \subset \mathbb{R}^n$. There exists a unique signed Radon measure ν satisfying*

$$L(f) = \int_{\mathbb{R}^n} f d\nu \quad (f \in C_c(\mathbb{R}^n)),$$

and $\nu \geq 0$ if $L(f) \geq 0$ for every $0 \leq f \in C_c(\mathbb{R}^n)$.

Proof. See [17, Chapter 7] and [5, Chapter 4]. □

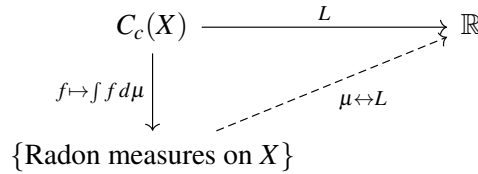


FIGURE 26. Riesz representation. A positive linear functional on compactly supported continuous functions is evaluation against one Radon measure, and equality on all test functions forces equality of measures. The dashed diagonal records the uniqueness and reconstruction step.

4.3. Covering theorems.

Lemma 4.3.1 (Five-times covering lemma). *Let $\mathcal{F}$ be a nonempty family of closed balls in $\mathbb{R}^n$ with $R := \sup_{B \in \mathcal{F}} r(B) < +\infty$. There exists a countable pairwise disjoint $\mathcal{G} \subset \mathcal{F}$ such that*

$$\bigcup_{B \in \mathcal{F}} B \subset \bigcup_{B \in \mathcal{G}} 5B.$$

Proof. For $k \geq 0$, use Zorn's lemma to choose a maximal disjoint $\mathcal{G}_k \subset \{B \in \mathcal{F} : 2^{-k-1}R < r(B) \leq 2^{-k}R\}$ disjoint from the preceding selections, put $\mathcal{G} := \bigcup_{k=0}^{\infty} \mathcal{G}_k$, and choose $d_G \in \mathbb{Q}^n \cap G^\circ$ for each $G \in \mathcal{G}$.

$$\begin{aligned} B = B_r(x) \in \mathcal{F} \setminus \mathcal{G} &\implies \exists \widehat{B} = B_s(y) \in \mathcal{G}, \quad B \cap \widehat{B} \neq \emptyset, \quad s \geq \frac{r}{2}, \\ z \in B &\implies |z - y| \leq r + (r + s) = 2r + s \leq 5s \implies B \subset 5\widehat{B} \quad (27). \\ G \neq G' &\implies G^\circ \cap (G')^\circ = \emptyset \implies d_G \neq d_{G'} \implies |\mathcal{G}| \leq |\mathbb{Q}^n| = \aleph_0. \end{aligned}$$

□

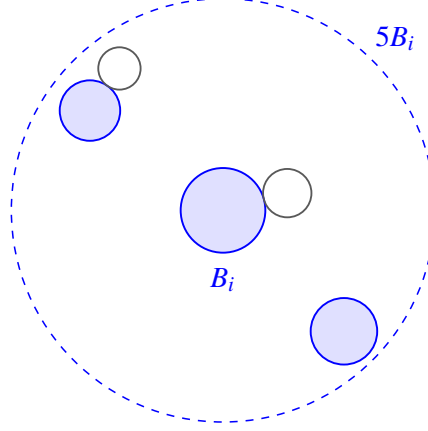


FIGURE 27. A discarded ball touches a selected ball and lies inside its fivefold dilation.

Definition 4.3.2 (Fine cover). A family $\mathcal{F}$ of closed balls is a fine cover of $A \subset \mathbb{R}^n$ when

$$\forall x \in A \quad \forall \delta > 0 \quad \exists B_r(x) \in \mathcal{F}, \quad 0 < r < \delta.$$

Theorem 4.3.3 (Vitali covering theorem). Let μ be a Radon measure on $\mathbb{R}^n$, let A be μ -measurable, and let $\mathcal{F}$ be a fine cover of A by closed balls with $\sup_{B \in \mathcal{F}} r(B) < +\infty$. There exists a countable pairwise disjoint $\{B_j\}_{j=1}^{\infty} \subset \mathcal{F}$ such that

$$\mu\left(A \setminus \bigcup_{j=1}^{\infty} B_j\right) = 0.$$

If $\mu(A) < +\infty$, every $\varepsilon > 0$ admits $J \in \mathbb{N}$ such that

$$\mu\left(A \setminus \bigcup_{j=1}^J B_j\right) < \varepsilon.$$

Proof. See [15, §2.8], [14, §1.5], [25, Chapter 2], and Figure 28.

□

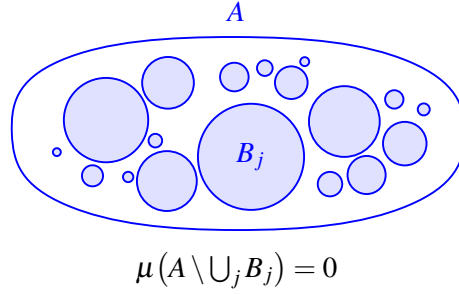


FIGURE 28. A fine cover admits pairwise disjoint balls of shrinking radii that leave only a null part of A uncovered.

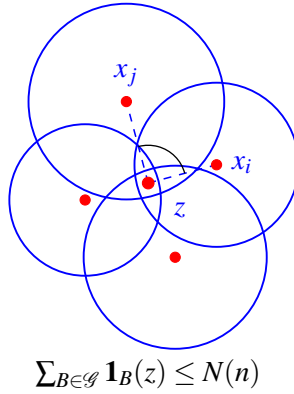
Theorem 4.3.4 (Besicovitch covering theorem). *There exists $N(n) \in \mathbb{N}$ such that every bounded $A \subset \mathbb{R}^n$ and every family $\mathcal{F}_x$ of closed balls centered at x , with arbitrarily small radii for each $x \in A$, admit a countable $\mathcal{G} \subset \bigcup_{x \in A} \mathcal{F}_x$ satisfying*

$$A \subset \bigcup_{B \in \mathcal{G}} B, \quad \mathcal{G} = \bigcup_{i=1}^{N(n)} \mathcal{G}_i, \quad B \cap B' = \emptyset \quad (B, B' \in \mathcal{G}_i, B \neq B'), \quad \sup_{y \in \mathbb{R}^n} \sum_{B \in \mathcal{G}} \mathbf{1}_B(y) \leq N(n).$$

Proof. See [15, §2.8], [14, §1.5], and [25, Chapter 2]. □

Corollary 4.3.5 (Finite-overlap integral bound). *For the family $\mathcal{G}$ in Theorem 4.3.4 and every positive Radon measure μ ,*

$$\sum_{B \in \mathcal{G}} \mu(B) = \int_{\mathbb{R}^n} \sum_{B \in \mathcal{G}} \mathbf{1}_B d\mu \leq N(n) \mu\left(\bigcup_{B \in \mathcal{G}} B\right).$$



$$\sum_{B \in \mathcal{G}} \mathbf{1}_B(z) \leq N(n)$$

FIGURE 29. Selected balls through z have centers separated by a definite angle, which caps how many can overlap there.

4.4. Differentiation of Radon measures.

Definition 4.4.1 (Relative upper and lower derivatives). Let μ, ν be Radon measures on $\mathbb{R}^n$. Using $0/0 := 0$ and $a/0 := +\infty$ for $a > 0$, define for every x the extreme derivatives and, whenever the limit exists, $D_\mu \nu(x)$ by

$$\overline{D}_\mu \nu(x) := \limsup_{r \downarrow 0} \frac{\nu(B_r(x))}{\mu(B_r(x))}, \quad \underline{D}_\mu \nu(x) := \liminf_{r \downarrow 0} \frac{\nu(B_r(x))}{\mu(B_r(x))}, \quad D_\mu \nu(x) := \lim_{r \downarrow 0} \frac{\nu(B_r(x))}{\mu(B_r(x))}.$$

Both extreme derivatives are Borel measurable, since $x \mapsto \mu(B_r(x))$ and $x \mapsto \nu(B_r(x))$ are Borel for every $r > 0$ and

$$\bar{D}_\mu \nu = \inf_{k \geq 1} \sup_{\substack{q \in \mathbb{Q} \\ 0 < q < 1/k}} \frac{\nu(B_q(\cdot))}{\mu(B_q(\cdot))}, \quad D_\mu \nu = \sup_{k \geq 1} \inf_{\substack{q \in \mathbb{Q} \\ 0 < q < 1/k}} \frac{\nu(B_q(\cdot))}{\mu(B_q(\cdot))}.$$

Lemma 4.4.2 (Density comparison inequalities). *For Radon measures μ, ν on $\mathbb{R}^n$, a Borel set $A \subset \mathbb{R}^n$, and $\alpha \geq 0$,*

$$\begin{aligned} \bar{D}_\mu \nu \leq \alpha \text{ on } A &\implies \nu(A) \leq \alpha \cdot \mu(A), & \bar{D}_\mu \nu \geq \alpha \text{ on } A &\implies \nu(A) \geq \alpha \cdot \mu(A), \\ \bar{D}_\mu \nu = 0 \text{ on } A &\implies \nu(A) = 0, & (\bar{D}_\mu \nu = +\infty \text{ on } A, \nu(A) < +\infty) &\implies \mu(A) = 0. \end{aligned}$$

Proof. Fix $m \in \mathbb{N}$, $A_m := A \cap B_m(0)$, $\varepsilon > 0$, and an open U with $A_m \subset U \subseteq B_{m+1}(0)^\circ$, restricting to balls centered on A_m , contained in U , and of radius less than 1, with ν used in the first selection and μ in the second.

$$\begin{aligned} \bar{D}_\mu \nu \leq \alpha \text{ on } A_m &\xrightarrow{4.3.3} \nu(A_m) \leq (\alpha + \varepsilon)\mu(U), \\ \alpha > 0, \quad 0 < \varepsilon < \alpha, &\quad \bar{D}_\mu \nu \geq \alpha \text{ on } A_m \xrightarrow{4.3.3} (\alpha - \varepsilon)\mu(A_m) \leq \nu(U), \\ \varepsilon \downarrow 0, \quad m \uparrow \infty &\xrightarrow{4.1.6} \nu(A) \leq \alpha\mu(A), \quad \nu(A) \geq \alpha\mu(A), \end{aligned}$$

□

Theorem 4.4.3 (Differentiation of measures). *For Radon measures μ, ν on $\mathbb{R}^n$, the derivative $D_\mu \nu(x)$ exists with $0 \leq D_\mu \nu(x) < +\infty$ for μ -almost every x , and the unique Lebesgue decomposition satisfies*

$$\begin{aligned} \nu &= \nu_{ac} + \nu_s, \quad \nu_{ac} \ll \mu, \quad \nu_s \perp \mu, \quad \nu_{ac}(A) = \int_A D_\mu \nu d\mu \quad (A \subset \mathbb{R}^n \text{ Borel}), \\ D_\mu \nu_s &= 0 \quad \mu\text{-almost everywhere}, \quad D_\mu \nu = \frac{d\nu_{ac}}{d\mu} \quad \mu\text{-almost everywhere}. \end{aligned}$$

Proof. By the Lebesgue decomposition and Radon–Nikodým theorem, write $\nu = f\mu + \nu_s$ with $\nu_s \perp \mu$, choose continuous $f_j \rightarrow f$ in $L^1_{loc}(\mu)$, fix $K \subset \text{int}(K')$ compact, and denote the local maximal error by M_j and the Lebesgue oscillation by L_f [17, Chapter 3].

$$\begin{aligned} M_j(x) &:= \limsup_{r \downarrow 0} \frac{1}{\mu(B_r(x))} \int_{B_r(x)} |f - f_j| d\mu, \quad L_f(x) := \limsup_{r \downarrow 0} \frac{1}{\mu(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| d\mu(y), \\ \mu(K \cap \{M_j > \varepsilon\}) &\xrightarrow{4.3.4, 4.3.5} \leq \frac{N(n)}{\varepsilon} \int_{K'} |f - f_j| d\mu, \\ L_f \leq M_j + |f_j - f|, \quad \mu(K \cap \{L_f > 2\varepsilon\}) &\leq \frac{N(n) + 1}{\varepsilon} \int_{K'} |f - f_j| d\mu \xrightarrow{j \rightarrow \infty} 0 \implies L_f = 0 \quad \mu\text{-a.e.}, \\ \mu(S) &= 0, \quad \nu_s(S^c) = 0, \quad S_j := S^c \cap \{\bar{D}_\mu \nu_s \geq j^{-1}\} \quad (j \geq 1), \\ \bar{D}_\mu \nu_s \geq j^{-1} \text{ on } S_j &\xrightarrow{4.4.2} 0 = \nu_s(S_j) \geq j^{-1}\mu(S_j) \implies \mu(S_j) = 0 \implies \bar{D}_\mu \nu_s = 0 \quad \mu\text{-a.e.}, \\ \bar{D}_\mu \nu_s = 0 &\implies D_\mu \nu = f \quad \mu\text{-a.e.}, \quad \nu_{ac} = f \cdot \mu \implies \nu_{ac}(A) = \int_A D_\mu \nu d\mu, \quad D_\mu \nu = \frac{d\nu_{ac}}{d\mu} \quad \mu\text{-a.e.}, \\ \nu &= \nu_{ac} + \nu_s = \tilde{\nu}_{ac} + \tilde{\nu}_s \implies \lambda := \nu_{ac} - \tilde{\nu}_{ac} = \tilde{\nu}_s - \nu_s, \quad \lambda \ll \mu, \quad \lambda \perp \mu \implies \lambda = 0 \quad (31). \end{aligned}$$

□

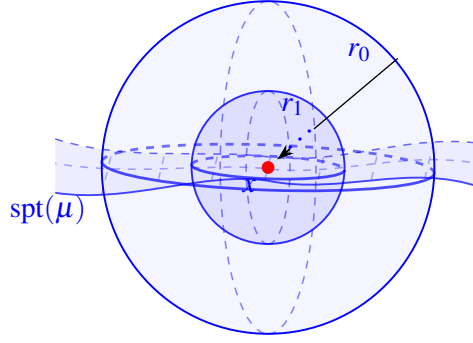


FIGURE 30. The ratios of ν -mass to μ -mass on balls shrinking to x recover the density $D_\mu \nu(x)$.

Corollary 4.4.4 (Lebesgue points and differentiation of sets). *For every Radon measure μ on $\mathbb{R}^n$, every $f \in L^1_{\text{loc}}(\mu)$, and every μ -measurable $E \subset \mathbb{R}^n$, at μ -almost every x ,*

$$\lim_{r \downarrow 0} \frac{1}{\mu(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| d\mu(y) = 0, \quad \lim_{r \downarrow 0} \frac{\mu(E \cap B_r(x))}{\mu(B_r(x))} = \mathbf{1}_E(x).$$

Proof. Replace E by a Borel representative and apply Theorem 4.4.3 to $|f - q|\mu$, $q \in \mathbb{Q}$, and to $\mu \llcorner E$.

$$\limsup_{r \downarrow 0} \frac{1}{\mu(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| d\mu(y) \leq 2 \inf_{q \in \mathbb{Q}} |f(x) - q| = 0.$$

□

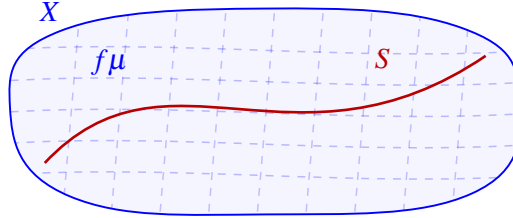


FIGURE 31. The measure splits into diffuse and μ -singular parts.

5. HAUSDORFF MEASURES, LIPSCHITZ MAPS, AND RECTIFIABILITY

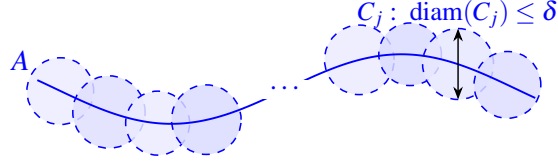
We use $\omega_s := \pi^{s/2}/\Gamma(1+s/2)$, Euclidean diameter diam , Lebesgue measure $\mathcal{L}^n$, orthogonal projection P_V , blow-up maps $\lambda_{x,r}(y) := (y-x)/r$, and the density conventions of §2 [15, 25, 28, 14].

5.1. Hausdorff measure, dimension, and densities.

Definition 5.1.1 (Hausdorff content and measure). For $s \geq 0$, $A \subset \mathbb{R}^n$, and $0 < \delta \leq +\infty$, give each nonempty covering set cost 1 when $s = 0$, give the empty set cost 0, and allow finite covers to be padded by empty sets. Define

$$\mathcal{H}_\delta^s(A) := \inf \left\{ \sum_{j=1}^{\infty} \omega_s \left(\frac{\text{diam}(C_j)}{2} \right)^s : A \subset \bigcup_{j=1}^{\infty} C_j, \text{diam}(C_j) \leq \delta \right\},$$

$$\mathcal{H}^s(A) := \lim_{\delta \downarrow 0} \mathcal{H}_\delta^s(A) = \sup_{\delta > 0} \mathcal{H}_\delta^s(A).$$


 FIGURE 32. Hausdorff content minimizes the s -cost of δ -covers.

Lemma 5.1.2 (Metric outer-measure structure). *For every $s \geq 0$, each $\mathcal{H}_\delta^s$ is an outer measure, $\mathcal{H}_{\delta_1}^s \geq \mathcal{H}_{\delta_2}^s$ for $\delta_1 \leq \delta_2$, and $\mathcal{H}^s$ is a metric outer measure.*

Proof. The empty-set and monotonicity axioms, together with $\mathcal{H}_{\delta_1}^s \geq \mathcal{H}_{\delta_2}^s$ for $\delta_1 \leq \delta_2$, follow from inclusion of admissible covers. Concatenating $\varepsilon 2^{-k}$ -optimal covers gives

$$A \subset \bigcup_{k=1}^{\infty} A_k \xrightarrow{\text{concatenate covers}} \mathcal{H}_\delta^s(A) \leq \sum_{k=1}^{\infty} \mathcal{H}_\delta^s(A_k),$$

$$0 < \delta < \text{dist}(A, B) \xrightarrow{32} \mathcal{H}_\delta^s(A \cup B) \geq \mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B) \implies \mathcal{H}^s(A \cup B) = \mathcal{H}^s(A) + \mathcal{H}^s(B).$$

□

Theorem 5.1.3 (Hausdorff measure is Borel regular). *For every $s \geq 0$, $\mathcal{H}^s$ is a Borel regular outer measure on $\mathbb{R}^n$.*

Proof. The metric outer-measure theorem (4.1.4) gives Borel measurability. If $\mathcal{H}^s(A) < +\infty$, choose 2^{-q} -covers $\{C_{qj}\}_j$ within 2^{-q} of the infimum and put

$$E := \bigcap_{q=1}^{\infty} \bigcup_{j=1}^{\infty} \overline{C_{qj}} \supset A, \quad \text{diam}(\overline{C_{qj}}) = \text{diam}(C_{qj}) \implies \mathcal{H}^s(E) \leq \mathcal{H}^s(A).$$

Outer monotonicity gives equality; for infinite measure take $E = \mathbb{R}^n$.

□

Proposition 5.1.4 (Basic properties). *For $A \subset \mathbb{R}^n$, $s \geq 0$, $\lambda > 0$, and every isometry O of $\mathbb{R}^n$,*

$$\mathcal{H}^0(A) = \#A, \quad \mathcal{H}^1 = \mathcal{L}^1 \text{ on } \mathbb{R}, \quad \mathcal{H}^n = \mathcal{L}^n \text{ on } \mathbb{R}^n,$$

$$\mathcal{H}^s(\lambda A) = \lambda^s \mathcal{H}^s(A), \quad \mathcal{H}^s(O(A)) = \mathcal{H}^s(A),$$

and $\mathcal{H}^s \equiv 0$ on $\mathbb{R}^n$ for $s > n$. Here $\#A = +\infty$ when A is infinite.

Proof.

$$\text{diam}(\lambda C) = \lambda \text{diam}(C), \quad \text{diam}(O(C)) = \text{diam}(C) \implies \mathcal{H}^s(\lambda A) = \lambda^s \mathcal{H}^s(A), \quad \mathcal{H}^s(O(A)) = \mathcal{H}^s(A),$$

$$s > n, \quad A \subset B_R(0), \quad 0 < \delta < 1 \implies \mathcal{H}_\delta^s(A) \leq C_{n,s,R} \delta^{s-n} \xrightarrow{\delta \downarrow 0} 0,$$

$$\mathbb{R}^n = \bigcup_{R=1}^{\infty} B_R(0) \implies \mathcal{H}^s \equiv 0 \quad (s > n).$$

The $s = 0$ convention gives $\mathcal{H}^0 = \#$; the normalized isodiametric theorem gives $\mathcal{H}^n = \mathcal{L}^n$, hence $\mathcal{H}^1 = \mathcal{L}^1$ on $\mathbb{R}$ [14, §2.2].

□

Definition 5.1.5 (Hausdorff dimension). For $0 \leq s < t$, every δ -cover gives

$$\mathcal{H}_\delta^t(A) \leq \frac{\omega_t}{\omega_s} \left(\frac{\delta}{2} \right)^{t-s} \mathcal{H}_\delta^s(A),$$

$$\mathcal{H}^s(A) < +\infty \implies \mathcal{H}^t(A) = 0, \quad \mathcal{H}^t(A) > 0 \implies \mathcal{H}^s(A) = +\infty,$$

and the Hausdorff dimension is

$$\dim_{\mathcal{H}}(A) := \inf\{s \geq 0 : \mathcal{H}^s(A) = 0\} = \sup\{s \geq 0 : \mathcal{H}^s(A) = +\infty\}$$

and is well defined, with $\sup \emptyset := 0$.

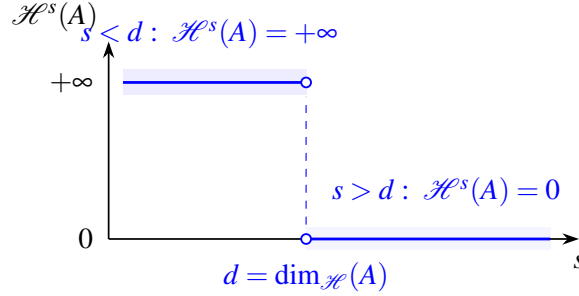


FIGURE 33. Hausdorff dimension is the critical exponent.

Example 5.1.6 (The four-corner Cantor set). Starting with $K_0 = [0, 1]^2$, replace every stage- j square by its four corner subsquares of one-quarter side length and let $K := \bigcap_{j=0}^{\infty} K_j$. Self-similarity and the open set condition give

$$0 < \mathcal{H}^1(K) < +\infty, \quad \dim_{\mathcal{H}}(K) = 1,$$

while K meets every rectifiable curve in $\mathcal{H}^1$ -measure zero [25, Chapter 18].

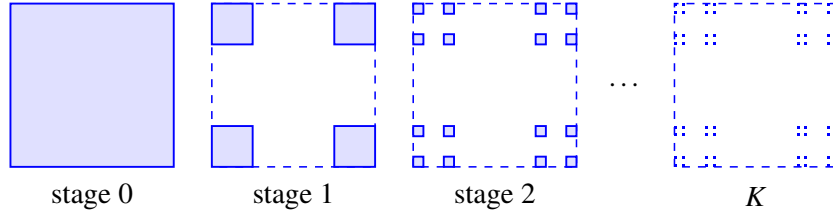


FIGURE 34. Each retained square is replaced by its four corner squares of one quarter the side, and the limit K is totally disconnected.

Definition 5.1.7 (Hausdorff densities). For a Radon measure μ on $\mathbb{R}^n$ and $s \geq 0$,

$$\Theta^{*s}(\mu, x) := \limsup_{r \downarrow 0} \frac{\mu(B_r(x))}{\omega_s r^s}, \quad \Theta_*^s(\mu, x) := \liminf_{r \downarrow 0} \frac{\mu(B_r(x))}{\omega_s r^s},$$

$\Theta^s(\mu, x) :=$ the common value when the two agree.

Proposition 5.1.8 (Density comparison with Hausdorff measure). Let μ be a Radon measure on $\mathbb{R}^n$. For every Borel $A \subset \mathbb{R}^n$, $0 \leq s \leq n$, and $t > 0$,

$$\Theta^{*s}(\mu, x) \geq t \quad (x \in A) \implies \mu(A) \geq t \mathcal{H}^s(A), \quad \Theta_*^s(\mu, x) \leq t \quad (x \in A) \implies \mu(A) \leq 2^s t \mathcal{H}^s(A).$$

Proof. The first implication is the upper-density estimate in [25, Theorem 6.9]. For the second fix $\varepsilon > 0$ and write

$$A_k := \{x \in A : \mu(B_r(x)) \leq (t + \varepsilon) \omega_s r^s \text{ for } 0 < r < k^{-1}\}.$$

Each A_k is Borel and $A_k \uparrow A$. If $\delta < k^{-1}$, $\{C_j\}$ is a δ -cover of A_k , and $x_j \in A_k \cap C_j$ whenever this intersection is nonempty, one has

$$C_j \cap A_k \subset B_{\text{diam}(C_j)}(x_j) \implies \mu^*(C_j \cap A_k) \leq 2^s(t + \varepsilon)\omega_s \left(\frac{\text{diam}(C_j)}{2} \right)^s,$$

$$\mu(A_k) \leq 2^s(t + \varepsilon)\mathcal{H}_\delta^s(A_k) \xrightarrow{\delta \downarrow 0, k \uparrow \infty, \varepsilon \downarrow 0} \mu(A) \leq 2^s t \mathcal{H}^s(A).$$

□

5.2. Lipschitz maps, Rademacher's theorem, and Whitney extension.

Definition 5.2.1 (Lipschitz maps and $W^{1,\infty}$). For $U \subset \mathbb{R}^n$ and $\mathbf{u} : U \rightarrow \mathbb{R}^m$,

$$\text{Lip}(\mathbf{u}; U) := \sup_{\substack{x, y \in U \\ x \neq y}} \frac{|\mathbf{u}(x) - \mathbf{u}(y)|}{|x - y|},$$

and $\mathbf{u}$ is (locally) Lipschitz when this quantity is finite (near every point). If U is open, a function $\mathbf{u} \in L^1_{\text{loc}}(U; \mathbb{R}^m)$ has weak derivative $D_i \mathbf{u} \in L^1_{\text{loc}}(U; \mathbb{R}^m)$ when, for every $\varphi \in C_c^\infty(U)$,

$$\int_U \mathbf{u} D_i \varphi d\mathcal{L}^n = - \int_U D_i \mathbf{u} \varphi d\mathcal{L}^n,$$

and $W^{1,\infty}_{\text{loc}}(U; \mathbb{R}^m)$ consists of the locally bounded $\mathbf{u}$ with locally bounded weak derivatives.

Proposition 5.2.2 (Lipschitz–Sobolev equivalence). *For open $U \subset \mathbb{R}^n$, after choosing continuous representatives,*

$$\text{Lip}_{\text{loc}}(U; \mathbb{R}^m) = W^{1,\infty}_{\text{loc}}(U; \mathbb{R}^m), \quad \|D\mathbf{u}\|_{L^\infty(U)} \leq \text{Lip}(\mathbf{u}; U),$$

with the reverse bound $\text{Lip}(\mathbf{u}; U) \leq \|D\mathbf{u}\|_{L^\infty(U)}$ on convex U , where matrix norms are operator norms.

Proof. For every rational unit vector $\mathbf{v}$, the restrictions of a Lipschitz map to almost every line parallel to $\mathbf{v}$ are one-dimensional Lipschitz maps. Fubini and one-dimensional integration by parts construct the weak directional derivatives, and density gives

$$|D\mathbf{u}| \leq \text{Lip}(\mathbf{u}; U) \quad \mathcal{L}^n\text{-a.e.}$$

Conversely, fix $B_{2r}(x_0) \Subset U$. For $x, y \in B_r(x_0)$, mollification gives

$$|(\mathbf{u} * \rho_\varepsilon)(x) - (\mathbf{u} * \rho_\varepsilon)(y)| \leq \|D\mathbf{u}\|_{L^\infty(B_{2r}(x_0))} |x - y| \implies |\mathbf{u}(x) - \mathbf{u}(y)| \leq \|D\mathbf{u}\|_{L^\infty(B_{2r}(x_0))} |x - y|.$$

If U is convex, the same argument along every segment gives the global reverse bound. [13, §5.8]. □

Proposition 5.2.3 (Difference quotients). *Let $U \subset \mathbb{R}^n$ be open, $V \Subset U$, $1 \leq i \leq n$, $1 \leq p \leq +\infty$, $\mathbf{u} \in W^{1,p}(U; \mathbb{R}^m)$, and $0 < |h| < \text{dist}(V, \partial U)$. Define $\delta_h^i \mathbf{u}(x) := h^{-1}(\mathbf{u}(x + h\mathbf{e}_i) - \mathbf{u}(x))$.*

$$\|\delta_h^i \mathbf{u}\|_{L^p(V)} \leq \|D_i \mathbf{u}\|_{L^p(U)}.$$

For $p < +\infty$,

$$\delta_h^i \mathbf{u} \xrightarrow{h \rightarrow 0} D_i \mathbf{u} \quad \text{in } L^p(V; \mathbb{R}^m).$$

Proof. For the ACL representative, the fundamental theorem holds for almost every x and gives

$$\delta_h^i \mathbf{u}(x) = \int_0^1 D_i \mathbf{u}(x + t h \mathbf{e}_i) dt \xrightarrow{\text{Jensen}} \|\delta_h^i \mathbf{u}\|_{L^p(V)} \leq \|D_i \mathbf{u}\|_{L^p(U)},$$

$$\|\delta_h^i \mathbf{u} - D_i \mathbf{u}\|_{L^p(V)} \leq \int_0^1 \|D_i \mathbf{u}(\cdot + t h \mathbf{e}_i) - D_i \mathbf{u}\|_{L^p(V)} dt \xrightarrow{h \rightarrow 0} 0.$$

□

Theorem 5.2.4 (Rademacher). *Every locally Lipschitz map $\mathbf{u} : U \rightarrow \mathbb{R}^m$ on an open $U \subset \mathbb{R}^n$ is Fréchet differentiable at $\mathcal{L}^n$ -almost every $x \in U$, meaning*

$$\lim_{h \rightarrow 0} \frac{|\mathbf{u}(x+h) - \mathbf{u}(x) - D\mathbf{u}(x) \cdot h|}{|h|} = 0,$$

with $D\mathbf{u}$ the weak derivative of (5.2.2); in particular every directional derivative at such x equals $D\mathbf{u}(x) \cdot \mathbf{v}$.

Proof. Fix a Lebesgue point x of $D\mathbf{u}$ (4.4.4) and a ball $B_{2r_0}(x) \Subset U$ on which $\mathbf{u}$ is Lipschitz. For $0 < r < r_0$, write $\mathbf{v}_{x,r}(y) := r^{-1}(\mathbf{u}(x+ry) - \mathbf{u}(x)) - D\mathbf{u}(x) \cdot y$ on $B_1(0)$.

$$\int_{B_1(0)} |D\mathbf{v}_{x,r}| d\mathcal{L}^n = r^{-n} \int_{B_r(x)} |D\mathbf{u} - D\mathbf{u}(x)| d\mathcal{L}^n \longrightarrow 0.$$

Poincaré supplies $\mathbf{c}_r \in \mathbb{R}^m$ with

$$\|\mathbf{v}_{x,r} - \mathbf{c}_r\|_{L^1(B_1(0))} \longrightarrow 0.$$

Since $\mathbf{v}_{x,r}(0) = 0$ and the family is uniformly Lipschitz, $\mathbf{c}_r \rightarrow 0$, whence

$$\sup_{B_{2/3}(0)} |\mathbf{v}_{x,r}| \longrightarrow 0 \quad (35).$$

For $0 < |h| < r_0/2$, let $r = 2|h|$ and $y = h/r$. Thus

$$\frac{|\mathbf{u}(x+h) - \mathbf{u}(x) - D\mathbf{u}(x) \cdot h|}{|h|} = 2|\mathbf{v}_{x,2|h|}(y)| \longrightarrow 0.$$

□

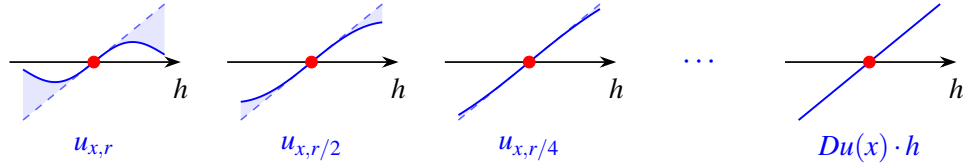


FIGURE 35. Successive blow-ups converge to the derivative $D\mathbf{u}(x)$.

Definition 5.2.5 (Whitney C^1 -jet). Let $K \subset \mathbb{R}^n$ be compact. A pair $\mathbf{f} : K \rightarrow \mathbb{R}^m$, $\mathbf{A} : K \rightarrow \mathbb{R}^{m \times n}$ with $\mathbf{A}$ continuous is a Whitney C^1 -jet when

$$\omega(r) := \sup_{\substack{x,y \in K \\ 0 < |x-y| \leq r}} \frac{|\mathbf{f}(y) - \mathbf{f}(x) - \mathbf{A}(x) \cdot (y-x)|}{|y-x|} \longrightarrow 0 \quad (r \downarrow 0).$$

Here $\sup \emptyset := 0$.

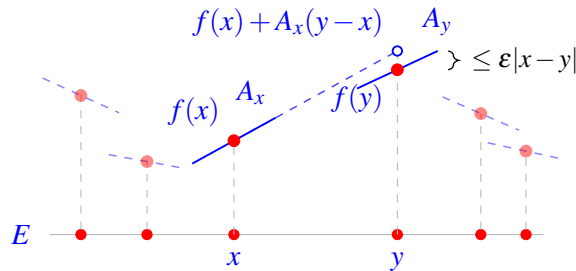


FIGURE 36. Whitney jets agree to first order along the closed set.

Theorem 5.2.6 (Whitney extension theorem). *Every Whitney C^1 -jet $(\mathbf{f}, \mathbf{A})$ on a compact $K \subset \mathbb{R}^n$ admits $\tilde{\mathbf{f}} \in C^1(\mathbb{R}^n; \mathbb{R}^m)$ with*

$$\tilde{\mathbf{f}}|_K = \mathbf{f}, \quad D\tilde{\mathbf{f}}|_K = \mathbf{A}.$$

Proof. [14, §6.5], [15, §3.1.14], [22, §5.1]; see (36). $\square$

Corollary 5.2.7 (Lusin approximation of Lipschitz maps). *Let $U \subset \mathbb{R}^n$ be bounded open, $\mathbf{u} : U \rightarrow \mathbb{R}^m$ Lipschitz, and $\varepsilon > 0$. There exist a compact $K \subset U$ and $\tilde{\mathbf{u}} \in C^1(\mathbb{R}^n; \mathbb{R}^m)$ with*

$$\mathcal{L}^n(U \setminus K) < \varepsilon, \quad \tilde{\mathbf{u}} = \mathbf{u} \text{ and } D\tilde{\mathbf{u}} = D\mathbf{u} \text{ on } K.$$

Proof. At every differentiability point x , define

$$\rho_r(x) := \sup_{\substack{h \in \mathbb{Q}^n, 0 < |h| \leq r \\ x+h \in U}} \frac{|\mathbf{u}(x+h) - \mathbf{u}(x) - D\mathbf{u}(x) \cdot h|}{|h|}.$$

Rademacher's theorem gives $\rho_r(x) \downarrow 0$ for $\mathcal{L}^n$ -almost every x . Egorov applied to $\rho_{1/j}$, Lusin applied to $D\mathbf{u}$, and inner regularity give a compact $K \Subset U$ with $\mathcal{L}^n(U \setminus K) < \varepsilon$ on which the convergence is uniform and $D\mathbf{u}$ is continuous. Hence

$$(\mathbf{u}|_K, D\mathbf{u}|_K) \text{ is a Whitney jet} \xrightarrow{5.2.6} \tilde{\mathbf{u}} \in C^1(\mathbb{R}^n; \mathbb{R}^m).$$

$\square$

5.3. The area formula and rectifiable sets.

Proposition 5.3.1 (Hausdorff measure under Lipschitz maps). *For $E \subset \mathbb{R}^n$, $\mathbf{f} : E \rightarrow \mathbb{R}^m$, and $L := \text{Lip}(\mathbf{f}; E) < +\infty$,*

$$\mathcal{H}^s(\mathbf{f}(A)) \leq L^s \mathcal{H}^s(A) \quad (A \subset E, s \geq 0).$$

Here $0^0 := 1$.

Proof. The case $L = 0$ is immediate. For $L > 0$, every δ -cover gives

$$\begin{aligned} A \subset \bigcup_i C_i &\implies \mathbf{f}(A) \subset \bigcup_i \mathbf{f}(C_i \cap E), \quad \text{diam}(\mathbf{f}(C_i \cap E)) \leq L \text{diam}(C_i), \\ \mathcal{H}_{L\delta}^s(\mathbf{f}(A)) &\leq L^s \mathcal{H}_\delta^s(A). \end{aligned}$$

Letting $\delta \downarrow 0$ proves the claim. $\square$

Definition 5.3.2 (Metric Jacobian). For linear $\mathbf{A} : \mathbb{R}^k \rightarrow \mathbb{R}^m$ with $k \leq m$, and at every differentiability point of $\mathbf{f} : \Omega \subset \mathbb{R}^k \rightarrow \mathbb{R}^m$,

$$J_k \mathbf{A} := \sqrt{\det(\mathbf{A}^\top \cdot \mathbf{A})}, \quad J_k \mathbf{f}(x) := J_k(D\mathbf{f}(x)).$$

For Lipschitz $\mathbf{f}$ on open Ω , put $J_k \mathbf{f} = 0$ on the $\mathcal{L}^k$ -null nondifferentiability set.

Lemma 5.3.3 (Low-rank images are null). *For open $\Omega \subset \mathbb{R}^k$, $\mathbf{f} \in C^1(\Omega; \mathbb{R}^m)$, and $k \leq m$,*

$$\mathcal{H}^k(\{x \in \Omega : \text{rank}(D\mathbf{f}(x)) < k\}) = 0.$$

Proof. Fix $0 \leq r < k$, a compact $K \Subset \{\text{rank}(D\mathbf{f}) = r\}$, and $0 < \eta < 1$. Uniform continuity of $D\mathbf{f}$ gives arbitrarily small balls $B_{\rho_j}(x)$ on whose fivefold dilates $\|D\mathbf{f} - D\mathbf{f}(x)\| \leq \eta$. The five-times lemma selects disjoint $B_{\rho_j}(x_j)$ with $K \subset \bigcup_j B_{5\rho_j}(x_j)$ and $\sum_j \rho_j^k \leq C(K)$. Rank- r affine approximation gives

$$\begin{aligned} \mathbf{f}(B_{5\rho_j}(x_j)) &\subset \bigcup_{\ell=1}^{C\eta^{-r}} B_{C\eta\rho_j}(y_{j,\ell}) \quad (37), \\ \mathcal{H}_{C\eta\delta}^k(\mathbf{f}(K)) &\leq C\eta^{k-r} \sum_j \rho_j^k \leq C(K)\eta^{k-r} \quad (\rho_j < \delta). \end{aligned}$$

$$\delta \downarrow 0, \quad \eta \downarrow 0 \implies \mathcal{H}^k(\mathbf{f}(K)) = 0.$$

Exhausting the rank strata by compact sets completes the proof. $\square$

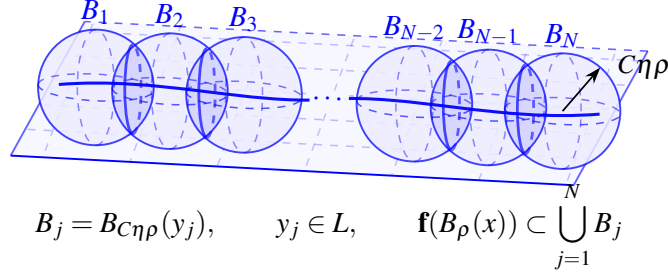


FIGURE 37. Near a rank- r point, $\mathbf{f}(B_\rho(x))$ is covered by $C\eta\rho$ -balls centered along an affine r -plane.

Theorem 5.3.4 (Area formula for an injective parametrization). *For open $\Omega \subset \mathbb{R}^k$, $k \leq m$, every injective $\mathbf{f} \in C^1(\Omega; \mathbb{R}^m)$, and every Borel $A \subset \Omega$,*

$$\mathcal{H}^k(\mathbf{f}(A)) = \int_A J_k \mathbf{f}(x) d\mathcal{L}^k(x) = \int_A \sqrt{\det(D\mathbf{f}(x)^\top \cdot D\mathbf{f}(x))} d\mathcal{L}^k(x).$$

Proof.

$$Z := \{\text{rank}(D\mathbf{f}) < k\} \implies J_k \mathbf{f}|_Z = 0, \quad \mathcal{H}^k(\mathbf{f}(Z)) = 0 \quad (5.3.3),$$

For $x \in \Omega \setminus Z$, choose an orthogonal map $\mathbf{Q}$ so that $\mathbf{p} := P_{\mathbb{R}^k} \circ \mathbf{Q} \circ \mathbf{f}$ has invertible derivative at x . On a neighborhood U_x , the inverse function theorem gives $\mathbf{Q} \circ \mathbf{f} \circ \mathbf{p}^{-1} = \mathbf{g}$, where $\mathbf{g}(z) = (z, \mathbf{h}(z))$, and

$$\mathbf{g}_\# \left(J_k \mathbf{g} \mathcal{L}^k \llcorner \mathbf{p}(U_x) \right) = \mathcal{H}^k \llcorner \mathbf{g}(\mathbf{p}(U_x)) \xrightarrow{\text{change of variables}} \mathcal{H}^k(\mathbf{f}(B)) = \int_B J_k \mathbf{f} d\mathcal{L}^k \quad (B \subset U_x \text{ Borel});$$

[15, §3.2.3]. Disjointizing a countable regular-chart cover of $\Omega \setminus Z$ and using injectivity gives the formula. $\square$

Definition 5.3.5 (Countably rectifiable set). An $\mathcal{H}^k$ -measurable set $E \subset \mathbb{R}^m$ is countably k -rectifiable when there exist Borel $A_i \subset \mathbb{R}^k$ and Lipschitz $\mathbf{f}_i : A_i \rightarrow \mathbb{R}^m$ with

$$\mathcal{H}^k \left(E \setminus \bigcup_{i=1}^{\infty} \mathbf{f}_i(A_i) \right) = 0,$$

and k -rectifiable when moreover $\mathcal{H}^k(E) < +\infty$.

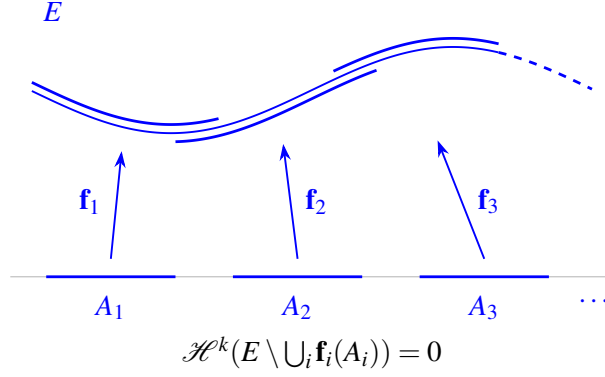


FIGURE 38. Each chart $\mathbf{f}_i$ carries a piece of $\mathbb{R}^k$ onto part of E , the patches overlap, and only the dashed remainder is missed.

Proposition 5.3.6 (Equivalent parametrizations). *For an $\mathcal{H}^k$ -measurable $E \subset \mathbb{R}^m$, the following are equivalent. The set E is countably k -rectifiable; $\mathcal{H}^k$ -almost all of E is covered by countably many Lipschitz graphs; $\mathcal{H}^k$ -almost all of E is covered by countably many embedded C^1 k -submanifolds.*

Proof. Extend each Lipschitz chart by Kirszbraun and exhaust its domain by bounded pieces. Lusin approximation and (5.3.1) replace it, up to a null image, by countably many C^1 patches. On each patch,

$$\mathcal{H}^k(\mathbf{f}_i(\{\text{rank}(D\mathbf{f}_i) < k\})) = 0 \quad (5.3.3).$$

At every full-rank point, the inverse function theorem makes the image locally a C^1 graph. Embedded C^1 manifolds have countable graph atlases, and every Lipschitz graph is a Lipschitz image. $\square$

Theorem 5.3.7 (Area formula with multiplicity). *Let $\Omega \subset \mathbb{R}^k$ be open, $k \leq m$, and let $\mathbf{f} : \Omega \rightarrow \mathbb{R}^m$ be Lipschitz. For Borel $A \subset \Omega$ and $N(\mathbf{f}, A, y) := \#(A \cap \mathbf{f}^{-1}(\{y\}))$,*

$$\int_A J_k \mathbf{f}(x) d\mathcal{L}^k(x) = \int_{\mathbb{R}^m} N(\mathbf{f}, A, y) d\mathcal{H}^k(y);$$

in particular $\mathcal{H}^k(\mathbf{f}(A)) = \int_A J_k \mathbf{f} d\mathcal{L}^k$ when $\mathbf{f}$ is injective on A .

Proof. Rademacher, Lusin approximation, low-rank nullity, and Borel disjointization give Borel $A_i, Z \subset A$ with

$$A = \bigsqcup_i A_i \sqcup Z, \quad \mathbf{f}|_{A_i} \text{ the restriction of a } C^1 \text{ embedding,} \quad \int_Z J_k \mathbf{f} d\mathcal{L}^k = 0, \quad \mathcal{H}^k(\mathbf{f}(Z)) = 0.$$

$$\sum_i \mathbf{1}_{\mathbf{f}(A_i)} = N(\mathbf{f}, A, \cdot) \quad \mathcal{H}^k\text{-a.e.} \xrightarrow{5.3.4} \int_A J_k \mathbf{f} d\mathcal{L}^k = \sum_i \mathcal{H}^k(\mathbf{f}(A_i)) = \int_{\mathbb{R}^m} N(\mathbf{f}, A, y) d\mathcal{H}^k(y) \quad (39).$$

$\square$

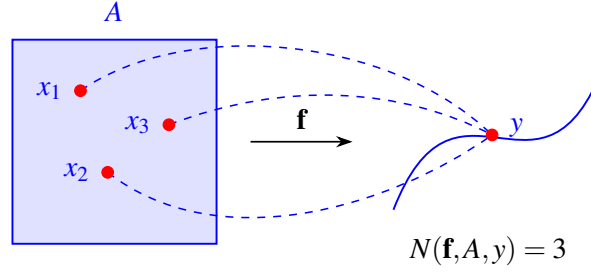


FIGURE 39. Three preimages make the image count y three times, which is what $N(\mathbf{f}, A, y)$ records.

5.4. Approximate tangent planes and the cone criterion.

Definition 5.4.1 (Blow-ups and approximate tangent planes). For a Radon measure μ on $\mathbb{R}^m$, $0 \leq k \leq m$, $x \in \mathbb{R}^m$, and $r > 0$, define

$$\lambda_{x,r}(y) := \frac{y-x}{r}, \quad \mu_{x,r} := r^{-k}(\lambda_{x,r})_{\#}\mu, \quad \int \varphi d\mu_{x,r} = r^{-k} \int \varphi\left(\frac{y-x}{r}\right) d\mu(y) \quad (\varphi \in C_c(\mathbb{R}^m)).$$

For locally $\mathcal{H}^k$ -finite, $\mathcal{H}^k$ -measurable $E \subset \mathbb{R}^m$, a k -plane P is the approximate tangent plane $T_x E$ when

$$r^{-k}(\lambda_{x,r})_{\#}(\mathcal{H}^k \llcorner E) \rightarrow \mathcal{H}^k \llcorner P \quad (r \downarrow 0).$$

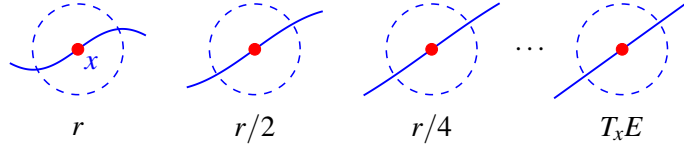


FIGURE 40. Magnifying E about x drives the rescaled measures to flat Hausdorff measure on the tangent plane $T_x E$.

Lemma 5.4.2 (Blow-up of a C^1 manifold). For $1 \leq k \leq m$, every embedded C^1 k -manifold $M \subset \mathbb{R}^m$, and $x \in M$,

$$r^{-k}(\lambda_{x,r})_{\#}(\mathcal{H}^k \llcorner M) \rightarrow \mathcal{H}^k \llcorner T_x M.$$

Proof. After a rigid motion, write $x = 0$, $T_x M = \mathbb{R}^k \times \{0\}$, and $M \cap U = \text{graph}(\mathbf{g})$ near 0, with $\mathbf{g}(0) = 0$ and $D\mathbf{g}(0) = 0$.

$$\mathbf{g}_r(z) := r^{-1}\mathbf{g}(rz) \implies D\mathbf{g}_r = D\mathbf{g}(r \cdot) \longrightarrow 0 \quad \text{locally uniformly} \xrightarrow{5.3.4} r^{-k}(\lambda_{x,r})_{\#}(\mathcal{H}^k \llcorner M) \rightarrow \mathcal{H}^k \llcorner T_x M.$$

□

Theorem 5.4.3 (Tangents of rectifiable sets). Let $1 \leq k \leq m$ and let $E \subset \mathbb{R}^m$ be $\mathcal{H}^k$ -measurable, countably k -rectifiable, and locally $\mathcal{H}^k$ -finite. For $\mathcal{H}^k$ -almost every $x \in E$ the approximate tangent plane $T_x E$ exists, is unique, and

$$\lim_{r \downarrow 0} \frac{\mathcal{H}^k(E \cap B_r(x))}{\omega_k r^k} = 1.$$

Proof. Write $\mu := \mathcal{H}^k \llcorner E$ and choose a Borel partition $E = \bigsqcup_i E_i$ up to a null set with $E_i \subset M_i$, where each M_i is an embedded C^1 k -manifold (5.3.6). Differentiation on E and M_i gives, for $\mathcal{H}^k$ -almost every $x \in E_i$ and every $R > 0$,

$$\frac{\mathcal{H}^k((E \triangle M_i) \cap B_{Rr}(x))}{r^k} \rightarrow 0 \xrightarrow{5.4.2} \mu_{x,r} \rightarrow \mathcal{H}^k \llcorner T_x M_i \quad (40),$$

$$\mathcal{H}^k \llcorner P = \mathcal{H}^k \llcorner Q \implies P = Q, \quad \mu_{x,r}(B_1(0)) \rightarrow \omega_k \implies \lim_{r \downarrow 0} \frac{\mathcal{H}^k(E \cap B_r(x))}{\omega_k r^k} = 1.$$

□

Definition 5.4.4 (Cones). For a k -plane $V \subset \mathbb{R}^m$ and $0 < \alpha < \pi/2$,

$$C(x, V, \alpha) := x + C(0, V, \alpha), \quad C(0, V, \alpha) := \{y : |P_{V^\perp} y| \leq (\tan \alpha) |P_V y|\},$$

the set of directions making angle at most α with V .

Theorem 5.4.5 (Cone criterion for rectifiability). *Let $1 \leq k \leq m$ and let μ be a Radon measure on $\mathbb{R}^m$ such that, for μ -almost every x ,*

$$0 < \Theta_*^k(\mu, x) \leq \Theta^{*k}(\mu, x) < +\infty, \quad \lim_{r \downarrow 0} \frac{\mu(B_r(x) \setminus C(x, V_x, \alpha_x))}{r^k} = 0$$

for some k -plane V_x and $0 < \alpha_x < \pi/2$. There are a countably k -rectifiable R and a positive locally integrable θ such that

$$\mu = \theta \mathcal{H}^k \llcorner R.$$

Proof. [25, Chapter 15], [8, Chapter 2], [15, §3.2.29]; see (41). □

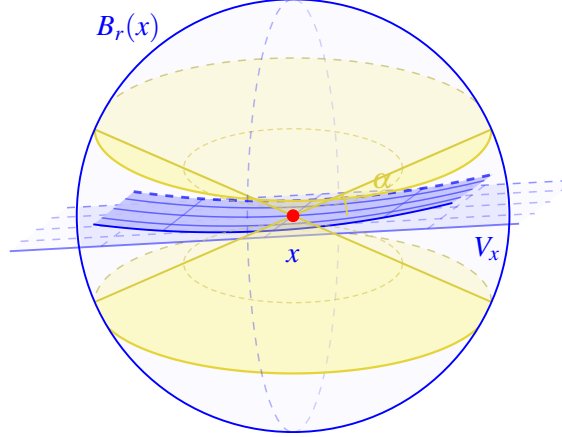


FIGURE 41. The support concentrates in the yellow cone about V_x .

Corollary 5.4.6 (Flat blow-ups imply rectifiability). *Let $1 \leq k \leq m$ and let μ be a Radon measure on $\mathbb{R}^m$. If, for μ -almost every x , there are $c(x) > 0$ and a k -plane P_x with*

$$r^{-k}(\lambda_{x,r})_\# \mu \rightarrow c(x) \mathcal{H}^k \llcorner P_x \quad (r \downarrow 0),$$

there are a countably k -rectifiable R and a positive locally integrable θ such that

$$\mu = \theta \mathcal{H}^k \llcorner R.$$

Proof. Fix $0 < \alpha < \pi/2$. Continuity-set convergence gives, for μ -almost every x ,

$$\Theta_*^k(\mu, x) = \Theta^{*k}(\mu, x) = c(x) > 0,$$

$$\lim_{r \downarrow 0} \frac{\mu(B_r(x) \setminus C(x, P_x, \alpha))}{r^k} = 0 \xrightarrow{5.4.5} \mu = \theta \mathcal{H}^k \llcorner R.$$

□

6. VARIFOLDS, FIRST VARIATION, AND ALLARD REGULARITY

Throughout this section, $U \subset \mathbb{R}^n$ is open, $1 \leq k < n$, $G(k, n)$ is the Grassmannian of unoriented k -planes, P_S is orthogonal projection onto $S \in G(k, n)$, and $\mathbf{X} \in C_c^1(U; \mathbb{R}^n)$. Write $\lambda_{x,r}(y) := (y - x)/r$ and

$$\operatorname{div}_S \mathbf{X}(x) := \sum_{j=1}^n \langle P_S(\nabla X^j(x)), \mathbf{e}_j \rangle = \operatorname{tr}(P_S \cdot D\mathbf{X}(x)), \quad \operatorname{div}_M := \operatorname{div}_{T_x M}.$$

Varifolds retain mass and tangent planes, but neither orientation nor a boundary operator [28, 9].

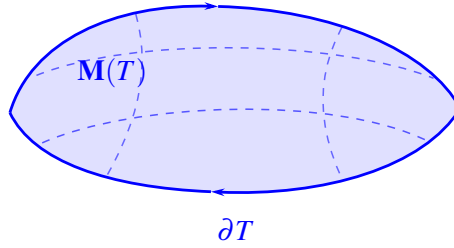


FIGURE 42. The sheet sweeps the mass $\mathbf{M}(T)$, while its oriented rim represents the boundary ∂T .

6.1. Varifolds and the first variation.

Definition 6.1.1 (Varifold, weight, rectifiable and integral varifolds). A k -varifold is a Radon measure V on $U \times G(k, n)$. For $\pi(x, S) := x$, its weight is

$$\|V\| := \pi_{\#} V, \quad \|V\|(A) = V(A \times G(k, n)) \quad (A \subset U \text{ Borel}).$$

For an $\mathcal{H}^k$ -measurable, countably k -rectifiable $M \subset U$ and $0 < \theta \in L_{\text{loc}}^1(\mathcal{H}^k \llcorner M)$, define

$$\int_{U \times G(k, n)} \varphi(x, S) dV(x, S) = \int_M \varphi(x, T_x M) \theta(x) d\mathcal{H}^k(x) \quad (\varphi \in C_c(U \times G(k, n))).$$

$$V = \mathbf{v}(M, \theta), \quad \|V\| = \theta \mathcal{H}^k \llcorner M, \quad V \text{ integral} \iff \theta \in \mathbb{N} \quad \mathcal{H}^k\text{-a.e. on } M.$$

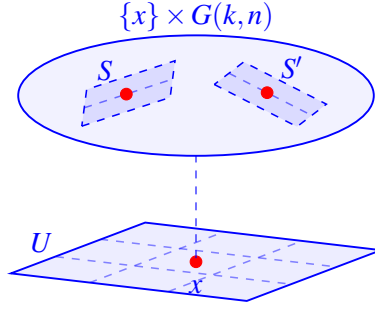


FIGURE 43. A varifold assigns unoriented planes to points, so one point may carry several and no boundary is defined.

Theorem 6.1.2 (Disintegration over the weight measure). *Every k -varifold admits a $\|V\|$ -measurable probability kernel $x \mapsto \mathbf{v}_x \in \mathcal{P}(G(k, n))$, unique $\|V\|$ -almost everywhere, such that*

$$\int \varphi(x, S) dV(x, S) = \int_U \left(\int_{G(k, n)} \varphi(x, S) d\mathbf{v}_x(S) \right) d\|V\|(x) \quad (\varphi \in C_c(U \times G(k, n))),$$

$$V = \mathbf{v}(M, \theta) \xLeftrightarrow{6.1.1} \|V\| = \theta \mathcal{H}^k \llcorner M, \quad \mathbf{v}_x = \delta_{T_x M} \quad \|V\| \text{-a.e.}$$

Proof.

$$\pi_{\#} V = \|V\|, \quad U, G(k, n) \text{ standard Borel,} \quad V \text{ } \sigma\text{-finite} \xRightarrow{[1, \S 2.5]} V(dx, dS) = \mathbf{v}_x(dS) \|V\|(dx).$$

$$V = \mathbf{v}(M, \theta) \xLeftrightarrow{6.1.1} V(dx, dS) = \delta_{T_x M}(dS) \theta(x) \mathcal{H}^k \llcorner M(dx) \iff \|V\| = \theta \mathcal{H}^k \llcorner M, \quad \mathbf{v}_x = \delta_{T_x M} \quad \|V\| \text{-a.e.} \quad \square$$

Definition 6.1.3 (Pushforward). For a C^1 -diffeomorphism $\phi : U \rightarrow U'$, set

$$J_S \phi(x) := \left| \bigwedge^k (D\phi(x)|_S) \right|, \quad \int_{U' \times G(k, n)} \varphi d(\phi_{\#} V) := \int_{U \times G(k, n)} \varphi(\phi(x), D\phi(x)[S]) J_S \phi(x) dV(x, S)$$

for every $\varphi \in C_c(U' \times G(k, n))$. In particular,

$$V = \mathbf{v}(M, \theta) \xRightarrow{5.3.7} \phi_{\#} V = \mathbf{v}(\phi(M), \theta \circ \phi^{-1}).$$

Definition 6.1.4 (First variation and stationarity). For $\mathbf{X} \in C_c^1(U; \mathbb{R}^n)$ and $\phi_t := \text{Id} + t\mathbf{X}$, define

$$\delta V(\mathbf{X}) := \left. \frac{d}{dt} \right|_{t=0} \int_{U \times G(k, n)} (J_S \phi_t(x) - 1) dV(x, S) = \int_{U \times G(k, n)} \text{div}_S \mathbf{X}(x) dV(x, S).$$

The varifold is stationary when $\delta V \equiv 0$, and has locally bounded first variation when

$$\|\delta V\|(K) := \sup \left\{ |\delta V(\mathbf{X})| : \mathbf{X} \in C_c^1(U; \mathbb{R}^n), \text{spt}(\mathbf{X}) \subset K, \|\mathbf{X}\|_{L^\infty} \leq 1 \right\} < +\infty \quad (K \Subset U).$$

Proposition 6.1.5 (Decomposition of first variation). *If V has locally bounded first variation, then there exist*

$$\mathbf{H} \in L_{\text{loc}}^1(\|V\|; \mathbb{R}^n), \quad \sigma_V \geq 0, \quad \sigma_V \perp \|V\|, \quad |\eta| = 1 \quad \sigma_V \text{-a.e.,}$$

unique up to the corresponding almost-everywhere equivalences, such that

$$\delta V = -\mathbf{H} \|V\| + \eta \sigma_V,$$

or equivalently

$$\delta V(\mathbf{X}) = - \int_U \mathbf{H} \cdot \mathbf{X} d\|V\| + \int_U \eta \cdot \mathbf{X} d\sigma_V.$$

Moreover,

$$\sigma_V = 0 \iff \delta V \ll \|V\|.$$

Proof.

$$\delta V \text{ locally bounded} \xrightarrow{4.2.5} \delta V \text{ is a vector Radon measure} \xrightarrow{4.2.4, 4.4.3} \delta V = -\mathbf{H}\|V\| + \eta \sigma_V.$$

Uniqueness of the absolutely continuous and singular parts gives the final equivalence. $\square$

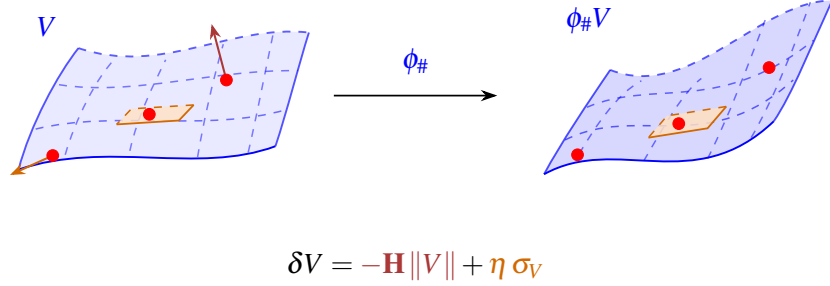


FIGURE 44. Pushforward transports tangent k -volume; first variation separates curvature from conormal flux.

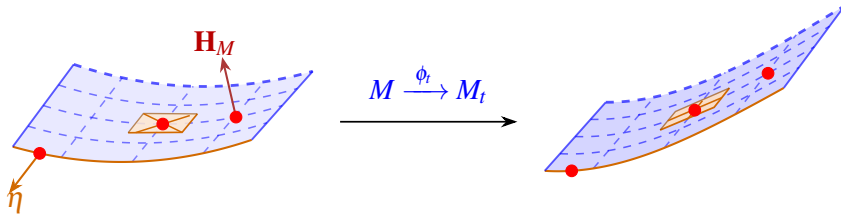
Proposition 6.1.6 (First variation of a smooth submanifold). *Let $M \subset U$ be a compact embedded C^2 k -submanifold with boundary, η its outward unit conormal, $V := \mathbf{v}(M, 1)$, and*

$$\mathbf{H}_M := \sum_{i=1}^k (D_{\tau_i} \tau_i)^\perp$$

for any local orthonormal frame $\{\tau_i\}_{i=1}^k$. For every $\mathbf{X} \in C_c^1(U; \mathbb{R}^n)$,

$$\delta V(\mathbf{X}) = \int_M \operatorname{div}_M \mathbf{X} d\mathcal{H}^k = - \int_M \mathbf{H}_M \cdot \mathbf{X} d\mathcal{H}^k + \int_{\partial M} \mathbf{X} \cdot \eta d\mathcal{H}^{k-1}.$$

Proof. Set $\phi_t := \operatorname{Id} + t\mathbf{X}$ and $M_t := \phi_t(M)$; the corresponding marked cells record the change of tangent k -volume, while $\mathbf{H}_M$ and η isolate the interior and boundary contributions.



$$J_t(x) := \det \left(\left[\langle (I + tD\mathbf{X})\tau_i, (I + tD\mathbf{X})\tau_j \rangle \right]_{i,j=1}^k \right)^{1/2}, \quad J_0(x) = \sum_{i=1}^k \langle D\mathbf{X}(x)\tau_i, \tau_i \rangle = \operatorname{div}_M \mathbf{X}(x).$$

$$\mathbf{X} = \mathbf{X}^\top + \mathbf{X}^\perp \implies \operatorname{div}_M \mathbf{X} = \operatorname{div}_M \mathbf{X}^\top - \mathbf{H}_M \cdot \mathbf{X}, \quad \int_M \operatorname{div}_M \mathbf{X}^\top d\mathcal{H}^k = \int_{\partial M} \mathbf{X} \cdot \eta d\mathcal{H}^{k-1}.$$

$$\delta V(\mathbf{X}) = \left. \frac{d}{dt} \right|_{t=0} \int_M J_t d\mathcal{H}^k = \int_M \operatorname{div}_M \mathbf{X} d\mathcal{H}^k = - \int_M \mathbf{H}_M \cdot \mathbf{X} d\mathcal{H}^k + \int_{\partial M} \mathbf{X} \cdot \eta d\mathcal{H}^{k-1}.$$

The area formula identifies the derivative with the tangential Jacobian, and Gauss–Green converts the tangential divergence into the conormal flux, proving the identity [28, §7, §9]. $\square$

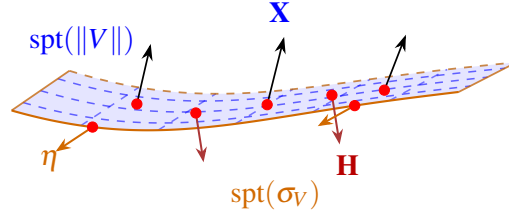


FIGURE 45. First variation separates interior mean curvature from the boundary conormal carried by $\text{spt}(\sigma_V)$.

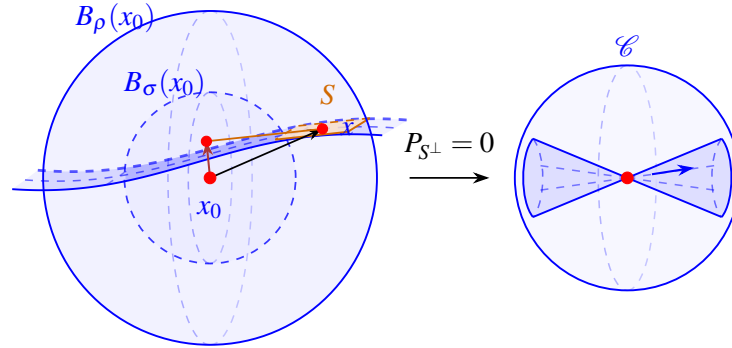
6.2. Monotonicity, tangent cones, and Allard regularity.

Theorem 6.2.1 (Monotonicity formula). *Let V be a stationary rectifiable k -varifold in U , $x_0 \in U$, and $0 < \sigma < \rho < \text{dist}(x_0, \partial U)$.*

$$\frac{\|V\|(B_\rho(x_0))}{\rho^k} - \frac{\|V\|(B_\sigma(x_0))}{\sigma^k} = \int_{(B_\rho(x_0) \setminus B_\sigma(x_0)) \times G(k,n)} \frac{|P_{S^\perp}(x - x_0)|^2}{|x - x_0|^{k+2}} dV(x, S).$$

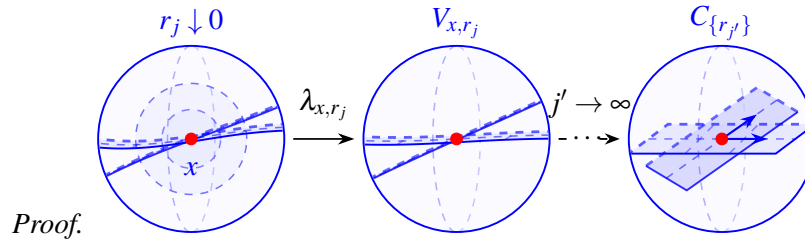
Hence $R_{x_0}(r) := r^{-k} \|V\|(B_r(x_0))$ is nondecreasing, $\Theta^k(\|V\|, x_0) = \omega_k^{-1} \lim_{r \downarrow 0} R_{x_0}(r)$ exists, $\Theta^k(\|V\|, \cdot)$ is upper semicontinuous, and R_{x_0} is constant on $(0, R)$ exactly when V is conical in $B_R(x_0)$, meaning $(D_{x_0, \lambda})_\#(V \llcorner B_r(x_0)) = V \llcorner B_{\lambda r}(x_0)$ for $D_{x_0, \lambda}(x) := x_0 + \lambda(x - x_0)$ and $0 < r, \lambda r < R$.

Proof. Consider the following geometric presentation:



The proof is induced. □

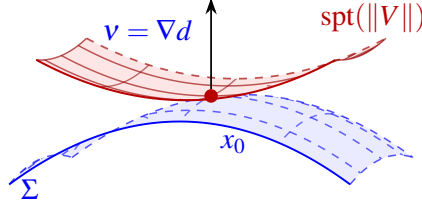
Theorem 6.2.2 (Tangent cones). *Let V be integral and stationary in U with $0 < \Theta^k(\|V\|, x) < +\infty$. Every $r_j \downarrow 0$ admits a subsequence with $V_{x, r_{j'}} := (\lambda_{x, r_{j'}})_\# V \rightarrow C$, where $C \neq 0$ is an integral stationary cone about 0 with $\Theta^k(\|C\|, 0) = \Theta^k(\|V\|, x)$, possibly depending on the subsequence.*



Proof. □

Theorem 6.2.3 (Maximum principle). *Let V be integral and stationary in U , let $\Omega \subset U$ have C^2 boundary Σ with $\text{spt}(\|V\|) \subset \overline{\Omega}$ locally along Σ , and let $\kappa_1 \leq \dots \leq \kappa_{n-1}$ be the eigenvalues of $-D^2d|_{T\Sigma}$ for the signed distance $d \geq 0$ in Ω . If $\kappa_1 + \dots + \kappa_k > 0$ on Σ (strict mean convexity when $k = n - 1$), then $\text{spt}(\|V\|) \cap \Sigma = \emptyset$.*

Proof. Assume a first contact x_0 , orient d into Ω , and choose $f := -d + Ad^2$ with $2A > \max\{0, \max_i \kappa_i(x_0)\}$, so $f \leq 0 = f(x_0)$ on the support near x_0 .



The positivity of the smallest k barrier curvatures gives $\text{tr}_S D^2 f(x_0) > 0$ for every $S \in G(k, n)$, contradicting the stationary-varifold maximum principle [12, Proposition A.1]. $\square$

Definition 6.2.4 (Excess and the Allard regime). For $x_0 \in U$, $0 < r < \text{dist}(x_0, \partial U)$, and $S_0 \in G(k, n)$,

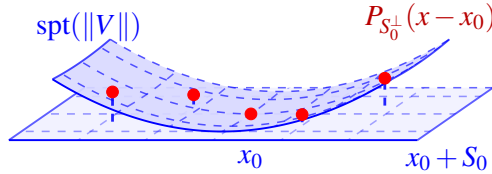
$$E_h(V, x_0, r, S_0) := r^{-k-2} \int_{B_r(x_0)} |P_{S_0^\perp}(x - x_0)|^2 d\|V\|(x), \quad E_t(V, x_0, r, S_0) := r^{-k} \int_{B_r(x_0) \times G(k, n)} \frac{|P_S - P_{S_0}|^2}{2} dV(x, S).$$

For $p > k$, $\alpha := 1 - k/p$, and $0 < \delta \leq \frac{1}{4}$, write $\mathcal{A}_{\delta, p}(V, x_0, r)$ when $V = \mathbf{v}(M, \theta)$ in $B_r(x_0) \Subset U$, $x_0 \in \text{spt}(\|V\|)$, $\theta \geq 1$ $\mathcal{H}^k$ -a.e., $\delta V = -\mathbf{H}\|V\|$, and

$$\|V\|(B_r(x_0)) \leq (1 + \delta)\omega_k r^k, \quad r^\alpha \|\mathbf{H}\|_{L^p(\|V\| \llcorner B_r(x_0))} \leq \delta;$$

for $y \in \text{spt}(\|V\|)$ and $B_s(y) \subset B_r(x_0)$, the scale-invariant combined excess is

$$\mathcal{E}_{*, \delta}(V, y, s, S) := \max \left\{ E_t(V, y, s, S), \delta^{-1} s^{2\alpha} \|\mathbf{H}\|_{L^p(\|V\| \llcorner B_s(y))}^2 \right\}.$$



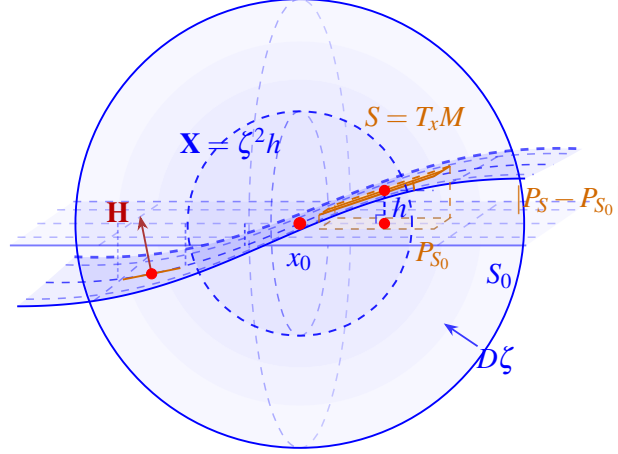
$$E_h = r^{-k-2} \int_{B_r(x_0)} |P_{S_0^\perp}(x - x_0)|^2 d\|V\|$$

FIGURE 46. Height excess averages the squared vertical drops from the support to the reference plane $x_0 + S_0$.

Lemma 6.2.5 (Caccioppoli height-to-tilt inequality). *For $B_r(x_0) \Subset U$, $S_0 \in G(k, n)$, and $\delta V = -\mathbf{H}\|V\|$ in $B_r(x_0)$ with $\mathbf{H} \in L^2(\|V\| \llcorner B_r(x_0))$,*

$$E_t(V, x_0, r/2, S_0) \leq C_{n,k} E_h(V, x_0, r, S_0) + C_{n,k} r^{2-k} \int_{B_r(x_0)} |\mathbf{H}|^2 d\|V\|.$$

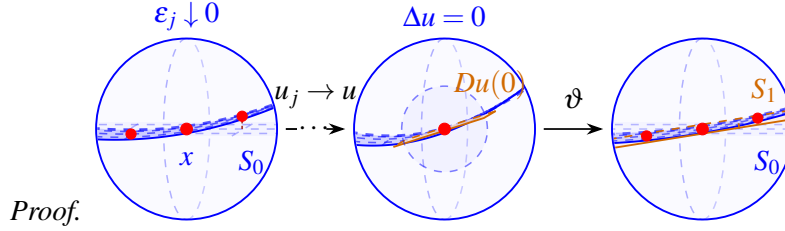
Proof. Take $\mathbf{X} := \zeta^2 h$, where $\zeta = 1$ on $B_{r/2}(x_0)$, $\zeta \searrow 0$ across the surrounding annulus, and the orange projection records the tilt from $S = T_x M$ to S_0 .



The first-variation identity and Young's inequality absorb the cutoff term, leaving the outer height and curvature energies to control the inner tilt. $\square$

Theorem 6.2.6 (One-step tilt-excess decay). *For $p > k$ and $\alpha := 1 - k/p$, there exist $\delta_0, \vartheta \in (0, \frac{1}{4}]$ depending on n, k, p such that, whenever $0 < \delta \leq \delta_0$, $\mathcal{A}_{\delta,p}(V, x_0, r)$, $x \in \text{spt}(\|V\|) \cap B_{\delta r/4}(x_0)$, $0 < \rho \leq \delta r/4$, and $S_0 \in G(k, n)$,*

$$\mathcal{E}_{*,\delta}(V, x, \rho, S_0) \leq \delta_0 \implies \exists S_1 \in G(k, n): \quad \mathcal{E}_{*,\delta}(V, x, \vartheta \rho, S_1) \leq \vartheta^{2\alpha} \mathcal{E}_{*,\delta}(V, x, \rho, S_0), \quad |P_{S_1} - P_{S_0}|^2 \leq C_{n,k,p} \vartheta^{-k} \mathcal{E}_{*,\delta}(V, x, \rho, S_0)$$



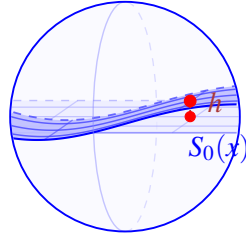
Proof.

$\square$

Lemma 6.2.7 (Initial plane). *For $p > k$, there exist $\epsilon_0, \rho_0, \gamma > 0$ such that $\mathcal{A}_{\epsilon_0,p}(V, x_0, r)$ forces every $x \in \text{spt}(\|V\|) \cap B_{\gamma r}(x_0)$ into a thin slab about some $S_0(x) \in G(k, n)$:*

$$\mathcal{E}_{*,\epsilon_0}(V, x, \rho_0 r, S_0(x)) \leq C_{n,k,p} \epsilon_0^{1/(2k)}, \quad 1 \leq \Theta^k(\|V\|, x) < 2.$$

Proof. Small mass excess leaves a nearly planar support.



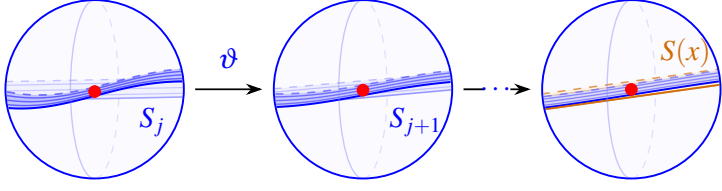
$$\mathcal{A}_{\epsilon_0,p} \xrightarrow{[28, \S 17 \text{ and Lemma 24.3}]} E_h \ll 1 \xrightarrow{6.2.5} E_0 \leq C \epsilon_0^{1/(2k)}$$

$\square$

Lemma 6.2.8 (Coherent tangent planes). *Under Lemma 6.2.7, with $\rho_j := \vartheta^j \rho_0 r$ and $E_j(x)$ the excess at scale ρ_j ,*

$$E_j(x) \leq C \vartheta^{2\alpha j} \varepsilon_0^{1/(2k)}, \quad S_j(x) \longrightarrow S(x), \quad |P_{S(x)} - P_{S(y)}| \leq C_{n,k,p} \varepsilon_0^{1/(4k)} \left(\frac{|x-y|}{r} \right)^\alpha.$$

Proof. Excess decay rotates the approximating planes by a summable amount.



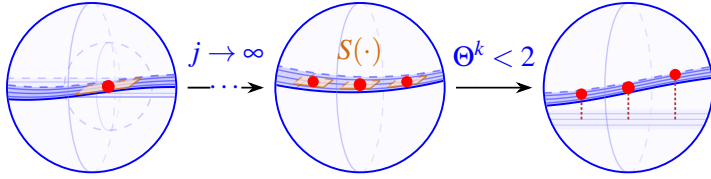
$$E_0 \ll 1 \xrightarrow{6.2.6} E_{j+1} \leq \vartheta^{2\alpha} E_j, \quad |P_{S_{j+1}} - P_{S_j}|^2 \leq C \vartheta^{-k} E_j \implies S_j \rightarrow S \in C^{0,\alpha}$$

□

Theorem 6.2.9 (Allard interior regularity). *For $p > k$ and $\alpha := 1 - k/p$, there exist $\varepsilon_0, \gamma > 0$ such that $\mathcal{A}_{\varepsilon_0,p}(V, x_0, r)$ turns the support into one $C^{1,\alpha}$ sheet: some $q \in O(n)$ and $u \in C^{1,\alpha}(B_{\gamma r}^k; \mathbb{R}^{n-k})$ with $u(0) = 0$, $Du(0) = 0$, and $M := x_0 + q(\text{graph}(u))$ satisfy*

$$V \llcorner B_{\gamma r}(x_0) = \mathbf{v}(M, 1) \llcorner B_{\gamma r}(x_0), \quad r^{-1} \|u\|_{L^\infty} + \|Du\|_{L^\infty} + r^\alpha [Du]_{C^{0,\alpha}} \leq C_{n,k,p} \varepsilon_0^{1/(4k)}.$$

Proof. The limiting planes assemble the support into a single multiplicity-one graph.



$$\mathcal{A}_{\varepsilon_0,p} \xrightarrow{6.2.7} E_0 \ll 1 \xrightarrow{6.2.8} S \in C^{0,\alpha} \xrightarrow{[28, \text{Theorem 24.2}]} V = \mathbf{v}(M, 1) \text{ in } B_{\gamma r}(x_0)$$

□

Part 3. Exercises

7. EXERCISES

Exercise 7.1 (1.1.4). For $v \in \Lambda_m(V)$, $w_i \in \Lambda^{n_i}(V)$, $v_i \in \Lambda_{m_i}(V)$, and $w \in \Lambda^q(V)$, with $n_1 + n_2 \leq m$ and $m_1 + m_2 \leq q$, show

$$(w_1 \wedge w_2) \lrcorner v = w_2 \lrcorner (w_1 \lrcorner v), \quad w \lrcorner (v_1 \wedge v_2) = (w \lrcorner v_1) \lrcorner v_2.$$

Exercise 7.2 (1.1.4). Set

$$\mathbf{E} := \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n, \quad \text{vol} := dx_1 \wedge \cdots \wedge dx_n, \quad * \omega := \omega \lrcorner \mathbf{E}, \quad * v := \text{vol} \lrcorner v,$$

$$\text{div } v := \sum_{\mathbf{i} \in J(k,n)} \sum_{j=1}^n \frac{\partial v_{\mathbf{i}}}{\partial x_j} dx_j \lrcorner \mathbf{e}_{\mathbf{i}}.$$

For $v \in C_c^\infty(\mathbb{R}^n, \Lambda_k(\mathbb{R}^n))$ and $\alpha \in C_c^\infty(\mathbb{R}^n, \Lambda^h(\mathbb{R}^n))$, with $0 \leq h \leq k-1$ and $1 \leq k \leq n$, show

$$\begin{aligned} ** &= (-1)^{k(n-k)} \text{id} \quad \text{on } \Lambda^k(\mathbb{R}^n) \text{ and } \Lambda_k(\mathbb{R}^n), \\ *(\alpha \lrcorner v) &= (-1)^{h(n-h)} (*v) \wedge \alpha, \quad *(d(*v)) = (-1)^{(n-k)(k-1)} \text{div } v, \\ \text{div}(\alpha \lrcorner v) &= (-1)^h (\alpha \lrcorner \text{div } v + d\alpha \lrcorner v). \end{aligned}$$

Exercise 7.3 (1.2.4). Show

$$\mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4 \in \Lambda_2(\mathbb{R}^4) \text{ is not simple,} \quad v \in \Lambda_{n-1}(\mathbb{R}^n) \implies v \text{ is simple.}$$

Use Exercise 7.2 for the second claim.

Exercise 7.4 (2.3.1). For $\varphi \in C_c^\infty(\mathbb{R}^n, \Lambda^k(\mathbb{R}^n))$, $\xi \in C_c^\infty(\mathbb{R}^n, \Lambda_p(\mathbb{R}^n))$, and $T \in \mathcal{D}_m(\mathbb{R}^n)$, with $1 \leq j \leq n$, show

$$\begin{aligned} T &= \sum_{\mathbf{i} \in J(m,n)} [T \lrcorner dx_{\mathbf{i}}] \wedge \mathbf{e}_{\mathbf{i}}, \quad D_{x_j}(T \lrcorner \varphi) = D_{x_j} T \lrcorner \varphi + T \lrcorner \frac{\partial \varphi}{\partial x_j}, \\ D_{x_j}(T \wedge \xi) &= D_{x_j} T \wedge \xi + T \wedge \frac{\partial \xi}{\partial x_j}, \quad \partial(\mathcal{L}^n \wedge \xi) = -\mathcal{L}^n \wedge \text{div } \xi, \\ m = n &\implies \partial T \wedge \mathbf{e}_j = -D_{x_j} T. \end{aligned}$$

Assume $0 \leq k \leq m$ for contractions and $m + p \leq n$ for $T \wedge \xi$; regard $\mathcal{L}^n$ as a 0-current.

Exercise 7.5 (5.2.4). For Lipschitz $f : \mathbb{R}^n \rightarrow \mathbb{R}$, show

$$\|\nabla(f * \rho_\varepsilon)\|_\infty \leq \text{Lip}(f),$$

and deduce Theorem 5.2.4 from weak-* compactness and Corollary 4.4.4.

Exercise 7.6 (3.4.3). After rescaling to a unit grid, construct the cubical radial projections and show

$$T = P + \partial Q + R, \quad \mathbf{M}(P) + \mathbf{M}(\partial P) + \mathbf{M}(Q) + \mathbf{M}(R) \leq C(\mathbf{M}(T) + \mathbf{M}(\partial T)).$$

Identify the grid-translation averaging step.

Exercise 7.7 (6.2.6). For a stationary varifold V in a cylinder over S_0 , mass-close to a multiplicity-one plane, show

$$\inf_{\substack{\ell: S_0 \rightarrow S_0^\perp \\ \ell \text{ affine}}} r^{-k-2} \int_{B_{r/2}(x_0)} \left| P_{S_0^\perp}(x - x_0) - \ell(P_{S_0}(x - x_0)) \right|^2 d\|V\|(x) \leq C E_t(V, x_0, r, S_0).$$

Exercise 7.8 (6.2.6). For stationary graphs V_j with height functions u_j , show

$$E_h(V_j, x_0, r, S_0) = \varepsilon_j^2 \rightarrow 0, \quad \varepsilon_j^{-1} u_j \rightharpoonup u \implies \Delta u = 0,$$

and

$$\exists \theta \in (0, \frac{1}{2}), \quad S_{1,j} \in G(k, n) : \quad E_h(V_j, x_0, \theta r, S_{1,j}) \leq C \theta^2 E_h(V_j, x_0, r, S_0).$$

Iterate the decay estimate to obtain the Hölder tangent-plane field of Theorem 6.2.6.

8. HOMEWORK

Exercise 8.1 (1.1.3). For $\omega_i \in \Lambda^1(V)$ and $\mathbf{v}_j \in \Lambda_1(V)$, show

$$(\omega_1 \wedge \cdots \wedge \omega_k)(\mathbf{v}_1, \dots, \mathbf{v}_k) = \det[\omega_i(\mathbf{v}_j)]_{i,j=1}^k.$$

Proof. Iterating Definition 1.1.2 gives

$$(\omega_1 \wedge \cdots \wedge \omega_k)(\mathbf{v}_1, \dots, \mathbf{v}_k) = \sum_{\sigma \in \mathfrak{S}_k} \text{sgn}(\sigma) \prod_{i=1}^k \omega_i(\mathbf{v}_{\sigma(i)}) = \det[\omega_i(\mathbf{v}_j)]_{i,j=1}^k.$$

□

Exercise 8.2 (1.3.2). For $\omega \in C_c^\infty(\mathbb{R}^n, \Lambda^k(\mathbb{R}^n))$ and $\omega' \in C_c^\infty(\mathbb{R}^n, \Lambda^h(\mathbb{R}^n))$, show

$$d(\omega \wedge \omega') = d\omega \wedge \omega' + (-1)^k \omega \wedge d\omega'.$$

Proof. Expand in the standard exterior basis and commute dx_j through the degree- k factor.

$$\omega = \sum_{\mathbf{i}} \omega_{\mathbf{i}} dx_{\mathbf{i}}, \quad \omega' = \sum_{\mathbf{j}} \omega'_{\mathbf{j}} dx_{\mathbf{j}} \implies d(\omega \wedge \omega') = \sum_{\mathbf{i}, \mathbf{j}} \left(\frac{\partial \omega_{\mathbf{i}}}{\partial x_j} \omega'_{\mathbf{j}} + \omega_{\mathbf{i}} \frac{\partial \omega'_{\mathbf{j}}}{\partial x_j} \right) dx_j \wedge dx_{\mathbf{i}} \wedge dx_{\mathbf{j}}.$$

$$dx_j \wedge dx_{\mathbf{i}} = (-1)^k dx_{\mathbf{i}} \wedge dx_j \implies d(\omega \wedge \omega') = d\omega \wedge \omega' + (-1)^k \omega \wedge d\omega'.$$

□

Exercise 8.3 (1.3.9). For $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, show

$$\int_V \nabla \cdot \mathbf{F} dx = \int_{\partial V} \mathbf{F} \cdot \mathbf{n} dS, \quad \int_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = \int_{\partial S} \mathbf{F} \cdot \boldsymbol{\tau} d\Gamma.$$

Proof. Set $\text{vol} := dx_1 \wedge dx_2 \wedge dx_3$ and use the flux and circulation forms.

$$\psi_2 := \sum_{j=1}^3 (-1)^{j-1} F_j dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_3.$$

$$d\psi_2 = (\nabla \cdot \mathbf{F}) \text{vol}, \quad \psi_2|_{\partial V} = (\mathbf{F} \cdot \mathbf{n}) dS \xrightarrow{1.3.9} \int_V \nabla \cdot \mathbf{F} dx = \int_{\partial V} \mathbf{F} \cdot \mathbf{n} dS.$$

$$\psi_1 := \sum_{j=1}^3 F_j dx_j.$$

$$d\psi_1|_S = ((\nabla \times \mathbf{F}) \cdot \mathbf{n}) dS, \quad \psi_1|_{\partial S} = (\mathbf{F} \cdot \boldsymbol{\tau}) d\Gamma \xrightarrow{1.3.9} \int_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = \int_{\partial S} \mathbf{F} \cdot \boldsymbol{\tau} d\Gamma.$$

□

Exercise 8.4 (2.3.4). For $T \in \mathcal{D}_n(U)$ and the mollification of Definition 2.3.3, set

$$F_\varepsilon(y) := T(\rho_\varepsilon(y - \cdot) dx_1 \wedge \cdots \wedge dx_n),$$

and show

$$F_\varepsilon \in C^\infty(U_\varepsilon), \quad T_\varepsilon = F_\varepsilon \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n, \quad D_{x_j} T_\varepsilon = (D_{x_j} T)_\varepsilon \quad (1 \leq j \leq n).$$

Proof. Set $\text{vol} := dx_1 \wedge \cdots \wedge dx_n$, write $D^\alpha := D_{x_1}^{\alpha_1} \cdots D_{x_n}^{\alpha_n}$, and differentiate the translated kernel.

$$\partial_y^\alpha F_\varepsilon(y) = (-1)^{|\alpha|} T(\partial_x^\alpha \rho_\varepsilon(y - \cdot) \text{vol}) = (D^\alpha T)(\rho_\varepsilon(y - \cdot) \text{vol}) \xrightarrow{2.1.3} F_\varepsilon \in C^\infty(U_\varepsilon).$$

$$\omega = g \text{vol} \implies T_\varepsilon(\omega) = T(\rho_\varepsilon * \omega) = \int_{U_\varepsilon} g(y) F_\varepsilon(y) dy \implies T_\varepsilon = F_\varepsilon \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n.$$

$$\frac{\partial F_\varepsilon}{\partial y_j}(y) = (D_{x_j} T)(\rho_\varepsilon(y - \cdot) \text{vol}) \implies D_{x_j} T_\varepsilon = (D_{x_j} T)_\varepsilon.$$

□

Exercise 8.5 (2.2.4). For $T = [\mathbb{S}^2] \in \mathcal{D}_2(\mathbb{R}^3)$ and $\pi(x_1, x_2, x_3) = (x_1, x_2)$, show

$$\pi_\# T = 0.$$

Proof. Write the sphere current as a boundary and push it forward.

$$T = \partial[B^3] \xrightarrow{2.2.4} \pi_\# T = \partial(\pi_\#[B^3]), \quad \Lambda^3(\mathbb{R}^2) = \{0\} \implies \pi_\#[B^3] = 0 \implies \pi_\# T = 0.$$

□

Exercise 8.6 (4.3.4). For a Radon measure μ on $\mathbb{R}^n$ and $f \in L^1(\mu)$, set

$$Mf(x) := \sup_{r>0} \frac{1}{\mu(B_r(x))} \int_{B_r(x)} |f| d\mu, \quad 0/0 := 0,$$

and show

$$\mu(\{Mf > \lambda\}) \leq \frac{N(n)}{\lambda} \|f\|_{L^1(\mu)},$$

and explain why Lemma 4.3.1 is insufficient.

Proof. Use the bounded-radius assigned-ball selection from the proof of Theorem 4.3.4. Set

$$A_{R,m} := \left\{ x \in B_R(0) : \exists 0 < r < m, \int_{B_r(x)} |f| d\mu > \lambda \mu(B_r(x)) \right\}.$$

$$x \in A_{R,m} \implies \exists B_x = B_{r_x}(x), \quad r_x < m, \quad \int_{B_x} |f| d\mu > \lambda \mu(B_x).$$

$$\{B_x : x \in A_{R,m}\} \xrightarrow{\text{proof of 4.3.4}} A_{R,m} \subset \bigcup_{B \in \mathcal{G}_{R,m}} B, \quad \sum_{B \in \mathcal{G}_{R,m}} \mathbf{1}_B \leq N(n).$$

$$\mu(A_{R,m}) \leq \sum_{B \in \mathcal{G}_{R,m}} \mu(B) < \frac{1}{\lambda} \sum_{B \in \mathcal{G}_{R,m}} \int_B |f| d\mu \leq \frac{N(n)}{\lambda} \|f\|_{L^1(\mu)}.$$

$$m, R \uparrow \infty \implies \mu(\{Mf > \lambda\}) \leq \frac{N(n)}{\lambda} \|f\|_{L^1(\mu)}.$$

$$A_{R,m} \subset \bigcup_i 5B_i, \quad B_i \text{ disjoint}, \quad \mu(A_{R,m}) \leq \sum_i \mu(5B_i), \quad \mu(5B_i) \leq C\mu(B_i) \quad \text{for general Radon } \mu.$$

□

Exercise 8.7 (2.1.7). Show

$$\partial[a, b] = \delta_b - \delta_a, \quad \partial[D] = [D], \quad \partial[v_0, \dots, v_k] = \sum_{j=0}^k (-1)^j [v_0, \dots, \widehat{v_j}, \dots, v_k].$$

Proof. Apply Stokes with the induced boundary orientations; ∂D is counterclockwise for $\tau_D = \mathbf{e}_1 \wedge \mathbf{e}_2$.

$$\partial \llbracket a, b \rrbracket(\psi) = \int_a^b \psi' = \psi(b) - \psi(a) \implies \partial \llbracket a, b \rrbracket = \delta_b - \delta_a, \quad \partial \llbracket D \rrbracket = \llbracket \partial D \rrbracket.$$

$$\partial \llbracket v_0, \dots, v_k \rrbracket(\psi) = \int_{[v_0, \dots, v_k]} d\psi \xrightarrow{1.3.9} \sum_{j=0}^k (-1)^j \int_{[v_0, \dots, \widehat{v}_j, \dots, v_k]} \psi.$$

$$\psi \text{ arbitrary} \implies \partial \llbracket v_0, \dots, v_k \rrbracket = \sum_{j=0}^k (-1)^j \llbracket v_0, \dots, \widehat{v}_j, \dots, v_k \rrbracket.$$

□

Exercise 8.8 (2.2.2). For T of locally finite mass, show

$$T(\omega) = \int_U \langle \omega, \tau_T \rangle d\|T\|, \quad \|\tau_T\| = 1 \quad \|T\| \text{-a.e.}, \quad \mathbf{M}_W(T) = \|T\|(W),$$

and, for $T = \llbracket \Sigma, \tau, \theta \rrbracket$, show

$$\|T\| = \theta \mathcal{H}^k \llcorner \Sigma, \quad \tau_T = \tau \quad \|T\| \text{-a.e.}$$

Proof. On each $W \Subset U$, mass continuity extends the coordinate functionals from $C_c^\infty(W)$ to $C_c(W)$; uniqueness glues their representing measures.

$$L_i(f) := T(f dx_i), \quad |L_i(f)| \leq \mathbf{M}_W(T) \|f\|_\infty \xrightarrow{4.2.5} \exists v_i : L_i(f) = \int_U f dv_i.$$

$$v := \sum_{i \in J(k,n)} v_i \mathbf{e}_i \xrightarrow{4.2.4} v = \sigma |v|_2, \quad |\sigma|_2 = 1 \quad |v|_2 \text{-a.e.}$$

$$d\|T\| := \|\sigma\| d|v|_2, \quad \tau_T := \frac{\sigma}{\|\sigma\|} \implies v = \tau_T \|T\|, \quad \|\tau_T\| = 1 \quad \|T\| \text{-a.e.}$$

$$\omega = \sum_i \omega_i dx_i \implies T(\omega) = \sum_i \int_U \omega_i dv_i = \int_U \langle \omega, \tau_T \rangle d\|T\|.$$

Radon regularity and smooth approximation of measurable norming covectors give

$$\mathbf{M}_W(T) = \sup_{\substack{\omega \in \mathcal{D}^k(U), \text{ spt}(\omega) \subset W \\ \|\omega\|_\infty \leq 1}} \left| \int_U \langle \omega, \tau_T \rangle d\|T\| \right| = \|T\|(W).$$

$$T = \llbracket \Sigma, \tau, \theta \rrbracket \xrightarrow{2.4.5} v = \theta \tau \mathcal{H}^k \llcorner \Sigma \implies \|T\| = \theta \mathcal{H}^k \llcorner \Sigma, \quad \tau_T = \tau \quad \|T\| \text{-a.e.}$$

□

Exercise 8.9 (2.4.3). Show that $\mathbf{F}$ is a norm on compactly supported flat chains and

$$\mathbf{F}(\partial T) \leq \mathbf{F}(T).$$

Proof. Use compatible decompositions on compactly supported normal currents; the estimates pass to their $\mathbf{F}$ -completion.

$$T = R + \partial S, \quad T' = R' + \partial S' \implies T + T' = (R + R') + \partial(S + S') \implies \mathbf{F}(T + T') \leq \mathbf{F}(T) + \mathbf{F}(T').$$

$$cT = cR + \partial(cS) \implies \mathbf{F}(cT) \leq |c| \mathbf{F}(T).$$

$$c \neq 0, \quad T = c^{-1}(cT) \implies \mathbf{F}(T) \leq |c|^{-1} \mathbf{F}(cT) \implies \mathbf{F}(cT) = |c| \mathbf{F}(T).$$

$$T = R + \partial S \implies |T(\omega)| \leq \mathbf{M}(R) \|\omega\|_\infty^* + \mathbf{M}(S) \|d\omega\|_\infty^* \leq C_\omega (\mathbf{M}(R) + \mathbf{M}(S)).$$

$$|T(\omega)| \leq C_\omega \mathbf{F}(T), \quad \mathbf{F}(T) = 0 \implies T(\omega) = 0 \implies T = 0.$$

$$T = R + \partial S \implies \partial T = \partial R = 0 + \partial R \implies \mathbf{F}(\partial T) \leq \mathbf{M}(R) \leq \mathbf{M}(R) + \mathbf{M}(S).$$

$$\mathbf{F}(\partial T) \leq \inf_{T=R+\partial S} (\mathbf{M}(R) + \mathbf{M}(S)) = \mathbf{F}(T).$$



REFERENCES

- [1] L. Ambrosio, N. Fusco, and D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems*, Oxford Mathematical Monographs, Oxford University Press, New York, NY, 2000.
- [2] W. K. Allard, *On the first variation of a varifold: boundary behavior*, Ann. of Math. (2) **101** (1975), 418–446.
- [3] F. J. Almgren, Jr., *Almgren's Big Regularity Paper*, World Scientific Monograph Series in Mathematics **1**, World Scientific, River Edge, NJ, 2000.
- [4] L. Ambrosio and B. Kirchheim, *Rectifiable sets in metric and Banach spaces*, Math. Ann. **318** (2000), 527–555.
- [5] H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, Universitext, Springer, New York, NY, 2011, Chapter 3.
- [6] S. X. Chang, *Two-dimensional area minimizing integral currents are classical minimal surfaces*, J. Amer. Math. Soc. **1** (1988), 699–778.
- [7] C. De Lellis, *Lecture B. Lecture notes*, Geometric Measure Theory.
- [8] C. De Lellis, *Rectifiable Sets, Densities and Tangent Measures*, Zürich Lectures in Advanced Mathematics, European Mathematical Society, Zürich, 2008, Chapters 1, 2, and 4.
- [9] C. De Lellis, *Allard's interior regularity theorem: an invitation to stationary varifolds*, in *Nonlinear Analysis in Geometry and Applied Mathematics, Part 2*, Harvard CMSA Series in Mathematics **2**, International Press, Somerville, MA, 2018, pp. 23–49.
- [10] C. De Lellis, J. Hirsch, A. Marchese, and S. Stuvard, *Regularity of area minimizing currents mod p* , Geom. Funct. Anal. **30** (2020), 1224–1336.
- [11] C. De Lellis, P. Minter, and A. Skorobogatova, *The fine structure of the singular set of area-minimizing integral currents*, preprint, 2023.
- [12] G. De Philippis, C. Gasparetto, and F. Schulze, *A short proof of Allard's and Brakke's regularity theorems*, Int. Math. Res. Not. IMRN **2024** (2024), no. 9, 7594–7613.
- [13] L. C. Evans, *Partial Differential Equations*, 2nd ed., Graduate Studies in Mathematics **19**, American Mathematical Society, Providence, RI, 2010, Chapters 2 and 5.
- [14] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, 2nd ed., Textbooks in Mathematics, CRC Press, Boca Raton, FL, 2025, Chapters 1–3.
- [15] H. Federer, *Geometric Measure Theory*, Grundlehren der mathematischen Wissenschaften **153**, Springer, New York, NY, 1969.
- [16] H. Federer and W. H. Fleming, *Normal and integral currents*, Ann. of Math. (2) **72** (1960), 458–520.
- [17] G. B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Pure and Applied Mathematics, Wiley, New York, NY, 1999, Chapter 9.
- [18] M. Giaquinta, G. Modica, and J. Souček, *Cartesian Currents in the Calculus of Variations I: Cartesian Currents*, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge **37**, Springer, Berlin, 1998.
- [19] M. Giaquinta, G. Modica, and J. Souček, *Cartesian Currents in the Calculus of Variations II: Variational Integrals*, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge **38**, Springer, Berlin, 1998.
- [20] R. Hardt and L. Simon, *Boundary regularity and embedded solutions for the oriented Plateau problem*, Ann. of Math. (2) **110** (1979), 439–486.
- [21] R. L. Jerrard and H. M. Soner, *Functions of bounded higher variation*, Indiana Univ. Math. J. **51** (2002), 645–677.
- [22] S. G. Krantz and H. R. Parks, *Geometric Integration Theory*, Cornerstones, Birkhäuser, Boston, MA, 2008.
- [23] F. Maggi, *Sets of Finite Perimeter and Geometric Variational Problems: An Introduction to Geometric Measure Theory*, Cambridge Studies in Advanced Mathematics **135**, Cambridge University Press, Cambridge, 2012.
- [24] A. Massaccesi, *Lecture A. Lecture notes*, Geometric Measure Theory.
- [25] P. Mattila, *Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability*, Cambridge Studies in Advanced Mathematics **44**, Cambridge University Press, Cambridge, 1995.
- [26] A. Naber and D. Valtorta, *The singular structure and regularity of stationary varifolds*, J. Eur. Math. Soc. (JEMS) **22** (2020), 3305–3382.
- [27] H. L. Royden and P. Fitzpatrick, *Real Analysis*, 5th ed., Pearson, London, 2023, Chapters 1–4, 7, 9–12, 18–19, 21–22.
- [28] L. Simon, *Lectures on Geometric Measure Theory*, Proceedings of the Centre for Mathematical Analysis **3**, Australian National University, Canberra, 1983.